Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Title:	**DESIGN CALCULATION** **3710 TAILINGS PASTE THICKENING** **DESIGN OF FOUNDATION**
Document Number:	**MD-216-3710-EG-CV-CAL-1391**
Revision:	**D01**
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Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	22.01.2020	FIRST ISSUE	AVG	SJM	AG	
A02	26.02.2020	REVISED AS PER EMPLOYER COMMENTS ON 18/02/2020	AVG	SJM	AG	
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D01	28.11.2023	AS BUILT	AVG	SJM	AG	

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document

1. INTENT

This document is to Prepare for design of **Tailings Paste Thickening (3710)** Structure's Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. SCOPE

The analysis has been carried out in Excel and Staad. For member strength calculation force values are directly taken from Load Data Provided & Output of Staad analysis.

3. DESIGN REQUIREMENTS

3.1 Design Load

All Design Loads are indicated in Foundation Load Plan (Doc. No. MD-216-3710-EG-ME-GAD- 1002) attached in Appendix A. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

3.2 Load Combination Load combinations are based on SBC – 301,2007 & SBC-304,2018.

Basic Load Cases are –

1. Tank Dead Load (D)
2. Max. Fluid Load (F1)
3. Normal Operational Fluid Load (F2)
4. Seismic Load (E)
5. Wind Load (W)

Allowable Stress Design

Load Case	Load Combination
201	D + F1
202	D + F1 + W
203	D + F2+ 0.7 E
204	0.6 D + W
205	0.6 (D + F2) + 0.7 E

Ultimate Strength Design

Load Case	Load Combination
301	1.4 (D + F1)
302	1.2 (D + F1) + 1.6 W
303	1.2 (D + F2) + E
304	0.9 D + 1.6 W
305	0.9 (D+ F2) + E

3.3 Stability against Overturning & Sliding

Overturning

Structures and buildings foundations, shall have the following minimum stability ratios against overturning:

- 1.5 For load combinations with wind or seismic loads.
- 2.0 For all other load combinations.

Sliding

A minimum factor of safety against sliding shall be 1.5.

4. CONCRETE FOUNDATIONS

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be shallow isolated type as per SBC-304, Clause 13.3. For design calculations as determined by the standard in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001).

4.4 Grouting

Grout shall be 30 mm thick cementitious under steel column bases of Strength 40 MPa.

4.5 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.6 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation and Column	50

4.7 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m^2 at 1.5m depth below EGL (ie. RL 910.50m) refer Table 6 - BH-14 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

5. DESIGN CALCULATION FOR FOUNDATION

MA'ADEN معادن		Outotec LARSEN & TOUBRO
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DESIGN OF FOUNDATION MKD. F1

Size of Columns	Along-X	Along-Z	ex	ez
Outer Column A	0.80m	0.80m	0.0000m	-5.10m
2nd Middle Column B	0.65m	0.65m	0.0000m	-1.60m
1st Middle Column C	0.50m	0.50m	0.0000m	2.05m
Inner Column D	0.50m	0.50m	0.0000m	6.00m
Size of fdn. raft:	1.80m	13.50m		
Thickness of raft:	0.75 m			
Original Gr. Level:	0.00 m			
Finished Gr. Level:	0.00 m			
U/S of B/Pl Above GL:	0.55 m			
U/S level of fdn:	-1.50 m			
Unit weight of conc:	25 kN/m^3			
Unit weight of soil:	18 kN/m^3			

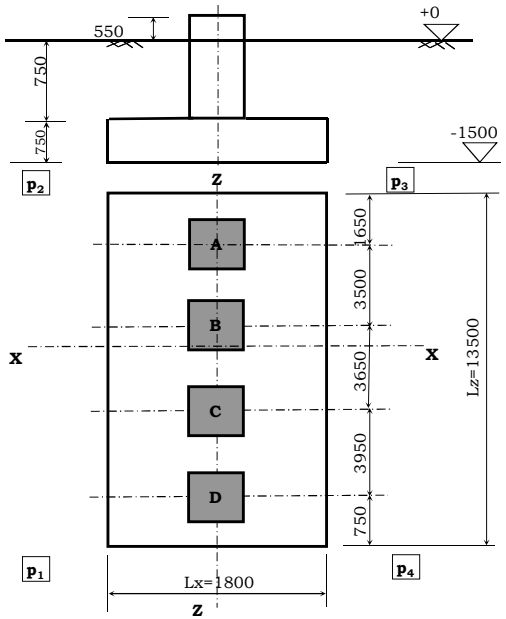
Calculation of Weight of Foundation :

Weight of raft = (1.8*13.5*0.75*25) = 455.63KN

Weight of soil = ((1.8*13.5)-2*(0.8*0.8))*0.75*18 = 310.77KN

Total = 766.40KN

Submerged wt. = 766.4-1.8*13.5*1.5*10 = 401.90KN



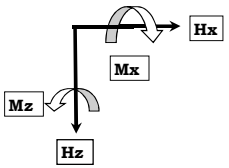
Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	25mm
Dia. of bars provided along Z direction at top=	32mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 676mm = 0.676m

Effective depth of slab along Z direction = 684mm = 0.684m

Clear projection along X direction = 0.65m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Basic Load Cases :

		Outer Column A		2nd Middle Column B		1st Middle Column C		Inner Column D	
Sl. No	Basic Load Case	V (kN)	Hx (kN)	V (kN)	Hx (kN)	V (kN)	Hx (kN)	V (kN)	Hx (kN)
1	Tank Dead Load (D)	90.0	0.0	23.0	0.0	24.0	0.0	20.0	0.0
2	Max. Fluid Load (F1)	2354.0	142.0	2714.0	0.0	2440.0	0.0	2005.0	0.0
3	Normal Operational Fluid Load (F2)	2018.0	122.0	2326.0	0.0	2091.0	0.0	1718.0	0.0
4	Wind Load (W)(+/-)	77.0	22.0	14.0	0.0	3.0	0.0	11.0	0.0
5	Earthquake Load (E)(+/-)	179.0	106.0	56.0	0.0	6.0	0.0	28.0	0.0

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	D + F1	9670.0	142.0	0.0	0.0	357.6	10436.40	142.00	291.10	0.00	357.60
202	D + F1 + W	9775.0	164.0	0.0	0.0	14.7	10541.40	164.00	336.20	0.00	14.65
203	D + F2+ 0.7 E	8498.3	196.2	0.0	0.0	-321.0	9264.70	196.20	402.21	0.00	-320.99
204	0.6 D + W	199.2	22.0	0.0	0.0	-538.9	965.60	22.00	45.10	0.00	-538.91
205	0.6(D+F2)+0.7 E	5174.3	147.4	0.0	0.0	-422.8	5940.70	147.40	302.17	0.00	-422.81
Ultimate Conditions											
301	1.4 (D + F1)	13538.0	198.8	0.0	0.0	500.6	14304.40	198.80	407.54	0.00	500.64
302	1.2(D + F1) + 1.6 W	11772.0	205.6	0.0	0.0	-119.6	12538.40	205.60	421.48	0.00	-119.60
303	1.2(D + F2) + E	10241.0	252.4	0.0	0.0	-516.7	11007.40	252.40	517.42	0.00	-516.74
304	0.9 D + 1.6 W	309.3	35.2	0.0	0.0	-842.7	1075.70	35.20	72.16	0.00	-842.66
305	0.9 (D+ F2) + E	7748.0	215.8	0.0	0.0	-593.1	8514.40	215.80	442.39	0.00	-593.11

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Check for overturning and sliding:

Load Cases:		Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D + F1	10436.40	142.00	291.10	0.00	357.60	32.27	197.00	29.40	
202	D + F1 + W	10541.40	164.00	336.20	0.00	14.65	28.22	4856.96	25.71	
203	D + F2+ 0.7 E	9264.70	196.20	402.21	0.00	320.99	20.73	194.82	18.89	
204	0.6 D + W	965.60	22.00	45.10	0.00	538.91	19.27	12.09	17.56	
205	0.6(D+F2)+0.7 E	5940.70	147.40	302.17	0.00	422.81	17.69	94.84	16.12	

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Load Combination & Calculation of Base pressure

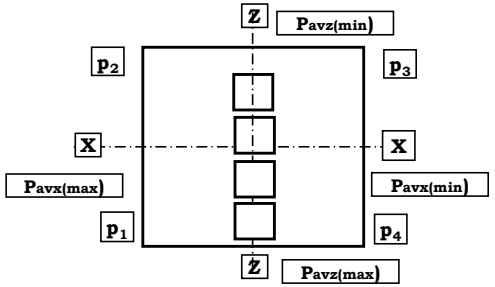
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m²

Net safe bearing capacity of soil for DL, LL= 500 kN/m² For Wind load/Seis. load cas 625 kN/m²

Gross safe bearing capacity of soil for DL, LL= 527 kN/m² For Wind load/Seis. load cas 652 kN/m²



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m²)												Remarks
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2.Lz$	6.Mx/ $Lx.(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	
201	D + F1	10436.40	291.10	357.60	429.48	39.93	6.54	475.95	462.87	396.09	383.01	448.95	469.41	389.55	436.02	422.94	**O.K**
202	D + F1 + W	10541.40	336.20	14.65	433.80	46.12	0.27	480.19	479.65	387.95	387.42	453.19	479.92	387.69	434.07	433.54	**O.K**
203	D + F2+ 0.7 E	9264.70	402.21	-320.99	381.26	55.17	5.87	442.31	430.57	331.96	320.22	415.31	436.44	326.09	387.14	375.40	**O.K**
204	0.6 D + W	965.60	45.10	-538.91	39.74	6.19	9.86	55.78	36.07	43.41	23.69	28.78	45.93	33.55	49.60	29.88	**O.K**
205	0.6(D+F2)+0.7 E	5940.70	302.17	-422.81	244.47	41.45	7.73	293.66	278.19	210.76	195.29	266.66	285.93	203.03	252.21	236.74	**O.K**
301	1.4 (D + F1)	14304.40	407.54	500.64	588.66	55.90	9.16	653.72	635.41	541.91	523.60	626.72	644.57	532.76	597.82	579.51	
302	1.2(D + F1) + 1.6 W	12538.40	421.48	-119.60	515.98	57.82	2.19	575.99	571.61	460.35	455.98	548.99	573.80	458.17	518.17	513.80	
303	1.2(D + F2) + E	11007.40	517.42	-516.74	452.98	70.98	9.45	533.41	514.50	391.45	372.55	506.41	523.96	382.00	462.43	443.53	
304	0.9 D + 1.6 W	1075.70	72.16	-842.66	44.27	9.90	15.41	69.58	38.75	49.78	18.96	42.58	54.17	34.37	59.68	28.86	
305	0.9 (D+ F2) + E	8514.40	442.39	-593.11	350.39	60.68	10.85	421.92	400.22	300.55	278.85	394.92	411.07	289.70	361.24	339.54	

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DESIGN OF FOUNDATION MKD. F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.8+2*0.5*0.684)+(0.8+2*0.5*0.684))*2

= 5.936 m

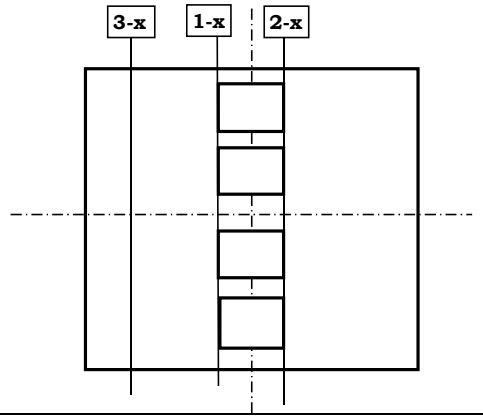
Punching Area= ((0.8+2*0.5*0.684)*(0.8+2*0.5*0.684))

= 2.2 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.75*25+0.75*0.75*18)

= 32.25 kN/m^2

Area of footing, A_f= 24.30 m^2



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DESIGN OF FOUNDATION MKD. F1

Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
301	1.4 (D + F1)	604.19	573.14	646.19	126.51	108.58	-15.94	2092.37
302	1.2(D + F1) + 1.6 W	532.04	499.93	575.47	111.46	92.92	-14.10	1839.30
303	1.2(D + F2) + E	472.70	433.26	526.01	100.26	77.49	-12.81	1668.67
304	0.9 D + 1.6 W	47.02	41.52	54.46	4.13	0.95	-0.57	54.56
305	0.9 (D+ F2) + E	367.24	333.53	412.82	76.94	57.47	-9.87	1268.27

Overhang = 0.65 m
Effective depth = 0.676 m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	126.51	0.28	0.37	0.001	597.84	1216.80	16ϕT	165.24	300	150	1340.41
At top	0.95	0.00	0.00	0.000	0.00	608.40	12ϕT	185.89	300	175	646.27

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= -0.57 kN/m

Shear stress , τ_v= (-0.57*1000)/(1000*676)

= 0.00 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 2092.37 kN

Strength of Concrete in shear Vc1 = 11869.83 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 8462.46 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 7913.22 kN **Table 22.6.5.2_ (c)**

Design Vc= 7913.22 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (2092.37*1000)/(5936*684)

= 0.52 N/mm^2 **No shear reinforcement reqd.**

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Calculation of distribution reinf in Z direction :

Effective depth = 0.684 m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	474.62	1.01	1.35	0.003	2255.30	1231.20	25ϕT	217.65	300	200	2454.37
At top	930.78	1.99	2.65	0.007	4531.91	615.60	32ϕT	177.46	300	175	4595.70

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 367.78 kN/m

Shear stress , τ_v= (367.78*1000)/(1000*684)

= 0.54 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

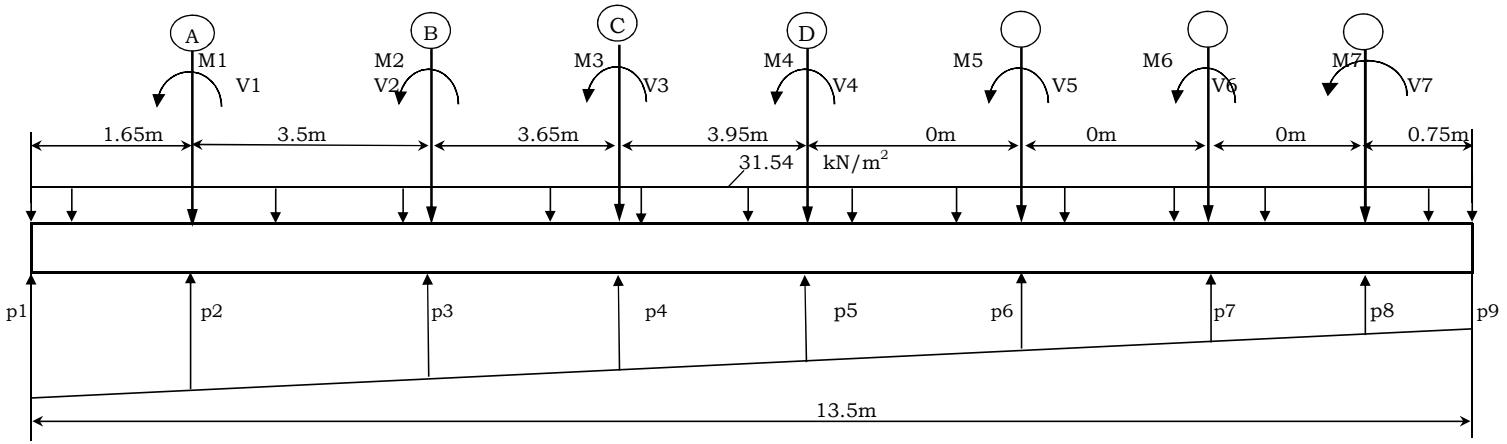
Along X Direction :			
At Bottom	**16ϕT**	**150**	c/c
At Top	**12ϕT**	**175**	c/c
Along Z Direction :			
At Bottom	**25ϕT**	**200**	c/c
At Top	**32ϕT**	**175**	c/c

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900707_GAU01_ST03_CED01_ 37101_OU100665185_R3	DESIGN CALCULATION 3710 TAILINGS PASTE THICKENING DESIGN OF FOUNDATION	Date: 28.11.2023
O19045_450_C_K_ 24150 _001_R3	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F1

CALCULATION OF BM & SF :

(Along Z direction)



Load Case	Load Combination	Soil pressure in UDL (kN/m)								
		p1	p2	p3	p4	p5	p6	p7	p8	p9
301	1.4 (D + F1)	1,160.22	By STAAD							958.96
302	1.2(D + F1) + 1.6 W	1,032.84								824.70
303	1.2(D + F2) + E	943.12								687.60
304	0.9 D + 1.6 W	97.50								61.87
305	0.9 (D+ F2) + E	739.93								521.46

From STAAD Output	
BM at top	1675.40 (kNm/m)
BM at bot.	854.31 (kNm/m)
SF (T. at top)	560.40 (kN/m)
SF (T. at bottom)	662.00 (kN/m)

Load Case	Load Combination	Vertical Load (kN/m)							Moment (kNm/m)						
		V1	V2	V3	V4	V5	V6	V7	M1	M2	M3	M4	M5	M6	M7
301	1.4 (D + F1)	3421.60	3831.80	3449.60	2835.00				407.54						
302	1.2(D + F1) + 1.6 W	3009.80	3298.40	2959.80	2441.00				421.48						
303	1.2(D + F2) + E	2708.60	2874.80	2544.00	2113.60				517.42						
304	0.9 D + 1.6 W	158.00	34.70	24.60	29.00				72.16						
305	0.9 (D+ F2) + E	2076.20	2170.10	1909.50	1592.20				442.39						

Software licensed to LNT ECC DIV	Job No 3710	Sheet No 1	Rev
	Part		
Job Title Mansourah Massarah Gold Project	Ref		
	By AvG	Date 21-Jan-20	Chd SJM
Client Ma'aden	File F1.std	Date/Time 06-Mar-2020 14:41	

Beam Force Detail

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
1	301:1.4 (D + F	0.000	0.00	0.00	0.00	0.00	0.00	-0.00
		0.165	0.00	181.23	0.00	0.00	0.00	-16.62
		0.330	0.00	362.07	0.00	0.00	0.00	-62.27
		0.495	0.00	542.51	0.00	0.00	0.00	-136.91
		0.660	0.00	722.55	0.00	0.00	0.00	-240.46
		0.825	0.00	902.20	0.00	0.00	0.00	-372.86
		0.990	0.00	1081.40	0.00	0.00	0.00	-538.15
		1.155	0.00	1260.21	0.00	0.00	0.00	-732.16
		1.320	0.00	1438.62	0.00	0.00	0.00	-954.82
		1.485	0.00	1616.64	0.00	0.00	0.00	-1206.07
		1.650	-0.00	1794.26	-0.00	-0.00	-0.00	-1485.85
	302:1.2(D + F1	0.000	0.00	-0.00	0.00	0.00	0.00	-0.00
		0.165	0.00	160.09	0.00	0.00	0.00	-14.68
		0.330	0.00	319.77	0.00	0.00	0.00	-55.01
		0.495	0.00	479.04	0.00	0.00	0.00	-120.91
		0.660	0.00	637.90	0.00	0.00	0.00	-212.34
		0.825	0.00	796.36	0.00	0.00	0.00	-329.22
		0.990	0.00	954.34	0.00	0.00	0.00	-475.10
		1.155	0.00	1111.93	0.00	0.00	0.00	-646.30
		1.320	0.00	1269.10	0.00	0.00	0.00	-842.74
		1.485	0.00	1425.86	0.00	0.00	0.00	-1064.36
		1.650	-0.00	1582.22	-0.00	-0.00	-0.00	-1311.10
	303:1.2(D + F2	0.000	0.00	-0.00	0.00	0.00	0.00	-0.00
		0.165	0.00	145.35	0.00	0.00	0.00	-13.33
		0.330	0.00	290.19	0.00	0.00	0.00	-49.93
		0.495	0.00	434.54	0.00	0.00	0.00	-109.73
		0.660	0.00	578.39	0.00	0.00	0.00	-192.65
		0.825	0.00	721.73	0.00	0.00	0.00	-298.60
		0.990	0.00	864.50	0.00	0.00	0.00	-430.78
		1.155	0.00	1006.77	0.00	0.00	0.00	-585.82
		1.320	0.00	1148.54	0.00	0.00	0.00	-763.64
		1.485	0.00	1289.81	0.00	0.00	0.00	-964.16
		1.650	-0.00	1430.58	-0.00	-0.00	-0.00	-1187.31
	304:0.9 D + 1.6	0.000	0.00	0.00	0.00	0.00	0.00	0.00
		0.165	0.00	5.95	0.00	0.00	0.00	-0.55
		0.330	0.00	11.83	0.00	0.00	0.00	-2.04
		0.495	0.00	17.64	0.00	0.00	0.00	-4.47
		0.660	0.00	23.38	0.00	0.00	0.00	-7.83
		0.825	0.00	29.05	0.00	0.00	0.00	-12.11
		0.990	0.00	34.64	0.00	0.00	0.00	-17.41
		1.155	0.00	40.16	0.00	0.00	0.00	-23.61
		1.320	0.00	45.61	0.00	0.00	0.00	-30.69
		1.485	0.00	50.99	0.00	0.00	0.00	-38.63

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d	Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		1.650	-0.00	56.30	-0.00	-0.00	-0.00	-47.44
	305:0.9 (D+ F2	0.000	0.00	0.00	0.00	0.00	0.00	-0.00
		0.165	0.00	111.86	0.00	0.00	0.00	-10.26
		0.330	0.00	223.30	0.00	0.00	0.00	-38.43
		0.495	0.00	334.31	0.00	0.00	0.00	-84.44
		0.660	0.00	444.89	0.00	0.00	0.00	-148.22
		0.825	0.00	555.04	0.00	0.00	0.00	-229.71
		0.990	0.00	664.70	0.00	0.00	0.00	-331.35
		1.155	0.00	773.93	0.00	0.00	0.00	-450.54
		1.320	0.00	882.74	0.00	0.00	0.00	-587.23
		1.485	0.00	991.12	0.00	0.00	0.00	-741.33
		1.650	-0.00	1099.06	-0.00	-0.00	-0.00	-912.79
2	301:1.4 (D + F	0.000	0.00	-1864.14	0.00	0.00	0.00	-1078.31
		0.350	0.00	-1488.86	0.00	0.00	0.00	-498.88
		0.700	0.00	-1115.36	0.00	0.00	0.00	-46.82
		1.050	0.00	-743.63	0.00	0.00	0.00	278.44
		1.400	0.00	-373.67	0.00	0.00	0.00	477.51
		1.750	0.00	-5.50	0.00	0.00	0.00	550.97
		2.100	0.00	360.65	0.00	0.00	0.00	481.65
		2.450	0.00	725.03	0.00	0.00	0.00	288.06
		2.800	0.00	1087.63	0.00	0.00	0.00	-29.22
		3.150	0.00	1448.45	0.00	0.00	0.00	-469.58
		3.500	-0.00	1807.50	-0.00	-0.00	-0.00	-1032.44
	302:1.2(D + F1	0.000	0.00	-1653.38	0.00	0.00	0.00	-889.62
		0.350	0.00	-1323.26	0.00	0.00	0.00	-375.18
		0.700	0.00	-994.97	0.00	0.00	0.00	27.26
		1.050	0.00	-668.52	0.00	0.00	0.00	318.31
		1.400	0.00	-343.91	0.00	0.00	0.00	498.58
		1.750	0.00	-21.14	0.00	0.00	0.00	568.69
		2.100	0.00	299.54	0.00	0.00	0.00	513.69
		2.450	0.00	618.38	0.00	0.00	0.00	349.89
		2.800	0.00	935.39	0.00	0.00	0.00	77.92
		3.150	0.00	1250.56	0.00	0.00	0.00	-301.61
		3.500	-0.00	1563.89	-0.00	-0.00	-0.00	-788.10
	303:1.2(D + F2	0.000	0.00	-1490.68	0.00	0.00	0.00	-669.89
		0.350	0.00	-1193.98	0.00	0.00	0.00	-205.91
		0.700	0.00	-899.53	0.00	0.00	0.00	157.52
		1.050	0.00	-607.34	0.00	0.00	0.00	421.15
		1.400	0.00	-317.40	0.00	0.00	0.00	585.72
		1.750	0.00	-29.71	0.00	0.00	0.00	652.00
		2.100	0.00	255.40	0.00	0.00	0.00	606.90
		2.450	0.00	538.26	0.00	0.00	0.00	465.19
		2.800	0.00	818.86	0.00	0.00	0.00	227.62
		3.150	0.00	1097.20	0.00	0.00	0.00	-105.06

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d	Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		3.500	-0.00	1373.30	-0.00	-0.00	-0.00	-532.09
	304:0.9 D + 1.6	0.000	0.00	-89.23	0.00	0.00	0.00	24.72
		0.350	0.00	-78.23	0.00	0.00	0.00	53.81
		0.700	0.00	-67.54	0.00	0.00	0.00	79.20
		1.050	0.00	-57.17	0.00	0.00	0.00	101.02
		1.400	0.00	-47.11	0.00	0.00	0.00	119.35
		1.750	0.00	-37.37	0.00	0.00	0.00	134.32
		2.100	0.00	-27.98	0.00	0.00	0.00	145.56
		2.450	0.00	-18.91	0.00	0.00	0.00	153.67
		2.800	0.00	-10.16	0.00	0.00	0.00	158.75
		3.150	0.00	-1.72	0.00	0.00	0.00	160.90
		3.500	-0.00	6.41	-0.00	-0.00	-0.00	160.23
	305:0.9 (D+ F2	0.000	0.00	-1144.94	0.00	0.00	0.00	-470.40
		0.350	0.00	-917.58	0.00	0.00	0.00	-113.93
		0.700	0.00	-692.14	0.00	0.00	0.00	165.52
		1.050	0.00	-468.64	0.00	0.00	0.00	368.59
		1.400	0.00	-247.06	0.00	0.00	0.00	495.93
		1.750	0.00	-27.41	0.00	0.00	0.00	548.18
		2.100	0.00	190.04	0.00	0.00	0.00	515.44
		2.450	0.00	405.56	0.00	0.00	0.00	409.05
		2.800	0.00	619.16	0.00	0.00	0.00	229.66
		3.150	0.00	830.82	0.00	0.00	0.00	-22.09
		3.500	-0.00	1040.56	-0.00	-0.00	-0.00	-345.56
3	301:1.4 (D + F	0.000	0.00	-2100.61	0.00	0.00	0.00	-1032.44
		0.365	0.00	-1728.34	0.00	0.00	0.00	-341.26
		0.730	0.00	-1358.00	0.00	0.00	0.00	218.18
		1.095	0.00	-989.58	0.00	0.00	0.00	646.54
		1.460	0.00	-623.10	0.00	0.00	0.00	944.51
		1.825	0.00	-258.55	0.00	0.00	0.00	1112.75
		2.190	0.00	103.79	0.00	0.00	0.00	1133.59
		2.555	0.00	464.20	0.00	0.00	0.00	1026.21
		2.920	0.00	822.69	0.00	0.00	0.00	791.29
		3.285	0.00	1179.24	0.00	0.00	0.00	429.49
		3.650	-0.00	1533.86	-0.00	-0.00	-0.00	-58.52
	302:1.2(D + F1	0.000	0.00	-1815.48	0.00	0.00	0.00	-788.10
		0.365	0.00	-1490.96	0.00	0.00	0.00	-191.32
		0.730	0.00	-1168.43	0.00	0.00	0.00	290.69
		1.095	0.00	-847.90	0.00	0.00	0.00	658.60
		1.460	0.00	-529.36	0.00	0.00	0.00	913.11
		1.825	0.00	-212.82	0.00	0.00	0.00	1054.92
		2.190	0.00	101.43	0.00	0.00	0.00	1068.81
		2.555	0.00	413.69	0.00	0.00	0.00	971.57
		2.920	0.00	723.95	0.00	0.00	0.00	763.88
		3.285	0.00	1032.22	0.00	0.00	0.00	446.44

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		3.650	-0.00	1338.49	-0.00	-0.00	-0.00	19.94
	303:1.2(D + F2	0.000	0.00	-1569.20	0.00	0.00	0.00	-532.09
		0.365	0.00	-1284.02	0.00	0.00	0.00	-17.23
		0.730	0.00	-1001.29	0.00	0.00	0.00	396.89
		1.095	0.00	-721.01	0.00	0.00	0.00	711.12
		1.460	0.00	-443.18	0.00	0.00	0.00	926.32
		1.825	0.00	-167.81	0.00	0.00	0.00	1043.34
		2.190	0.00	104.77	0.00	0.00	0.00	1049.25
		2.555	0.00	374.89	0.00	0.00	0.00	958.89
		2.920	0.00	642.56	0.00	0.00	0.00	773.12
		3.285	0.00	907.78	0.00	0.00	0.00	492.79
		3.650	-0.00	1170.55	-0.00	-0.00	-0.00	118.75
	304:0.9 D + 1.6	0.000	0.00	-24.37	0.00	0.00	0.00	160.23
		0.365	0.00	-16.28	0.00	0.00	0.00	167.47
		0.730	0.00	-8.53	0.00	0.00	0.00	171.91
		1.095	0.00	-1.12	0.00	0.00	0.00	173.66
		1.460	0.00	5.95	0.00	0.00	0.00	172.83
		1.825	0.00	12.68	0.00	0.00	0.00	169.56
		2.190	0.00	19.01	0.00	0.00	0.00	163.64
		2.555	0.00	25.00	0.00	0.00	0.00	155.53
		2.920	0.00	30.66	0.00	0.00	0.00	145.36
		3.285	0.00	35.97	0.00	0.00	0.00	133.25
		3.650	-0.00	40.93	-0.00	-0.00	-0.00	119.30
	305:0.9 (D+ F2	0.000	0.00	-1181.27	0.00	0.00	0.00	-345.56
		0.365	0.00	-964.89	0.00	0.00	0.00	41.67
		0.730	0.00	-750.60	0.00	0.00	0.00	352.50
		1.095	0.00	-538.41	0.00	0.00	0.00	587.68
		1.460	0.00	-328.32	0.00	0.00	0.00	747.91
		1.825	0.00	-120.32	0.00	0.00	0.00	833.95
		2.190	0.00	85.28	0.00	0.00	0.00	836.11
		2.555	0.00	288.79	0.00	0.00	0.00	765.71
		2.920	0.00	490.20	0.00	0.00	0.00	623.48
		3.285	0.00	689.51	0.00	0.00	0.00	410.13
		3.650	-0.00	886.73	-0.00	-0.00	-0.00	126.41
4	301:1.4 (D + F	0.000	0.00	-1824.67	0.00	0.00	0.00	-58.52
		0.395	0.00	-1443.39	0.00	0.00	0.00	578.49
		0.790	0.00	-1064.37	0.00	0.00	0.00	1069.53
		1.185	0.00	-687.62	0.00	0.00	0.00	1415.46
		1.580	0.00	-313.12	0.00	0.00	0.00	1617.13
		1.975	0.00	59.12	0.00	0.00	0.00	1675.40
		2.370	0.00	428.77	0.00	0.00	0.00	1570.86
		2.765	0.00	796.16	0.00	0.00	0.00	1324.82
		3.160	0.00	1161.28	0.00	0.00	0.00	938.14
		3.555	0.00	1524.15	0.00	0.00	0.00	411.67

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d	Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		3.950	-0.00	1884.75	-0.00	-0.00	-0.00	-253.74
	302:1.2(D + F	0.000	0.00	-1551.25	0.00	0.00	0.00	19.94
		0.395	0.00	-1222.38	0.00	0.00	0.00	560.44
		0.790	0.00	-895.85	0.00	0.00	0.00	975.12
		1.185	0.00	-571.66	0.00	0.00	0.00	1264.87
		1.580	0.00	-249.81	0.00	0.00	0.00	1430.56
		1.975	0.00	69.71	0.00	0.00	0.00	1473.06
		2.370	0.00	386.55	0.00	0.00	0.00	1375.93
		2.765	0.00	701.05	0.00	0.00	0.00	1157.59
		3.160	0.00	1013.22	0.00	0.00	0.00	818.93
		3.555	0.00	1323.04	0.00	0.00	0.00	360.84
		3.950	-0.00	1630.53	-0.00	-0.00	-0.00	-215.82
	303:1.2(D + F	0.000	0.00	-1289.98	0.00	0.00	0.00	118.75
		0.395	0.00	-1008.77	0.00	0.00	0.00	566.49
		0.790	0.00	-730.43	0.00	0.00	0.00	906.83
		1.185	0.00	-454.96	0.00	0.00	0.00	1140.83
		1.580	0.00	-182.37	0.00	0.00	0.00	1269.59
		1.975	0.00	87.36	0.00	0.00	0.00	1294.19
		2.370	0.00	353.80	0.00	0.00	0.00	1201.12
		2.765	0.00	617.37	0.00	0.00	0.00	1006.32
		3.160	0.00	878.08	0.00	0.00	0.00	710.86
		3.555	0.00	1135.91	0.00	0.00	0.00	315.82
		3.950	-0.00	1390.87	-0.00	-0.00	-0.00	-177.71
	304:0.9 D + 1.6	0.000	0.00	11.34	0.00	0.00	0.00	119.30
		0.395	0.00	16.28	0.00	0.00	0.00	113.73
		0.790	0.00	20.81	0.00	0.00	0.00	106.34
		1.185	0.00	24.95	0.00	0.00	0.00	97.29
		1.580	0.00	28.68	0.00	0.00	0.00	86.73
		1.975	0.00	32.01	0.00	0.00	0.00	74.80
		2.370	0.00	34.89	0.00	0.00	0.00	61.51
		2.765	0.00	37.37	0.00	0.00	0.00	47.20
		3.160	0.00	39.44	0.00	0.00	0.00	32.01
		3.555	0.00	41.12	0.00	0.00	0.00	16.10
		3.950	-0.00	42.39	-0.00	-0.00	-0.00	-0.37
	305:0.9 (D+ F	0.000	0.00	-953.46	0.00	0.00	0.00	126.41
		0.395	0.00	-742.73	0.00	0.00	0.00	456.70
		0.790	0.00	-534.46	0.00	0.00	0.00	706.57
		1.185	0.00	-328.64	0.00	0.00	0.00	876.94
		1.580	0.00	-125.28	0.00	0.00	0.00	968.73
		1.975	0.00	75.63	0.00	0.00	0.00	982.87
		2.370	0.00	273.74	0.00	0.00	0.00	909.45
		2.765	0.00	469.39	0.00	0.00	0.00	760.44
		3.160	0.00	662.58	0.00	0.00	0.00	536.79
		3.555	0.00	853.32	0.00	0.00	0.00	239.40

Beam Force Detail Cont...

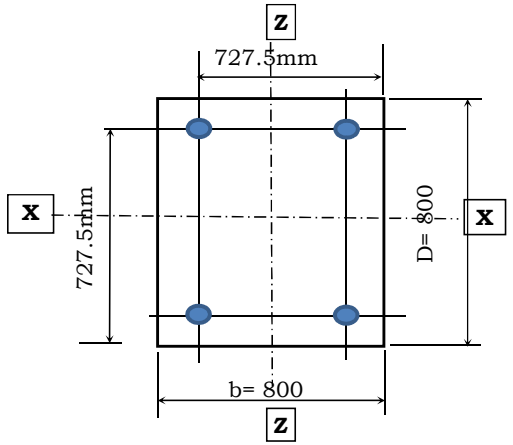
			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		3.950	-0.00	1041.60	-0.00	-0.00	-0.00	-130.79
5	301:1.4 (D + F	0.000	0.00	-678.05	0.00	0.00	0.00	-253.74
		0.075	0.00	-609.87	0.00	0.00	0.00	-205.73
		0.150	0.00	-541.77	0.00	0.00	0.00	-162.69
		0.225	0.00	-473.76	0.00	0.00	0.00	-124.61
		0.300	0.00	-405.83	0.00	0.00	0.00	-91.48
		0.375	0.00	-337.97	0.00	0.00	0.00	-63.30
		0.450	0.00	-270.22	0.00	0.00	0.00	-40.78
		0.525	0.00	-202.54	0.00	0.00	0.00	-23.19
		0.600	0.00	-134.95	0.00	0.00	0.00	-10.54
		0.675	0.00	-67.43	0.00	0.00	0.00	-2.81
		0.750	-0.00	0.00	-0.00	-0.00	-0.00	0.00
	302:1.2(D + F1	0.000	0.00	-576.96	0.00	0.00	0.00	-215.82
		0.075	0.00	-518.88	0.00	0.00	0.00	-174.97
		0.150	0.00	-460.88	0.00	0.00	0.00	-138.35
		0.225	0.00	-402.97	0.00	0.00	0.00	-105.95
		0.300	0.00	-345.14	0.00	0.00	0.00	-77.78
		0.375	0.00	-287.40	0.00	0.00	0.00	-53.82
		0.450	0.00	-229.75	0.00	0.00	0.00	-34.67
		0.525	0.00	-172.19	0.00	0.00	0.00	-19.71
		0.600	0.00	-114.71	0.00	0.00	0.00	-8.96
		0.675	0.00	-57.31	0.00	0.00	0.00	-2.39
		0.750	-0.00	0.00	-0.00	-0.00	-0.00	-0.00
	303:1.2(D + F2	0.000	0.00	-475.66	0.00	0.00	0.00	-177.71
		0.075	0.00	-427.62	0.00	0.00	0.00	-144.04
		0.150	0.00	-379.69	0.00	0.00	0.00	-113.86
		0.225	0.00	-331.86	0.00	0.00	0.00	-87.18
		0.300	0.00	-284.13	0.00	0.00	0.00	-63.98
		0.375	0.00	-236.50	0.00	0.00	0.00	-44.26
		0.450	0.00	-188.99	0.00	0.00	0.00	-28.50
		0.525	0.00	-141.59	0.00	0.00	0.00	-16.21
		0.600	0.00	-94.29	0.00	0.00	0.00	-7.36
		0.675	0.00	-47.09	0.00	0.00	0.00	-1.96
		0.750	-0.00	0.00	-0.00	-0.00	-0.00	-0.00
	304:0.9 D + 1.6	0.000	0.00	-1.24	0.00	0.00	0.00	-0.37
		0.075	0.00	-1.05	0.00	0.00	0.00	-0.29
		0.150	0.00	-0.88	0.00	0.00	0.00	-0.22
		0.225	0.00	-0.72	0.00	0.00	0.00	-0.16
		0.300	0.00	-0.57	0.00	0.00	0.00	-0.11
		0.375	0.00	-0.44	0.00	0.00	0.00	-0.07
		0.450	0.00	-0.32	0.00	0.00	0.00	-0.04
		0.525	0.00	-0.22	0.00	0.00	0.00	-0.02
		0.600	0.00	-0.13	0.00	0.00	0.00	-0.01
		0.675	0.00	-0.06	0.00	0.00	0.00	-0.00

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		0.750	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00
	305:0.9 (D+ F2	0.000	0.00	-350.29	0.00	0.00	0.00	-130.79
		0.075	0.00	-314.85	0.00	0.00	0.00	-105.99
		0.150	0.00	-279.51	0.00	0.00	0.00	-83.78
		0.225	0.00	-244.25	0.00	0.00	0.00	-64.14
		0.300	0.00	-209.08	0.00	0.00	0.00	-47.07
		0.375	0.00	-174.01	0.00	0.00	0.00	-32.55
		0.450	0.00	-139.03	0.00	0.00	0.00	-20.96
		0.525	0.00	-104.14	0.00	0.00	0.00	-11.92
		0.600	0.00	-69.34	0.00	0.00	0.00	-5.41
		0.675	0.00	-34.62	0.00	0.00	0.00	-1.44
		0.750	-0.00	0.00	-0.00	-0.00	-0.00	-0.00

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900707_GAU01_ST03_CED01_ 37101_OU100665185_R3	DESIGN CALCULATION 3710 TAILINGS PASTE THICKENING DESIGN OF FOUNDATION	Date: 28.11.2023
O19045_450_C_K_ 24150 _001_R3	AS BUILT	Rev: D01

DESIGN OF COLUMN MKD C1 **(Outer Column)**



Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	72.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%

Table 20.6.1.3.1

For Compression-controlled sections

Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1300mm
Unsupproted length of column about Z (l_{uz}) =	1300mm

For Columns not braced against sidesway,

Where, Effective length factor K= 2

Slenderness ratio, $K\,l_{ux} / r_x$ = 11.26 < 22 SECTION 6.2.5a Radius of gyration r_x= 231 mm

$K\,l_{uz} / r_z$ = 11.26 < 22 SECTION 6.2.5a Radius of gyration r_z= 231 mm

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	90.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	2354.0	142.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	2018.0	122.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	77.0	22.0	0.0	0.0	0.0
5	Earthquake Load (E)(+/-)	179.0	106.0	0.0	0.0	0.0
			γ_x =	0.84		
			γ_z =	0.84		

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = P_n/f_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	342.16	25.8	0.0	526.4	39.8	0.0	0.24
302	1.2(D + F1) + 1.6 W	305.6	26.7	0.0	470.2	41.1	0.0	0.21
303	1.2(D + F2) + E	311.2	32.8	0.0	478.7	50.5	0.0	0.21
304	0.9 D + 1.6 W	20.4	4.6	0.0	31.4	7.0	0.0	0.01
305	0.9 (D+ F2) + E	207.6	28.1	0.0	319.4	43.2	0.0	0.14

Reinforcement: Provide,

4-Nos	25 mm Dia
4-Nos	25 mm Dia
4-Nos	25 mm Dia

1964.29 Sq mm (At Corner)
1964.29 Sq mm (Along Z axis)
1964.29 Sq mm (Along Y axis)

A_{sc} pvd. = 5892.86 Sq mm

Total A_{sc} = 0.9 %

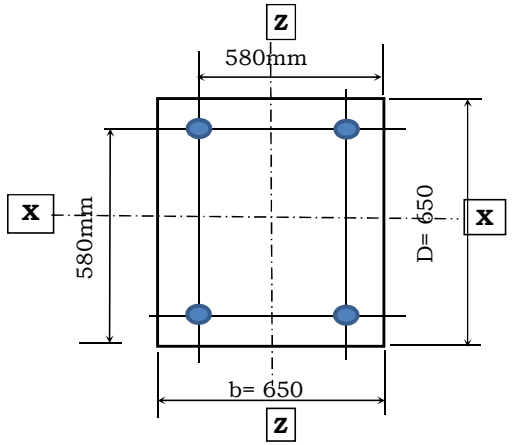
β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 2133.969 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cb^2D$						
301	0.10	0.10	179.2	179.2	0.55	1.16	0.17	O.K
302	0.09	0.09	161.3	161.3	0.55	1.16	0.21	O.K
303	0.09	0.09	161.3	161.3	0.55	1.16	0.26	O.K
304	0.05	0.05	89.6	89.6	0.55	1.16	0.05	O.K
305	0.08	0.08	143.4	143.4	0.55	1.16	0.25	O.K

Links, 10mm dia @ 300 C/C

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900707_GAU01_ST03_CED01_ 37101_OU100665185_R3	DESIGN CALCULATION 3710 TAILINGS PASTE THICKENING DESIGN OF FOUNDATION	Date: 28.11.2023
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DESIGN OF COLUMN MKD C2 **(2nd Middle Columns)**



Input:	Clear cover to reinf.=	50mm	Table 20.6.1.3.1
	Dia. of main reinf.=	20mm	
	Dia. of link=	10mm	
	Effective cover =	70mm	
	Grade of Concrete (f_c') =	M35	
	Grade of Steel (f_y) =	Fy420	
	Grade of Reinf(links).=	Fy420	
	Min. Steel Ratio	0.50%	
	Max. Steel Ratio	4.00%	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1300mm
Unsupproted length of column about Z (l_{uz}) =	1300mm

For Columns not braced against sidesway,

Where, Effective length factor K= 2

Slenderness ratio,	$K\,l_{ux} / r_x$ =	13.86	< 22 SECTION 6.2.5a	Radius of gyration r_x=	188	mm
	$K\,l_{uz} / r_z$ =	13.86	< 22 SECTION 6.2.5a	Radius of gyration r_z=	188	mm

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	23.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	2714.0	0.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	2326.0	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	14.0	0.0	0.0	0.0	0.0
5	Earthquake Load (E)(+/-)	56.0	0.0	0.0	0.0	0.0
			γ_x =	0.82		
			γ_z =	0.82		

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = P_n/f_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	383.18	0.0	0.0	589.5	0.0	0.0	0.40
302	1.2(D + F1) + 1.6W	330.7	0.0	0.0	508.7	0.0	0.0	0.34
303	1.2(D + F2) + E	334.0	0.0	0.0	513.9	0.0	0.0	0.35
304	0.9 D + 1.6W	4.3	0.0	0.0	6.6	0.0	0.0	0.00
305	0.9 (D+ F2) + E	217.0	0.0	0.0	333.9	0.0	0.0	0.23

Reinforcement: Provide,

4-Nos	20 mm Dia
4-Nos	20 mm Dia
4-Nos	20 mm Dia

1257.14 Sq mm (At Corner)
1257.14 Sq mm (Along Z axis)
1257.14 Sq mm (Along Y axis)

A_{sc} pvd. = 3771.43 Sq mm

Total A_{sc} = 0.9 %

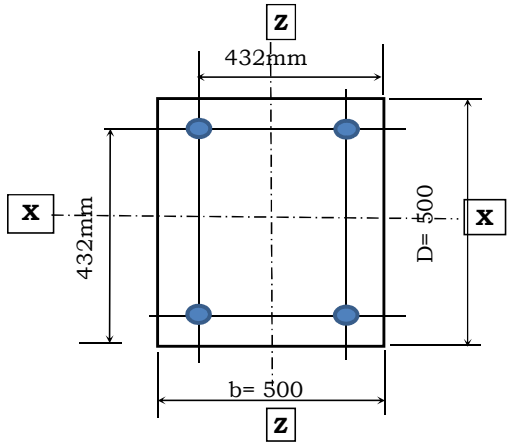
β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1404.118 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cb^2D$						
301	0.06	0.06	57.7	57.7	0.55	1.16	0.00	O.K
302	0.06	0.06	57.7	57.7	0.55	1.16	0.00	O.K
303	0.06	0.06	57.7	57.7	0.55	1.16	0.00	O.K
304	0.05	0.05	48.1	48.1	0.55	1.16	0.00	O.K
305	0.06	0.06	57.7	57.7	0.55	1.16	0.00	O.K

Links, 10mm dia @ 300 C/C

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O19045_450_C_K_ 24150 _001_R3	AS BUILT	Rev: D01

DESIGN OF COLUMN MKD C3 **(1st Middle Columns)**



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	16mm	
Dia. of link=	10mm	
Effective cover =	68mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1300mm
Unsupproted length of column about Z (l_{uz}) =	1300mm

For Columns not braced against sidesway, Where, Effective length factor K= 2

Slenderness ratio,	$K l_{ux} / r_x$ =	18.02	< 22 SECTION 6.2.5a	Radius of gyration r_x=	144	mm
	$K l_{uz} / r_z$ =	18.02	< 22 SECTION 6.2.5a	Radius of gyration r_z=	144	mm

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O19045_450_C_K_ 24150 _001_R3	AS BUILT

Date: 28.11.2023

Rev: D01

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	24.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	2440.0	0.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	2091.0	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	3.0	0.0	0.0	0.0	0.0
5	Earthquake Load (E)(+/-)	6.0	0.0	0.0	0.0	0.0

γ_x = 0.77

γ_z = 0.77

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = P_n/f_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	344.96	0.0	0.0	530.7	0.0	0.0	0.61
302	1.2(D + F1) + 1.6W	296.2	0.0	0.0	455.6	0.0	0.0	0.52
303	1.2(D + F2) + E	296.3	0.0	0.0	455.8	0.0	0.0	0.52
304	0.9 D + 1.6W	2.6	0.0	0.0	4.1	0.0	0.0	0.00
305	0.9 (D+ F2) + E	191.0	0.0	0.0	293.8	0.0	0.0	0.34

Reinforcement: Provide,

4-Nos	16 mm Dia
4-Nos	16 mm Dia
4-Nos	16 mm Dia

804.57 Sq mm (At Corner)

804.57 Sq mm (Along Z axis)

804.57 Sq mm (Along Y axis)

A_{sc} pvd. = 2413.71 Sq mm

Total A_{sc} = 1.0 %

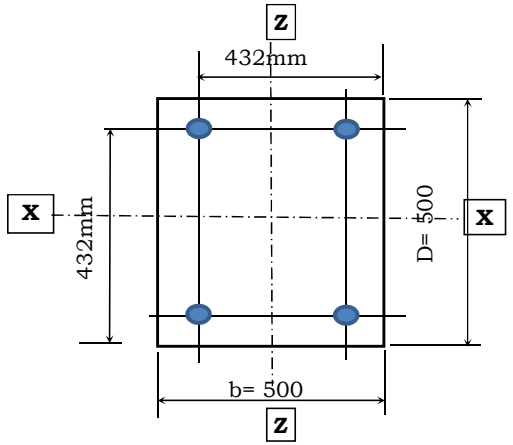
β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 837.9452 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cb^2D$						
301	0.06	0.06	26.3	26.3	0.55	1.16	0.00	O.K
302	0.06	0.06	26.3	26.3	0.55	1.16	0.00	O.K
303	0.06	0.06	26.3	26.3	0.55	1.16	0.00	O.K
304	0.05	0.05	21.9	21.9	0.55	1.16	0.00	O.K
305	0.06	0.06	26.3	26.3	0.55	1.16	0.00	O.K

Links, 10mm dia @ 300 C/C

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O19045_450_C_K_ 24150 _001_R3	AS BUILT	Rev: D01

DESIGN OF COLUMN MKD C4 **(Inner Columns)**



Input:		
Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	16mm	
Dia. of link=	10mm	
Effective cover =	68mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1300mm
Unsupproted length of column about Z (l_{uz}) =	1300mm

For Columns not braced against sideway, Where, Effective length factor K= 2

Slenderness ratio,	$K\,l_{ux} / r_x$ =	18.02	< 22 SECTION 6.2.5a	Radius of gyration r_x=	144	mm
	$K\,l_{uz} / r_z$ =	18.02	< 22 SECTION 6.2.5a	Radius of gyration r_z=	144	mm

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	20.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	2005.0	0.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	1718.0	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	11.0	0.0	0.0	0.0	0.0
5	Earthquake Load (E)(+/-)	28.0	0.0	0.0	0.0	0.0
			γ_x =	0.77		
			γ_z =	0.77		

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = P_n/f_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	283.50	0.0	0.0	436.2	0.0	0.0	0.50
302	1.2(D + F1) + 1.6W	244.8	0.0	0.0	376.6	0.0	0.0	0.43
303	1.2(D + F2) + E	245.8	0.0	0.0	378.2	0.0	0.0	0.43
304	0.9 D + W	3.6	0.0	0.0	5.5	0.0	0.0	0.01
305	0.9 (D+ F2) + E	159.2	0.0	0.0	245.0	0.0	0.0	0.28

Reinforcement: Provide,

4-Nos	16 mm Dia
4-Nos	16 mm Dia
4-Nos	16 mm Dia

804.57 Sq mm (At Corner)
804.57 Sq mm (Along Z axis)
804.57 Sq mm (Along Y axis)

A_{sc} pvd. = 2413.71 Sq mm

Total A_{sc} = 1.0 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 837.9452 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cb^2D$						
301	0.06	0.06	26.3	26.3	0.55	1.16	0.00	O.K
302	0.06	0.06	26.3	26.3	0.55	1.16	0.00	O.K
303	0.06	0.06	26.3	26.3	0.55	1.16	0.00	O.K
304	0.05	0.05	21.9	21.9	0.55	1.16	0.00	O.K
305	0.06	0.06	26.3	26.3	0.55	1.16	0.00	O.K

Links, 10mm dia @ 300 C/C

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DESIGN OF FOUNDATION MKD. F1A

Size of Columns	Along-X	Along-Z	ex	ez
Outer Column A	0.80m	0.80m	0.0000m	-5.00m
2nd Middle Column B	0.65m	0.65m	0.0000m	-1.50m
1st Middle Column C	0.50m	0.50m	0.0000m	2.15m
Inner Column D	0.50m	0.50m	0.0000m	6.10m
Size of fdn. raft:	1.80m	13.70m		
Thickness of raft:	0.85 m			
Original Gr. Level:	0.00 m			
Finished Gr. Level:	0.00 m			
U/S of B/Pl Above GL:	0.55 m			
U/S level of fdn:	-1.50 m			
Unit weight of conc:	25 kN/m^3			
Unit weight of soil:	18 kN/m^3			

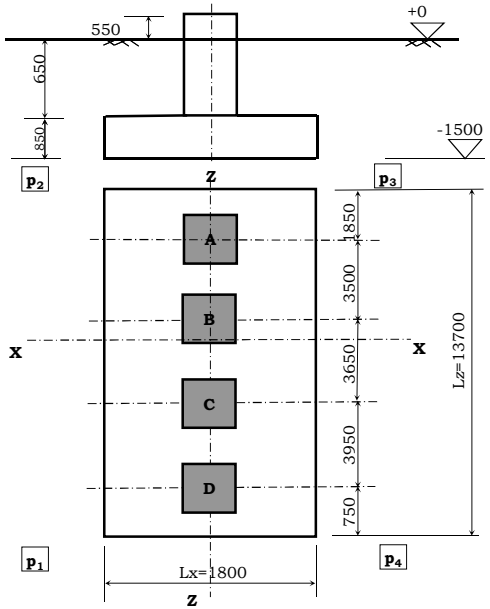
Calculation of Weight of Foundation :

Weight of raft = (1.8*13.7*0.85*25) = 524.03KN

Weight of soil = ((1.8*13.7)-2*(0.8*0.8))*0.65*18 = 273.55KN

Total = 797.57KN

Submerged wt. = 797.57-1.8*13.7*1.5*10 = 427.67KN



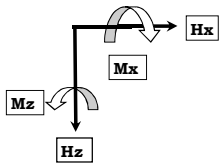
Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	25mm
Dia. of bars provided along Z direction at top=	32mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 774mm = 0.774m

Effective depth of slab along Z direction = 784mm = 0.784m

Clear projection along X direction = 0.65m



H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Basic Load Cases :

		Outer Column A		2nd Middle Column B		1st Middle Column C		Inner Column D	
Sl. No	Basic Load Case	V (kN)	Hx (kN)	V (kN)	Hx (kN)	V (kN)	Hx (kN)	V (kN)	Hx (kN)
1	Tank Dead Load (D)	290.0	0.0	23.0	0.0	24.0	0.0	20.0	0.0
2	Max. Fluid Load (F1)	2354.0	142.0	2714.0	0.0	2440.0	0.0	2005.0	0.0
3	Normal Operational Fluid Load (F2)	2018.0	122.0	2326.0	0.0	2091.0	0.0	1718.0	0.0
4	Wind Load (W)(+/-)	77.0	22.0	14.0	0.0	3.0	0.0	11.0	0.0
5	Earthquake Load (E)(+/-)	179.0	106.0	56.0	0.0	6.0	0.0	28.0	0.0

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	D + F1	9870.0	142.0	0.0	0.0	324.6	10667.57	142.00	291.10	0.00	324.60
202	D + F1 + W	9975.0	164.0	0.0	0.0	-7.9	10772.57	164.00	336.20	0.00	-7.85
203	D + F2+ 0.7 E	8698.3	196.2	0.0	0.0	-471.2	9495.87	196.20	402.21	0.00	-471.16
204	0.6 D + W	319.2	22.0	0.0	0.0	-1119.0	1116.77	22.00	45.10	0.00	-1118.99
205	0.6(D+F2)+0.7 E	5294.3	147.4	0.0	0.0	-505.4	6091.87	147.40	302.17	0.00	-505.38
Ultimate Conditions											
301	1.4 (D + F1)	13818.0	198.8	0.0	0.0	454.4	14615.57	198.80	407.54	0.00	454.44
302	1.2(D + F1) + 1.6 W	12012.0	205.6	0.0	0.0	-142.4	12809.57	205.60	421.48	0.00	-142.40
303	1.2(D + F2) + E	10481.0	252.4	0.0	0.0	-692.6	11278.57	252.40	517.42	0.00	-692.64
304	0.9 D + 1.6 W	489.3	35.2	0.0	0.0	-1711.7	1286.87	35.20	72.16	0.00	-1711.73
305	0.9 (D+ F2) + E	7928.0	215.8	0.0	0.0	-718.3	8725.57	215.80	442.39	0.00	-718.31

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Check for overturning and sliding:

Load Cases:		Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D + F1	10667.57	142.00	291.10	0.00	324.60	32.98	225.12	30.05	
202	D + F1 + W	10772.57	164.00	336.20	0.00	7.85	28.84	9400.27	26.27	
203	D + F2+ 0.7 E	9495.87	196.20	402.21	0.00	471.16	21.25	138.06	19.36	
204	0.6 D + W	1116.77	22.00	45.10	0.00	1118.99	22.29	6.84	20.30	
205	0.6(D+F2)+0.7 E	6091.87	147.40	302.17	0.00	505.38	18.14	82.57	16.53	

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DESIGN OF FOUNDATION MKD. F1A

Load Combination & Calculation of Base pressure

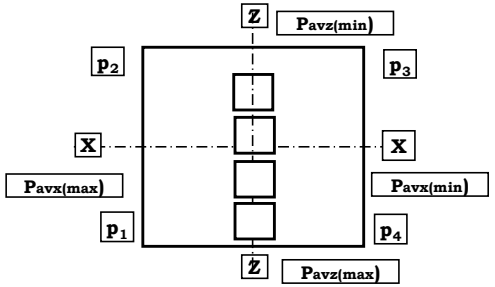
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 20 kN/m^2

Net safe bearing capacity of soil for DL, LL= 500 kN/m^2 For Wind load/Seis. load cas 625 kN/m^2

Gross safe bearing capacity of soil for DL, LL= 527 kN/m^2 For Wind load/Seis. load cas 652 kN/m^2



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	D + F1	10667.57	291.10	324.60	432.59	39.35	5.76	477.70	466.17	399.00	387.47	450.70	471.94	393.24	438.35	426.82	**O.K**
202	D + F1 + W	10772.57	336.20	-7.85	436.84	45.44	0.14	482.43	482.15	391.54	391.26	455.43	482.29	391.40	436.99	436.71	**O.K**
203	D + F2+ 0.7 E	9495.87	402.21	-471.16	385.07	54.37	8.37	447.81	431.07	339.07	322.34	420.81	439.44	330.71	393.44	376.71	**O.K**
204	0.6 D + W	1116.77	45.10	-1118.99	45.29	6.10	19.87	71.26	31.51	59.06	19.32	44.26	51.39	39.19	65.16	25.42	**O.K**
205	0.6(D+F2)+0.7 E	6091.87	302.17	-505.38	247.03	40.84	8.98	296.85	278.90	215.17	197.21	269.85	287.88	206.19	256.01	238.06	**O.K**
301	1.4 (D + F1)	14615.57	407.54	454.44	592.68	55.09	8.07	655.84	639.70	545.67	529.52	628.84	647.77	537.60	600.76	584.61	
302	1.2(D + F1) + 1.6 W	12809.57	421.48	-142.40	519.45	56.97	2.53	578.95	573.89	465.00	459.95	551.95	576.42	462.48	521.98	516.92	
303	1.2(D + F2) + E	11278.57	517.42	-692.64	457.36	69.94	12.30	539.60	515.00	399.72	375.12	512.60	527.30	387.42	469.66	445.06	
304	0.9 D + 1.6 W	1286.87	72.16	-1711.73	52.18	9.75	30.40	92.34	31.54	72.83	12.03	65.34	61.94	42.43	82.59	21.79	
305	0.9 (D+ F2) + E	8725.57	442.39	-718.31	353.83	59.80	12.76	426.39	400.88	306.79	281.28	399.39	413.64	294.04	366.59	341.08	

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DESIGN OF FOUNDATION MKD. F1A

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.8+2*0.5*0.784)+(0.8+2*0.5*0.784))*2

= 6.336 m

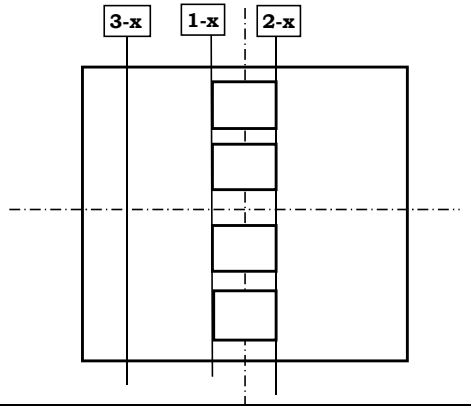
Punching Area= ((0.8+2*0.5*0.784)*(0.8+2*0.5*0.784))

= 2.51 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.85*25+0.65*0.85*18)

= 32.95 kN/m^2

Area of footing, A_f= 24.66 m^2



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DESIGN OF FOUNDATION MKD. F1A

Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
301	1.4 (D + F1)	607.99	577.38	655.36	127.08	109.41	-76.71	2175.55
302	1.2(D + F1) + 1.6 W	535.28	503.63	584.27	111.91	93.64	-67.88	1906.26
303	1.2(D + F2) + E	476.79	437.93	536.94	100.87	78.44	-61.90	1751.86
304	0.9 D + 1.6 W	54.89	49.48	63.28	5.63	2.50	-3.68	200.21
305	0.9 (D+ F2) + E	370.45	337.23	421.88	77.38	58.20	-47.72	1326.37

Overhang = 0.65 m
Effective depth = 0.774 m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	127.08	0.21	0.28	0.001	523.72	1393.20	20ϕT	225.49	300	200	1570.80
At top	2.50	0.00	0.01	0.000	0.00	696.60	12ϕT	162.36	300	150	753.98

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi Vc)/bd$ = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= -3.68 kN/m

Shear stress , τ_v= (-3.68*1000)/(1000*774)

= 0.00 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 2175.55 kN

Strength of Concrete in shear Vc1 = 14506.42 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 10742.44 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 9670.94 kN **Table 22.6.5.2_ (c)**

Design Vc= 9670.94 kN

Allowable shear stress, $(\phi Vc)/bd$ = 1.48 N/mm^2

Design punching shear stress, τ = (2175.55*1000)/(6336*784)

= 0.44 N/mm^2 **No shear reinforcement reqd.**

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DESIGN OF FOUNDATION MKD. F1A

Calculation of distribution reinf in Z direction :

Effective depth = 0.784 m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	759.94	1.24	1.65	0.004	3167.51	1411.20	25ϕT	154.97	300	150	3272.49
At top	916.66	1.49	1.99	0.005	3844.88	705.60	32ϕT	209.17	300	200	4021.24

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 556.44 kN/m

Shear stress , τ_v= (556.44*1000)/(1000*784)

= 0.71 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

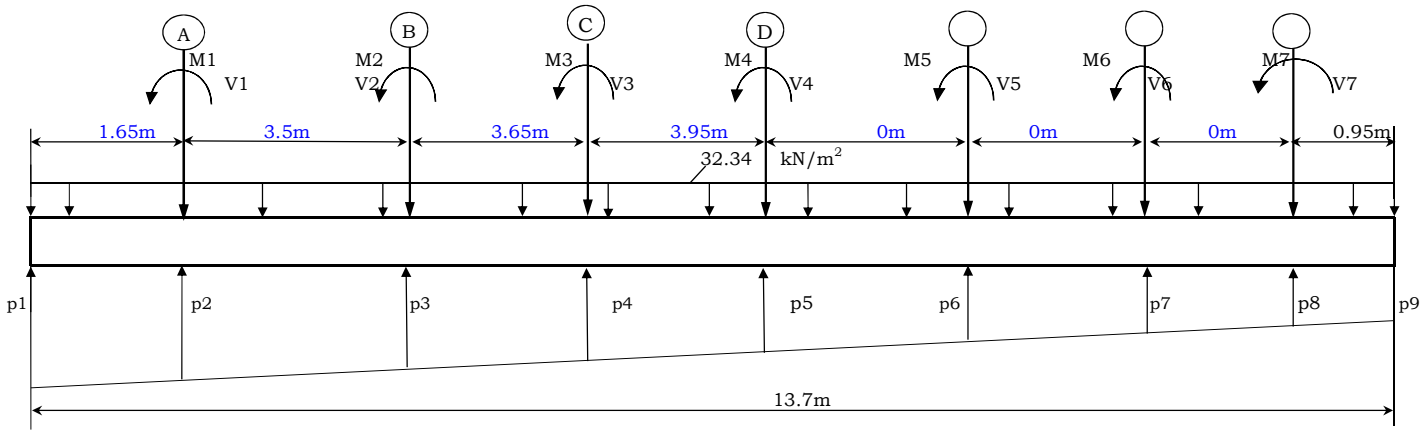
Along X Direction :			
At Bottom	**20ϕT**	**200**	c/c
At Top	**12ϕT**	**150**	c/c
Along Z Direction :			
At Bottom	**25ϕT**	**150**	c/c
At Top	**32ϕT**	**200**	c/c

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DESIGN OF FOUNDATION MKD. F1A

CALCULATION OF BM & SF :

(Along Z direction)



Load Case	Load Combination	Soil pressure in UDL (kN/m)								
		p1	p2	p3	p4	p5	p6	p7	p8	p9
301	1.4 (D + F1)	1,165.99	By STAAD							967.67
302	1.2(D + F1) + 1.6 W	1,037.56								832.46
303	1.2(D + F2) + E	949.14								697.36
304	0.9 D + 1.6 W	111.49								76.37
305	0.9 (D+ F2) + E	744.54								529.26

From STAAD Output	
BM at top	1649.99 (kNm/m)
BM at bot.	1367.89 (kNm/m)
SF (T. at top)	1001.60 (kN/m)
SF (T. at bottom)	975.00 (kN/m)

Load Case	Load Combination	Vertical Load (kN/m)							Moment (kNm/m)						
		V1	V2	V3	V4	V5	V6	V7	M1	M2	M3	M4	M5	M6	M7
301	1.4 (D + F1)	3701.60	3831.80	3449.60	2835.00				407.54						
302	1.2(D + F1) + 1.6 W	3249.80	3298.40	2959.80	2441.00				421.48						
303	1.2(D + F2) + E	2948.60	2874.80	2544.00	2113.60				517.42						
304	0.9 D + 1.6 W	338.00	34.70	24.60	29.00				72.16						
305	0.9 (D+ F2) + E	2256.20	2170.10	1909.50	1592.20				442.39						

Software licensed to LNT ECC DIV	Job No 3710	Sheet No 1	Rev
	Part		
Job Title Mansourah Massarah Gold Project	Ref		
	By AvG	Date 21-Jan-20	Chd SJM
Client Ma'aden	File F1A.std	Date/Time 06-Mar-2020 11:42	

Beam Force Detail

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
1	301:1.4 (D + F	0.000	0.00	-0.00	0.00	0.00	0.00	-0.00
		0.185	0.00	203.85	0.00	0.00	0.00	-20.96
		0.370	0.00	407.21	0.00	0.00	0.00	-78.53
		0.555	0.00	610.09	0.00	0.00	0.00	-172.64
		0.740	0.00	812.50	0.00	0.00	0.00	-303.20
		0.925	0.00	1014.42	0.00	0.00	0.00	-470.12
		1.110	0.00	1215.78	0.00	0.00	0.00	-678.49
		1.295	0.00	1416.67	0.00	0.00	0.00	-923.03
		1.480	0.00	1617.08	0.00	0.00	0.00	-1203.67
		1.665	0.00	1817.00	0.00	0.00	0.00	-1520.30
		1.850	-0.00	2016.44	-0.00	-0.00	-0.00	-1872.85
	302:1.2(D + F1	0.000	0.00	-0.00	0.00	0.00	0.00	0.00
		0.185	0.00	180.34	0.00	0.00	0.00	-18.54
		0.370	0.00	360.19	0.00	0.00	0.00	-69.47
		0.555	0.00	539.53	0.00	0.00	0.00	-152.71
		0.740	0.00	718.38	0.00	0.00	0.00	-268.15
		0.925	0.00	896.73	0.00	0.00	0.00	-415.72
		1.110	0.00	1074.51	0.00	0.00	0.00	-599.90
		1.295	0.00	1251.79	0.00	0.00	0.00	-816.00
		1.480	0.00	1428.57	0.00	0.00	0.00	-1063.94
		1.665	0.00	1604.86	0.00	0.00	0.00	-1343.64
		1.850	-0.00	1780.64	-0.00	-0.00	-0.00	-1655.00
	303:1.2(D + F2	0.000	0.00	-0.00	0.00	0.00	0.00	0.00
		0.185	0.00	163.66	0.00	0.00	0.00	-16.83
		0.370	0.00	326.70	0.00	0.00	0.00	-63.04
		0.555	0.00	489.14	0.00	0.00	0.00	-138.51
		0.740	0.00	650.96	0.00	0.00	0.00	-243.15
		0.925	0.00	812.17	0.00	0.00	0.00	-376.84
		1.110	0.00	972.68	0.00	0.00	0.00	-543.60
		1.295	0.00	1132.58	0.00	0.00	0.00	-739.17
		1.480	0.00	1291.87	0.00	0.00	0.00	-963.44
		1.665	0.00	1450.55	0.00	0.00	0.00	-1216.31
		1.850	-0.00	1608.61	-0.00	-0.00	-0.00	-1497.66
	304:0.9 D + 1.6	0.000	0.00	-0.00	0.00	0.00	0.00	0.00
		0.185	0.00	9.25	0.00	0.00	0.00	-0.95
		0.370	0.00	18.42	0.00	0.00	0.00	-3.56
		0.555	0.00	27.51	0.00	0.00	0.00	-7.81
		0.740	0.00	36.51	0.00	0.00	0.00	-13.69
		0.925	0.00	45.42	0.00	0.00	0.00	-21.18
		1.110	0.00	54.24	0.00	0.00	0.00	-30.49
		1.295	0.00	62.97	0.00	0.00	0.00	-41.37
		1.480	0.00	71.61	0.00	0.00	0.00	-53.83
		1.665	0.00	80.18	0.00	0.00	0.00	-67.82

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		1.850	-0.00	88.65	-0.00	-0.00	-0.00	-83.35
	305:0.9 (D+ F2	0.000	0.00	0.00	0.00	0.00	0.00	0.00
		0.185	0.00	125.85	0.00	0.00	0.00	-12.94
		0.370	0.00	251.19	0.00	0.00	0.00	-48.47
		0.555	0.00	376.00	0.00	0.00	0.00	-106.50
		0.740	0.00	500.28	0.00	0.00	0.00	-186.92
		0.925	0.00	624.04	0.00	0.00	0.00	-289.66
		1.110	0.00	747.21	0.00	0.00	0.00	-417.77
		1.295	0.00	869.85	0.00	0.00	0.00	-567.99
		1.480	0.00	991.97	0.00	0.00	0.00	-740.22
		1.665	0.00	1113.57	0.00	0.00	0.00	-934.36
		1.850	-0.00	1234.64	-0.00	-0.00	-0.00	-1150.34
2	301:1.4 (D + F	0.000	0.00	-1918.55	0.00	0.00	0.00	-1465.31
		0.350	0.00	-1542.73	0.00	0.00	0.00	-866.94
		0.700	0.00	-1168.64	0.00	0.00	0.00	-396.14
		1.050	0.00	-796.27	0.00	0.00	0.00	-52.34
		1.400	0.00	-425.63	0.00	0.00	0.00	165.04
		1.750	0.00	-56.70	0.00	0.00	0.00	256.57
		2.100	0.00	310.25	0.00	0.00	0.00	205.02
		2.450	0.00	675.47	0.00	0.00	0.00	28.91
		2.800	0.00	1038.97	0.00	0.00	0.00	-271.17
		3.150	0.00	1400.75	0.00	0.00	0.00	-694.66
		3.500	-0.00	1760.81	-0.00	-0.00	-0.00	-1240.98
	302:1.2(D + F1	0.000	0.00	-1701.79	0.00	0.00	0.00	-1233.52
		0.350	0.00	-1370.78	0.00	0.00	0.00	-702.30
		0.700	0.00	-1041.55	0.00	0.00	0.00	-283.40
		1.050	0.00	-714.11	0.00	0.00	0.00	23.78
		1.400	0.00	-388.44	0.00	0.00	0.00	219.84
		1.750	0.00	-64.56	0.00	0.00	0.00	305.36
		2.100	0.00	257.28	0.00	0.00	0.00	265.33
		2.450	0.00	577.34	0.00	0.00	0.00	116.10
		2.800	0.00	895.62	0.00	0.00	0.00	-141.73
		3.150	0.00	1212.11	0.00	0.00	0.00	-507.56
		3.500	-0.00	1526.83	-0.00	-0.00	-0.00	-980.81
	303:1.2(D + F2	0.000	0.00	-1540.23	0.00	0.00	0.00	-980.24
		0.350	0.00	-1243.10	0.00	0.00	0.00	-499.00
		0.700	0.00	-948.16	0.00	0.00	0.00	-118.47
		1.050	0.00	-655.40	0.00	0.00	0.00	162.09
		1.400	0.00	-364.84	0.00	0.00	0.00	343.38
		1.750	0.00	-76.46	0.00	0.00	0.00	426.16
		2.100	0.00	209.41	0.00	0.00	0.00	397.27
		2.450	0.00	493.09	0.00	0.00	0.00	271.51
		2.800	0.00	774.59	0.00	0.00	0.00	49.59
		3.150	0.00	1053.90	0.00	0.00	0.00	-267.75

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		3.500	-0.00	1331.02	-0.00	-0.00	-0.00	-679.78
	304:0.9 D + 1.6	0.000	0.00	-173.26	0.00	0.00	0.00	-11.19
		0.350	0.00	-157.49	0.00	0.00	0.00	46.37
		0.700	0.00	-142.03	0.00	0.00	0.00	98.63
		1.050	0.00	-126.87	0.00	0.00	0.00	145.68
		1.400	0.00	-112.02	0.00	0.00	0.00	187.62
		1.750	0.00	-97.47	0.00	0.00	0.00	224.55
		2.100	0.00	-83.28	0.00	0.00	0.00	255.90
		2.450	0.00	-69.39	0.00	0.00	0.00	282.47
		2.800	0.00	-55.80	0.00	0.00	0.00	304.37
		3.150	0.00	-42.52	0.00	0.00	0.00	321.69
		3.500	-0.00	-29.54	-0.00	-0.00	-0.00	334.55
	305:0.9 (D+ F2	0.000	0.00	-1179.95	0.00	0.00	0.00	-707.95
		0.350	0.00	-952.52	0.00	0.00	0.00	-339.24
		0.700	0.00	-726.97	0.00	0.00	0.00	-47.58
		1.050	0.00	-503.29	0.00	0.00	0.00	167.65
		1.400	0.00	-281.48	0.00	0.00	0.00	307.08
		1.750	0.00	-61.54	0.00	0.00	0.00	371.34
		2.100	0.00	156.26	0.00	0.00	0.00	350.47
		2.450	0.00	372.19	0.00	0.00	0.00	255.83
		2.800	0.00	586.25	0.00	0.00	0.00	88.04
		3.150	0.00	798.44	0.00	0.00	0.00	-152.27
		3.500	-0.00	1008.75	-0.00	-0.00	-0.00	-464.49
3	301:1.4 (D + F	0.000	0.00	-2144.43	0.00	0.00	0.00	-1240.98
		0.365	0.00	-1771.04	0.00	0.00	0.00	-534.03
		0.730	0.00	-1399.53	0.00	0.00	0.00	40.77
		1.095	0.00	-1029.90	0.00	0.00	0.00	484.07
		1.460	0.00	-662.13	0.00	0.00	0.00	796.53
		1.825	0.00	-296.25	0.00	0.00	0.00	978.80
		2.190	0.00	67.50	0.00	0.00	0.00	1013.12
		2.555	0.00	429.37	0.00	0.00	0.00	918.71
		2.920	0.00	789.36	0.00	0.00	0.00	696.23
		3.285	0.00	1147.48	0.00	0.00	0.00	346.32
		3.650	-0.00	1503.73	-0.00	-0.00	-0.00	-130.36
	302:1.2(D + F1	0.000	0.00	-1858.04	0.00	0.00	0.00	-980.81
		0.365	0.00	-1532.01	0.00	0.00	0.00	-368.79
		0.730	0.00	-1207.92	0.00	0.00	0.00	127.90
		1.095	0.00	-885.77	0.00	0.00	0.00	509.93
		1.460	0.00	-565.56	0.00	0.00	0.00	777.97
		1.825	0.00	-247.28	0.00	0.00	0.00	932.72
		2.190	0.00	68.77	0.00	0.00	0.00	958.83
		2.555	0.00	382.89	0.00	0.00	0.00	873.15
		2.920	0.00	695.07	0.00	0.00	0.00	676.36
		3.285	0.00	1005.31	0.00	0.00	0.00	369.12

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		3.650	-0.00	1313.61	-0.00	-0.00	-0.00	-47.89
	303:1.2(D + F2	0.000	0.00	-1604.89	0.00	0.00	0.00	-679.78
		0.365	0.00	-1318.56	0.00	0.00	0.00	-152.13
		0.730	0.00	-1034.61	0.00	0.00	0.00	274.37
		1.095	0.00	-753.04	0.00	0.00	0.00	600.53
		1.460	0.00	-473.85	0.00	0.00	0.00	827.19
		1.825	0.00	-197.05	0.00	0.00	0.00	955.18
		2.190	0.00	77.04	0.00	0.00	0.00	971.45
		2.555	0.00	348.75	0.00	0.00	0.00	890.91
		2.920	0.00	618.07	0.00	0.00	0.00	714.38
		3.285	0.00	885.02	0.00	0.00	0.00	442.70
		3.650	-0.00	1149.59	-0.00	-0.00	-0.00	76.68
	304:0.9 D + 1.6	0.000	0.00	-40.69	0.00	0.00	0.00	334.55
		0.365	0.00	-27.53	0.00	0.00	0.00	346.72
		0.730	0.00	-14.70	0.00	0.00	0.00	354.28
		1.095	0.00	-2.20	0.00	0.00	0.00	357.36
		1.460	0.00	9.96	0.00	0.00	0.00	356.05
		1.825	0.00	21.79	0.00	0.00	0.00	350.49
		2.190	0.00	33.25	0.00	0.00	0.00	340.20
		2.555	0.00	44.37	0.00	0.00	0.00	325.91
		2.920	0.00	55.16	0.00	0.00	0.00	307.74
		3.285	0.00	65.61	0.00	0.00	0.00	285.79
		3.650	-0.00	75.74	-0.00	-0.00	-0.00	260.19
	305:0.9 (D+ F2	0.000	0.00	-1208.06	0.00	0.00	0.00	-464.49
		0.365	0.00	-991.01	0.00	0.00	0.00	-67.62
		0.730	0.00	-776.00	0.00	0.00	0.00	252.61
		1.095	0.00	-563.02	0.00	0.00	0.00	496.91
		1.460	0.00	-352.07	0.00	0.00	0.00	665.98
		1.825	0.00	-143.16	0.00	0.00	0.00	760.54
		2.190	0.00	63.42	0.00	0.00	0.00	770.85
		2.555	0.00	267.97	0.00	0.00	0.00	708.23
		2.920	0.00	470.48	0.00	0.00	0.00	573.39
		3.285	0.00	670.96	0.00	0.00	0.00	367.04
		3.650	-0.00	869.40	-0.00	-0.00	-0.00	89.89
4	301:1.4 (D + F	0.000	0.00	-1852.49	0.00	0.00	0.00	-130.36
		0.395	0.00	-1469.38	0.00	0.00	0.00	517.24
		0.790	0.00	-1088.46	0.00	0.00	0.00	1018.15
		1.185	0.00	-709.74	0.00	0.00	0.00	1373.21
		1.580	0.00	-333.22	0.00	0.00	0.00	1583.25
		1.975	0.00	41.11	0.00	0.00	0.00	1649.09
		2.370	0.00	412.93	0.00	0.00	0.00	1551.19
		2.765	0.00	782.55	0.00	0.00	0.00	1310.95
		3.160	0.00	1149.97	0.00	0.00	0.00	929.19
		3.555	0.00	1515.21	0.00	0.00	0.00	406.75

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		3.950	-0.00	1878.24	-0.00	-0.00	-0.00	-255.56
	302:1.2(D + F	0.000	0.00	-1580.22	0.00	0.00	0.00	-47.89
		0.395	0.00	-1249.07	0.00	0.00	0.00	503.56
		0.790	0.00	-920.20	0.00	0.00	0.00	928.30
		1.185	0.00	-593.60	0.00	0.00	0.00	1227.19
		1.580	0.00	-269.27	0.00	0.00	0.00	1401.08
		1.975	0.00	52.79	0.00	0.00	0.00	1450.83
		2.370	0.00	372.25	0.00	0.00	0.00	1359.80
		2.765	0.00	689.45	0.00	0.00	0.00	1146.56
		3.160	0.00	1004.37	0.00	0.00	0.00	811.94
		3.555	0.00	1317.02	0.00	0.00	0.00	356.81
		3.950	-0.00	1627.40	-0.00	-0.00	-0.00	-217.97
	303:1.2(D + F	0.000	0.00	-1311.47	0.00	0.00	0.00	76.68
		0.395	0.00	-1028.22	0.00	0.00	0.00	532.46
		0.790	0.00	-747.76	0.00	0.00	0.00	880.04
		1.185	0.00	-470.09	0.00	0.00	0.00	1120.46
		1.580	0.00	-195.21	0.00	0.00	0.00	1254.78
		1.975	0.00	76.88	0.00	0.00	0.00	1284.03
		2.370	0.00	345.79	0.00	0.00	0.00	1194.56
		2.765	0.00	611.91	0.00	0.00	0.00	1002.40
		3.160	0.00	875.24	0.00	0.00	0.00	708.58
		3.555	0.00	1135.78	0.00	0.00	0.00	314.16
		3.950	-0.00	1393.54	-0.00	-0.00	-0.00	-179.81
	304:0.9 D + 1.6	0.000	0.00	19.91	0.00	0.00	0.00	260.19
		0.395	0.00	30.44	0.00	0.00	0.00	250.01
		0.790	0.00	40.58	0.00	0.00	0.00	235.86
		1.185	0.00	50.33	0.00	0.00	0.00	217.89
		1.580	0.00	59.69	0.00	0.00	0.00	196.25
		1.975	0.00	68.66	0.00	0.00	0.00	171.08
		2.370	0.00	77.19	0.00	0.00	0.00	142.08
		2.765	0.00	85.33	0.00	0.00	0.00	109.87
		3.160	0.00	93.08	0.00	0.00	0.00	74.62
		3.555	0.00	100.45	0.00	0.00	0.00	36.47
		3.950	-0.00	107.42	-0.00	-0.00	-0.00	-4.45
	305:0.9 (D+ F	0.000	0.00	-970.35	0.00	0.00	0.00	89.89
		0.395	0.00	-758.22	0.00	0.00	0.00	426.55
		0.790	0.00	-548.47	0.00	0.00	0.00	682.23
		1.185	0.00	-341.10	0.00	0.00	0.00	857.83
		1.580	0.00	-136.12	0.00	0.00	0.00	954.25
		1.975	0.00	66.48	0.00	0.00	0.00	972.37
		2.370	0.00	266.35	0.00	0.00	0.00	902.18
		2.765	0.00	463.84	0.00	0.00	0.00	755.71
		3.160	0.00	658.95	0.00	0.00	0.00	533.87
		3.555	0.00	851.67	0.00	0.00	0.00	237.55

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		3.950	-0.00	1042.01	-0.00	-0.00	-0.00	-132.35
5	301:1.4 (D + F	0.000	0.00	-682.85	0.00	0.00	0.00	-255.56
		0.075	0.00	-614.20	0.00	0.00	0.00	-207.21
		0.150	0.00	-545.64	0.00	0.00	0.00	-163.86
		0.225	0.00	-477.15	0.00	0.00	0.00	-125.50
		0.300	0.00	-408.74	0.00	0.00	0.00	-92.14
		0.375	0.00	-340.41	0.00	0.00	0.00	-63.76
		0.450	0.00	-272.17	0.00	0.00	0.00	-41.08
		0.525	0.00	-204.01	0.00	0.00	0.00	-23.36
		0.600	0.00	-135.93	0.00	0.00	0.00	-10.62
		0.675	0.00	-67.92	0.00	0.00	0.00	-2.83
		0.750	-0.00	0.00	-0.00	-0.00	-0.00	-0.00
	302:1.2(D + F1	0.000	0.00	-582.65	0.00	0.00	0.00	-217.97
		0.075	0.00	-524.02	0.00	0.00	0.00	-176.71
		0.150	0.00	-465.46	0.00	0.00	0.00	-139.73
		0.225	0.00	-406.98	0.00	0.00	0.00	-107.02
		0.300	0.00	-348.59	0.00	0.00	0.00	-78.56
		0.375	0.00	-290.27	0.00	0.00	0.00	-54.36
		0.450	0.00	-232.06	0.00	0.00	0.00	-35.02
		0.525	0.00	-173.92	0.00	0.00	0.00	-19.91
		0.600	0.00	-115.86	0.00	0.00	0.00	-9.05
		0.675	0.00	-57.89	0.00	0.00	0.00	-2.41
		0.750	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00
	303:1.2(D + F2	0.000	0.00	-481.22	0.00	0.00	0.00	-179.81
		0.075	0.00	-432.63	0.00	0.00	0.00	-145.74
		0.150	0.00	-384.15	0.00	0.00	0.00	-115.22
		0.225	0.00	-335.77	0.00	0.00	0.00	-88.22
		0.300	0.00	-287.49	0.00	0.00	0.00	-64.75
		0.375	0.00	-239.32	0.00	0.00	0.00	-44.79
		0.450	0.00	-191.25	0.00	0.00	0.00	-28.85
		0.525	0.00	-143.29	0.00	0.00	0.00	-16.40
		0.600	0.00	-95.42	0.00	0.00	0.00	-7.45
		0.675	0.00	-47.66	0.00	0.00	0.00	-1.99
		0.750	-0.00	-0.00	-0.00	-0.00	-0.00	0.00
	304:0.9 D + 1.6	0.000	0.00	-12.10	0.00	0.00	0.00	-4.45
		0.075	0.00	-10.82	0.00	0.00	0.00	-3.59
		0.150	0.00	-9.56	0.00	0.00	0.00	-2.83
		0.225	0.00	-8.32	0.00	0.00	0.00	-2.16
		0.300	0.00	-7.09	0.00	0.00	0.00	-1.58
		0.375	0.00	-5.87	0.00	0.00	0.00	-1.09
		0.450	0.00	-4.67	0.00	0.00	0.00	-0.70
		0.525	0.00	-3.48	0.00	0.00	0.00	-0.40
		0.600	0.00	-2.31	0.00	0.00	0.00	-0.18
		0.675	0.00	-1.15	0.00	0.00	0.00	-0.05

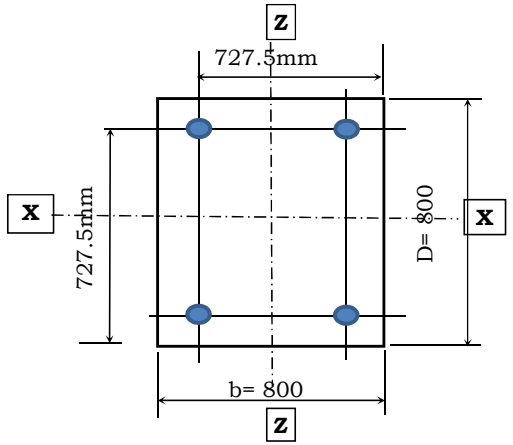
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	Part		
Job Title Mansourah Massarah Gold Project	Ref		
	By AvG	Date 21-Jan-20	Chd SJM
Client Ma'aden	File F1A.std	Date/Time 06-Mar-2020 11:42	

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		0.750	-0.00	-0.00	-0.00	-0.00	-0.00	0.00
	305:0.9 (D+ F	0.000	0.00	-354.40	0.00	0.00	0.00	-132.35
		0.075	0.00	-318.57	0.00	0.00	0.00	-107.26
		0.150	0.00	-282.82	0.00	0.00	0.00	-84.78
		0.225	0.00	-247.16	0.00	0.00	0.00	-64.91
		0.300	0.00	-211.58	0.00	0.00	0.00	-47.63
		0.375	0.00	-176.09	0.00	0.00	0.00	-32.95
		0.450	0.00	-140.70	0.00	0.00	0.00	-21.22
		0.525	0.00	-105.40	0.00	0.00	0.00	-12.06
		0.600	0.00	-70.18	0.00	0.00	0.00	-5.48
		0.675	0.00	-35.05	0.00	0.00	0.00	-1.46
		0.750	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00

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DESIGN OF COLUMN MKD C1A (Outer Column)



Input:	Clear cover to reinf.=	50mm	Table 20.6.1.3.1
	Dia. of main reinf.=	25mm	
	Dia. of link=	10mm	
	Effective cover =	72.5mm	
	Grade of Concrete (f_c') =	M35	
	Grade of Steel (f_y) =	Fy420	
	Grade of Reinf(links).=	Fy420	
	Min. Steel Ratio	0.50%	
	Max. Steel Ratio	4.00%	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1300mm
Unsupproted length of column about Z (l_{uz}) =	1300mm

For Columns not braced against sidesway,

Where, Effective length factor K= 2

Slenderness ratio,	$K l_{ux} / r_x$ =	11.26	< 22 SECTION 6.2.5a	Radius of gyration r_x=	231	mm
	$K l_{uz} / r_z$ =	11.26	< 22 SECTION 6.2.5a	Radius of gyration r_z=	231	mm

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	290.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	2354.0	142.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	2018.0	122.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	77.0	22.0	0.0	0.0	0.0
5	Earthquake Load (E)(+/-)	179.0	106.0	0.0	0.0	0.0

Including Bridge Load = 90+200 = 290kN

γ_x =	0.84
γ_z =	0.84

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = P_n/f_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	370.16	25.8	0.0	569.5	39.8	0.0	0.25
302	1.2(D + F1) + 1.6W	329.6	26.7	0.0	507.1	41.1	0.0	0.23
303	1.2(D + F2) + E	335.2	32.8	0.0	515.7	50.5	0.0	0.23
304	0.9 D + 1.6W	38.4	4.6	0.0	59.1	7.0	0.0	0.03
305	0.9 (D+ F2) + E	225.6	28.1	0.0	347.1	43.2	0.0	0.15

Reinforcement: Provide,

4-Nos	25 mm Dia
4-Nos	25 mm Dia
4-Nos	25 mm Dia

1964.29 Sq mm (At Corner)
1964.29 Sq mm (Along Z axis)
1964.29 Sq mm (Along Y axis)

A_{sc} pvd. = 5892.86 Sq mm

Total A_{sc} = 0.9 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 2133.969 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cb^2D$						
301	0.10	0.10	179.2	179.2	0.55	1.16	0.17	O.K
302	0.09	0.09	161.3	161.3	0.55	1.16	0.21	O.K
303	0.09	0.09	161.3	161.3	0.55	1.16	0.26	O.K
304	0.05	0.05	89.6	89.6	0.55	1.16	0.05	O.K
305	0.08	0.08	143.4	143.4	0.55	1.16	0.25	O.K

Links, 10mm dia @ 300 C/C

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DESIGN OF FOUNDATION MKD. F2 (Central Column)

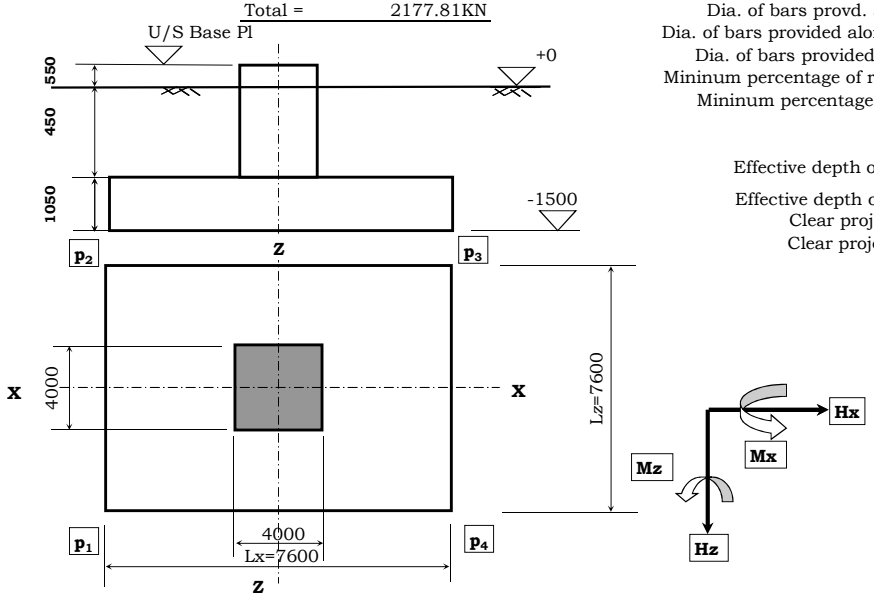
	Along-X	Along-Z
Size of column :	4.00m	4.00m
Size of fdn. raft :	7.60m	7.60m
Thickness of raft :	1.05 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.55 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Tank Dead Load	D
Max. Fluid Load	F1
Normal Operational Fluid Load	F2
Wind Load	W
Earthquake Load	E

Calculation of Weight of Foundation :

Weight of raft & Col.=	(7.6*7.6*1.05*24)+(4*4*1*24) =	1839.55KN
Weight of soil =	((7.6*7.6)-(4*4))*0.45*18 =	338.26KN
	Total =	2177.81KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	25mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	25mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 987.5mm = 0.9875m

Effective depth of slab along Z direction = 975mm = 0.975m

Clear projection along Z direction = 1.8m

Clear projection along X direction = 1.8m

H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F2

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mz (kNm)
1	Tank Dead Load (D)	2898.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	23413.0	0.0	0.0	0.0	0.0
3	Normal Operational Fluid Load (F2)	20066.0	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	0.0	388.0	0.0	1331.0	0.0
5	Earthquake Load (E)(+/-)	0.0	1547.0	0.0	6463.0	0.0

Included Bridge Load,
V= 962+1936=2898kN

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	D + F1	26311.0	0.0	0.0	0.0	0.0	28488.81	0.00	0.00	0.00	0.00
202	D + F1 + W	26311.0	388.0	0.0	0.0	0.0	28488.81	388.00	795.40	0.00	0.00
203	D + F2+ 0.7 E	22964.0	1082.9	0.0	0.0	0.0	25141.81	1082.90	2219.95	0.00	0.00
204	0.6 D + W	1738.8	388.0	1331.0	0.0	0.0	3916.61	388.00	2126.40	0.00	0.00
205	0.6(D+F2)+0.7 E	13778.4	1082.9	6463.0	0.0	0.0	15956.21	1082.90	8682.95	0.00	0.00
Ultimate Conditions											
301	1.4 (D + F1)	36835.4	0.0	0.0	0.0	0.0	39013.21	0.00	0.00	0.00	0.00
302	1.2(D + F1) + 1.6W	31573.2	620.8	0.0	0.0	0.0	33751.01	620.80	1272.64	0.00	0.00
303	1.2(D + F2) + E	31573.2	1547.0	0.0	0.0	0.0	33751.01	1547.00	3171.35	0.00	0.00
304	0.9 D + 1.6 W	2608.2	620.8	0.0	0.0	0.0	4786.01	620.80	1272.64	0.00	0.00
305	0.9 (D+ F2) + E	20667.6	1547.0	0.0	0.0	0.0	22845.41	1547.00	3171.35	0.00	0.00

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Check for overturning and sliding:

Load Cases:		Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D + F1	28488.81	0.00	0.00	0.00	0.00	-		-	
202	D + F1 + W	28488.81	388.00	795.40	0.00	0.00	122.49		26.43	
203	D + F2+ 0.7 E	25141.81	1082.90	2219.95	0.00	0.00	38.73		8.36	
204	0.6 D + W	3916.61	388.00	2126.40	0.00	0.00	6.30		3.63	
205	0.6(D+F2)+0.7 E	15956.21	1082.90	8682.95	0.00	0.00	6.28		5.30	

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DESIGN OF FOUNDATION MKD. F2

Load Combination & Calculation of Base pressure

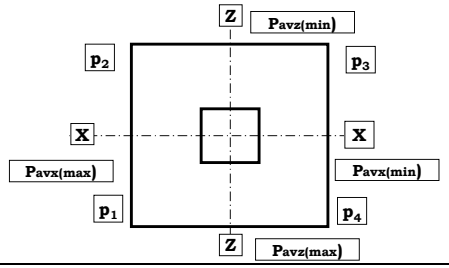
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 500 kN/m^2 For WL/SL case = 625 kN/m^2

Gross SBC of soil for DL, LL= 527 kN/m^2 For WL/SL case = 652 kN/m^2



Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	Base Pressure (kN/m^2) V/(Lx.Lz)	6.Mz/(Lx)^2.Lz	6.Mx/Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	D + F1	#######	0.00	0.00	493.23	0.00	0.00	493.23	493.23	493.23	493.23	466.23	493.23	493.23	493.23	493.23	O.K
202	D + F1 + W	#######	795.40	0.00	493.23	10.87	0.00	504.10	504.10	482.36	482.36	477.10	504.10	482.36	493.23	493.23	O.K
203	D + F2+ 0.7 E	#######	2219.95	0.00	435.28	30.34	0.00	465.62	465.62	404.94	404.94	438.62	465.62	404.94	435.28	435.28	O.K
204	0.6 D + W	3916.61	2126.40	0.00	67.81	29.06	0.00	96.87	96.87	38.74	38.74	69.87	96.87	38.74	67.81	67.81	O.K
205	0.6(D+F2)+0.7 E	#######	8682.95	0.00	276.25	118.68	0.00	394.93	394.93	157.57	157.57	367.93	394.93	157.57	276.25	276.25	O.K
301	1.4 (D + F1)	#######	0.00	0.00	675.44	0.00	0.00	675.44	675.44	675.44	675.44	648.44	675.44	675.44	675.44	675.44	
302	1.2(D + F1) + 1.6W	#######	1272.64	0.00	584.33	17.39	0.00	601.73	601.73	566.94	566.94	574.73	601.73	566.94	584.34	584.34	
303	1.2(D + F2) + E	#######	3171.35	0.00	584.33	43.35	0.00	627.68	627.68	540.99	540.99	600.68	627.68	540.99	584.34	584.34	
304	0.9 D + 1.6 W	4786.01	1272.64	0.00	82.86	17.39	0.00	100.25	100.25	65.47	65.47	73.25	100.25	65.47	82.86	82.86	
305	0.9 (D+ F2) + E	#######	3171.35	0.00	395.52	43.35	0.00	438.87	438.87	352.18	352.18	411.87	438.87	352.18	395.53	395.53	
	Max :	39013	8683	0													
	Min :	3917	0	0													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 466.23 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 476.23 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 477.10 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 487.10 kN/m^2

Sample calculation of values of base pressure:

Load case : D + F1 + W

Maximum pressure, p_1 (pmax) = 493.23+10.87+0.00= 504.10 kN/m^2

p_2= 493.23+10.87-0.00= 504.10 kN/m^2

p_3= 493.23-(10.87+0.00)= 482.36 kN/m^2

p_4= 493.23-10.87+0.00= 482.36 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 504.10-1.5*18= 477.10 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (504.10+504.10)/2= 504.10 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (482.36+482.36)/2= 482.36 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (504.10+482.36)/2= 493.23 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (482.36+504.10)/2= 493.23 kN/m^2

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O19045_450_C_K_ 24150 _001_R3	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F2

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

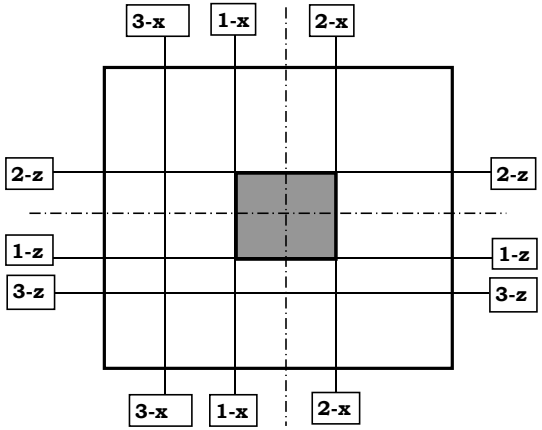
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((4+2*0.5*0.975)+(4+2*0.5*0.975))*2
= 19.9 m

Punching Area= ((4+2*0.5*0.975)*(4+2*0.5*0.975))
= 24.75 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (1.05*24+0.45*18)
= 33.30 kN/m^2

Area of footing, A_f= 57.76 m^2



Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
Ultimate Conditions								
301	1.4 (D + F1)	675.44	675.44	675.44	1040.27	1040.27	521.74	21196.92
302	1.2(D + F1) + 1.6W	593.49	575.18	598.01	916.41	868.95	460.34	18189.56
303	1.2(D + F2) + E	607.15	561.52	618.41	951.81	833.54	479.17	18189.56
304	0.9 D + 1.6 W	92.01	73.71	96.53	104.01	56.56	52.89	1635.98
305	0.9 (D+ F2) + E	418.34	372.71	429.60	645.94	527.67	325.76	11956.98

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DESIGN OF FOUNDATION MKD. F2

Sample calculation of BM & SF

Load case : 301

p_{1x} = 675.44-(675.44-675.44)*1.8/7.6 = 675.44 kN/m^2 Overhang = 1.8 m

p_{2x} = 675.44+(675.44-675.44)*1.8/7.6 = 675.44 kN/m^2 Effective depth = 0.9875 m

p_{3x} = 675.44-(675.44-675.44)*(1.8-0.9875)/7.6 = 675.44 kN/m^2

BM_{1x} = (675.44-33.3)*1.8*1.8/2+0.5*(675.44-675.44)*1.8*1.8*0.67 = 1040.27 kN-m/m

BM_{2x} = (675.44-33.3)*1.8*1.8/2-0.5*(-675.44+675.44)*1.8*1.8*0.67 = 1040.27 kN-m/m

SF_{3x} = (675.44-33.3)*(1.8-0.9875)+0.5*(675.44-675.44)*(1.8-0.9875) = 521.74 kN/m

$S.F_{(punching)}$ = (57.76-24.75)*(675.44-33.3) = 21196.92 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	1040.27	1.07	1.42	0.003	3428.25	1777.50	25ϕT	143.19	300	125	3926.99
At top	0.00	0.00	0.00	0.000	0.00	888.75	16ϕT	226.23	300	225	893.61

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 521.74 kN/m

Shear stress , τ_v= (521.74*1000)/(1000*987.5)

= 0.53 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 21196.92 kN

Strength of Concrete in shear Vc1 = 58129.18 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 28991.56 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 38752.79 kN **Table 22.6.5.2_ (c)**

Design Vc= 28991.56 kN

Allowable shear stress, (ϕ Vc)/bd = 1.11 N/mm^2

Design punching shear stress, τ = (21196.92*1000)/(19900*987.5)

= 1.08 N/mm^2 **No shear reinforcement reqd.**

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DESIGN OF FOUNDATION MKD. F2

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z}	p_{2z}	p_{3z}	$B.M_{1z}$	$B.M_{2z}$	$S.F_{3z}$
Ultimate Conditions							
301	1.4 (D + F1)	675.44	675.44	675.44	1040.27	1040.27	529.77
302	1.2(D + F1) + 1.6W	584.34	584.34	584.34	892.68	892.68	454.61
303	1.2(D + F2) + E	584.34	584.34	584.34	892.68	892.68	454.61
304	0.9 D + 1.6 W	82.86	82.86	82.86	80.29	80.29	40.89
305	0.9 (D+ F2) + E	395.53	395.53	395.53	586.81	586.81	298.84

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DESIGN OF FOUNDATION MKD. F2

Sample calculation of BM & SF

Load case : 301

p_{1z} = 675.44-(675.44-675.44)*1.8/7.6= 675.44 kN/m^2

p_{2z} = 675.44+(675.44-675.44)*1.8/7.6= 675.44 kN/m^2

p_{3z} = 675.44-(675.44-675.44)*(1.8-0.975)/7.6= 675.44 kN/m^2

Overhang = 1.8 m

Effective depth = 0.975 m

BM_{1z} = (675.44-33.3)*1.8*1.8/2+0.5*(675.44-675.44)*1.8*1.8*0.67 = 1040.27 kN-m/m

BM_{2z} = (675.44-33.3)*1.8*1.8/2-0.5*(-675.44+675.44)*1.8*1.8*0.67 = 1040.27 kN-m/m

SF_{3z} = (675.44-33.3)*(1.8-0.975)+0.5*(675.44-675.44)*(1.8-0.975) = 529.77 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	1040.27	1.094	1.459	0.0036	3519.06	1755.00	25ϕT	139.49	300	125	3927.0
At top	0.00	0.000	0.000	0.0000	0.00	877.50	16ϕT	229.13	300	225	893.6

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 529.77 kN/m

Shear stress , τ_v= (529.77*1000)/(1000*975)

= 0.54 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **25ϕT** **125** c/c

At Top **16ϕT** **225** c/c

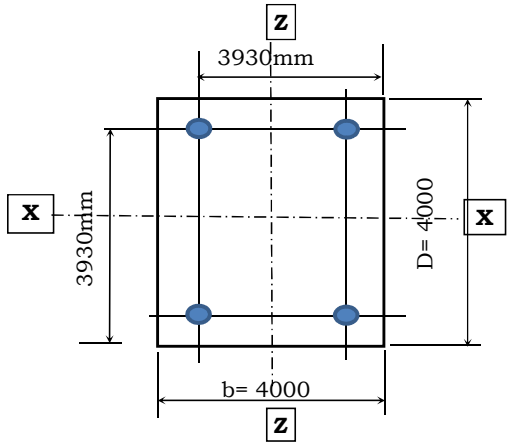
Along Z Direction :

At Bottom **25ϕT** **125** c/c

At Top **16ϕT** **225** c/c

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DESIGN OF COLUMN MKD C5 (Inner Column)



Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	20mm
Dia. of link=	10mm
Effective cover =	70mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.15%
Max. Steel Ratio	4.00%

Table 20.6.1.3.1

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1000mm
Unsupproted length of column about Z (l_{uz}) =	1000mm

For Columns not braced against sideway,

Where, Effective length factor K= 2

Slenderness ratio,	$K\,l_{ux} / r_x$ =	1.73	< 22 SECTION 6.2.5a	Radius of gyration r_x=	1154	mm
	$K\,l_{uz} / r_z$ =	1.73	< 22 SECTION 6.2.5a	Radius of gyration r_z=	1154	mm

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	2898.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	23413.0	0.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	20066.0	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	0.0	388.0	0.0	1331.0	0.0
5	Earthquake Load (E)(+/-)	0.0	1547.0	0.0	6463.0	0.0
			γ_x =	0.97		
			γ_z =	0.97		

Included Bridge Load, V= 962+1936=2898kN

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = P_n/f_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	3683.54	0.0	0.0	5667.0	0.0	0.0	0.10
302	1.2(D + F1) + 1.6W	3157.3	275.0	0.0	4857.4	423.1	0.0	0.09
303	1.2(D + F2) + E	3157.3	801.0	0.0	4857.4	1232.3	0.0	0.09
304	0.9 D + 1.6W	260.8	275.0	0.0	401.3	423.1	0.0	0.01
305	0.9 (D+ F2) + E	2066.8	801.0	0.0	3179.6	1232.3	0.0	0.06

Reinforcement: Provide,

4-Nos	20 mm Dia
40-Nos	20 mm Dia
40-Nos	20 mm Dia

1257.14 Sq mm (At Corner)
12571.43 Sq mm (Along Z axis)
12571.43 Sq mm (Along Y axis)

A_{sc} pvd. = 26400.00 Sq mm

Total A_{sc} = 0.17 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 48630.26 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cb^2D$						
301	0.05	0.05	11200.0	11200.0	0.55	1.16	0.00	O.K
302	0.05	0.05	11200.0	11200.0	0.55	1.16	0.02	O.K
303	0.05	0.05	11200.0	11200.0	0.55	1.16	0.08	O.K
304	0.05	0.05	11200.0	11200.0	0.55	1.16	0.02	O.K
305	0.05	0.05	11200.0	11200.0	0.55	1.16	0.08	O.K

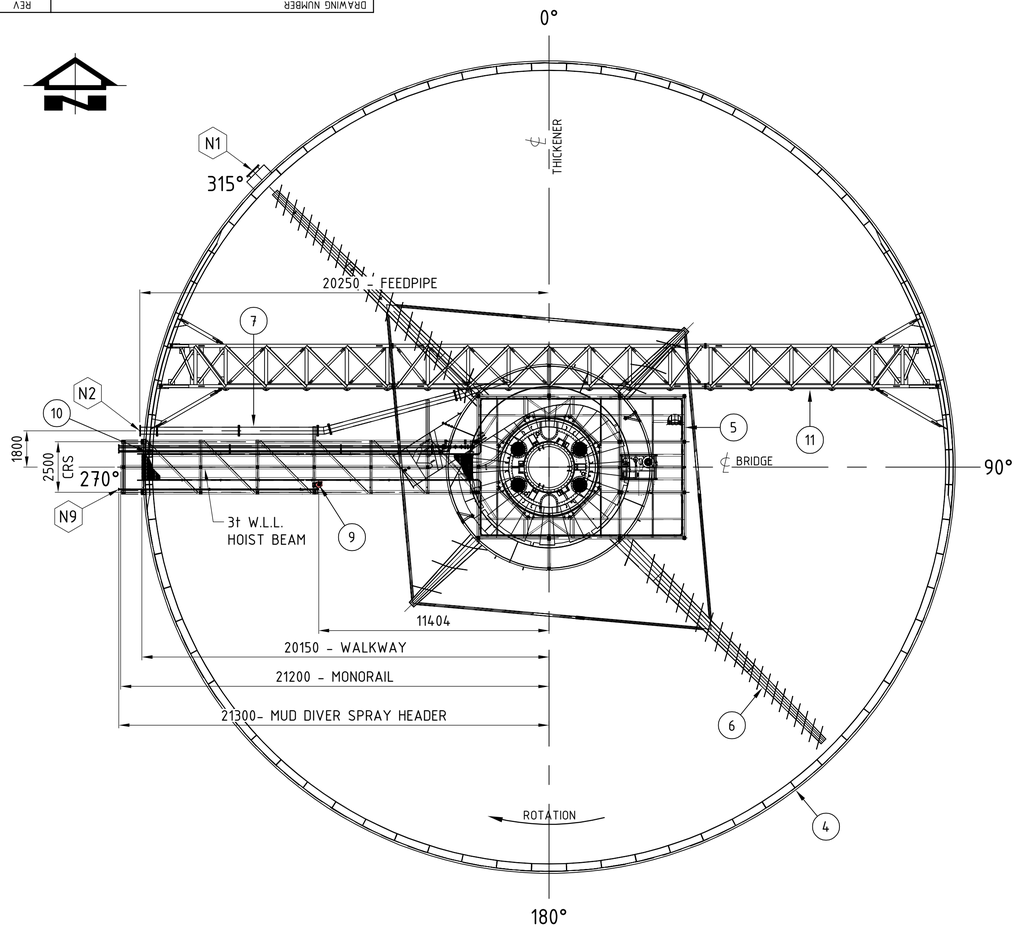
Links, 12mm dia @ 150 C/C

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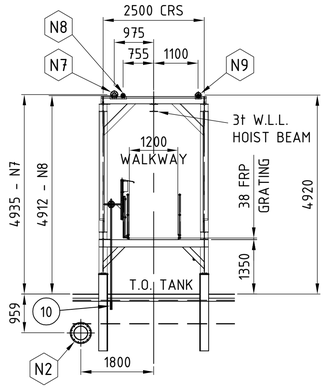
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

ATTACHMENT 1 LOAD DATA DRAWINGS

DRAWING NUMBER MD- 216- 3710- EG- ME- GAD- 1001 REV A02



PLAN

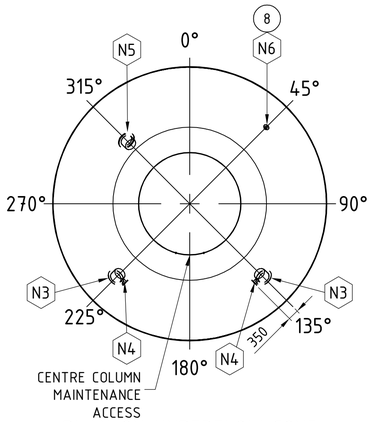


SECTION A
SCALE 1:75

N9	MUD DIVER SPRAY WATER	DN50	1	2" ASME B16.5 CLASS 150 F.F.		
N8	CONCENTRATED FLOC	DN32	1	1-1/4" ASME B16.5 CLASS 150 F.F.		
N7	DILUTION WATER	DN80	1	3" ASME B16.5 CLASS 150 F.F.		
N6	BED MASS	DN80	1	3" ASME B16.5 CLASS 150 F.F.		
N5	DRAIN (BLANKED)	DN400	1	16" ASME B16.5 CLASS 150 F.F.		
N4	FLUSH	DN100	2	4" ASME B16.5 CLASS 150 F.F.		
N3	UNDERFLOW (1 DUTY / 1 STANDBY)	DN400	2	16" ASME B16.5 CLASS 150 F.F.		
N2	FEEDPIPE	DN350	1	14" ASME B16.5 CLASS 150 F.F.		
N1	OVERFLOW	DN600	1	24" ASME B16.5 CLASS 150 F.F.		
ITEM	DESCRIPTION	SIZE	QTY	CONNECTION TYPE		

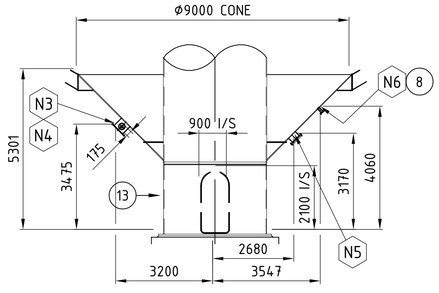
NOZZLE LIST

13	UNDERFLOW MONORAIL - W.L.L. 3t	-	2	-	-	-
12	CENTRE COLUMN c/w INTERNAL ACCESS	-	1	-	-	-
11	STATIC PICKETS c/w SUPPORT STRUCTURE	-	1	-	-	-
10	TURBIDITY SENSOR	-	1	-	-	-
9	BED LEVEL DETECTOR - MUD DIVER	-	1	-	-	-
8	BED MASS SENSOR - RETRACTABLE	-	1	-	-	-
7	**FEED SYSTEM c/w AUTODIL (6mm RUBBER LINED PARTIAL)**	-	1	-	-	-
6	RAKE MECHANISM ASSEMBLY - STRUTLESS SYSTEM	-	1	-	-	-
5	BRIDGE & PLATFORM ASSEMBLY	-	1	-	-	-
4	TANK & SUPPORT STRUCTURE	-	1	-	-	-
3	HYDRAULIC POWER PACK - 75 kW SIZE 4	-	1	-	-	-
2	CONTROL PANEL - PLC	-	1	-	-	-
1	DRIVE SYSTEM - SCD4004 - 300 LIFT - S.M.A.R.T.	-	1	-	-	-
PART	DESCRIPTION	SUB-DWG.NO.\ MATERIAL	QTY	WEIGHT (KG)	ITEM NO.	REMARKS

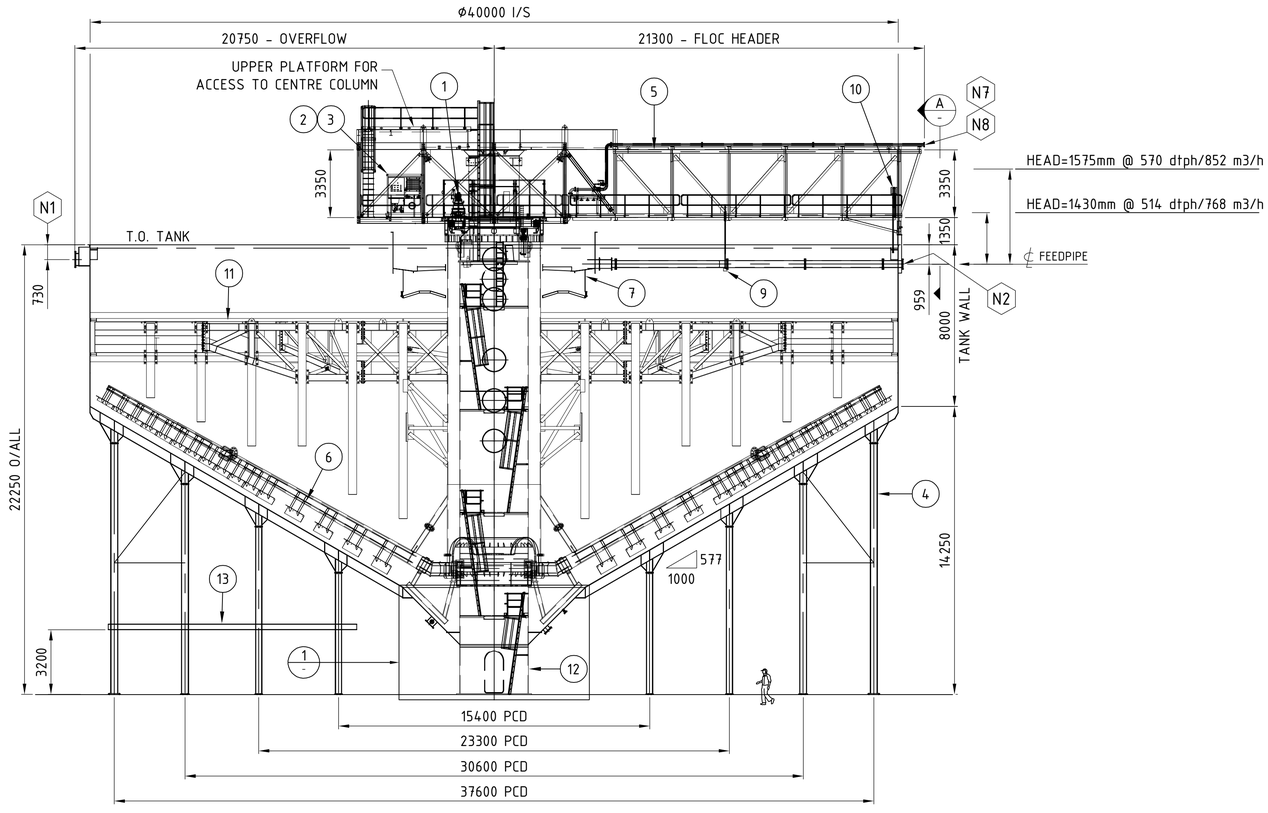


PLAN ON UNDERFLOW CONE

SCALE 1:100



DETAIL 1
SCALE 1:75
REFER TO PLAN FOR TRUE ORIENTATION



SECTIONAL ELEVATION
REFER TO PLAN FOR TRUE ORIENTATION

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-ME-DED-1001	GENERAL NOTES								
MD-216-3000-EG-ME-SPC-3001	TECHNICAL SPECIFICATION								
MD-216-3710-EG-ME-DAT-1001	EQUIPMENT DATASHEET	A02	22.11.2019	REVISED AS PER CLIENT COMMENTS	MB	EJ	SC	RL	
MD-216-3710-EG-ME-GAD-1002	FOUNDATION AND LOAD PLAN	A01	04.10.2019	ISSUED FOR APPROVAL	MB	EJ	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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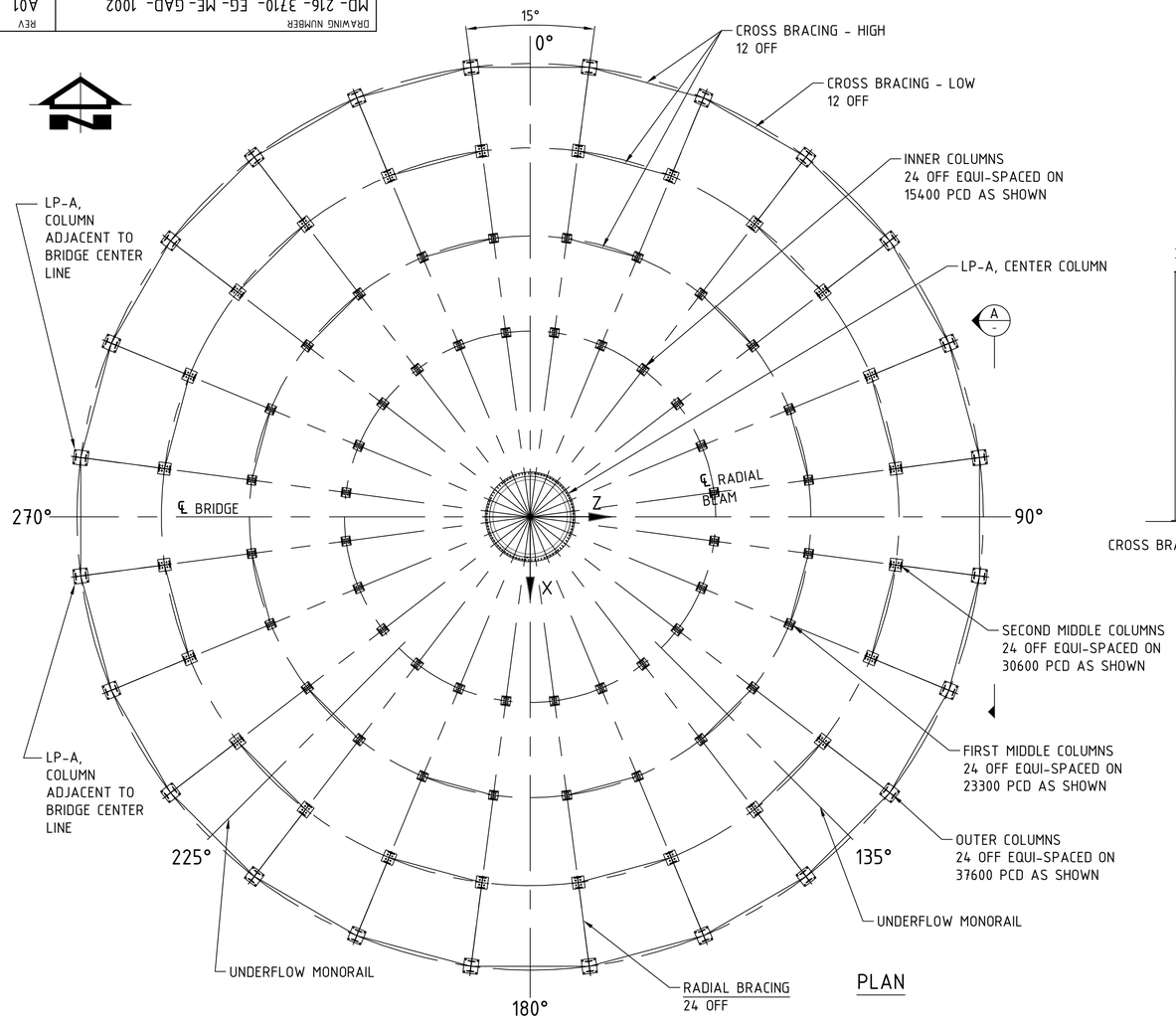
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CHECKED	edwjar	20.11.19
ENGINEER	soucha	22.11.19
APPROVED	ricleh	22.11.19
EMPLOYER		
SCALE		A1

LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-G-03151-001 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 __GAU01 __ST03__MTL03 __37101 — DOCUMENT ID: OU601243914 — REVISION: 2

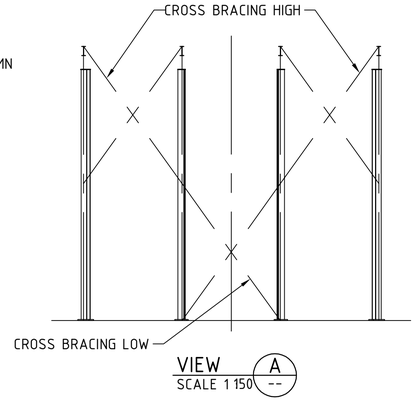
MANSOURAH-MASSARAH GOLD PROJECT
EQUIPMENT GENERAL ARRANGEMENT DRAWING
TAILINGS PASTE THICKENING
3710-THK-001 T10-TAILS PASTE THK, 40m

DRAWING NUMBER: MD - 216 - 3710 - EG- ME - GAD - 1001 — REV: A02

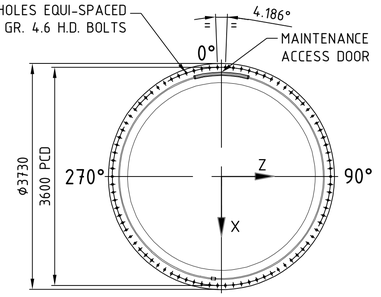
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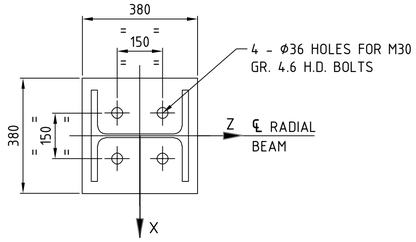
PLAN



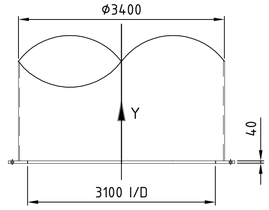
VIEW A
SCALE 1 150



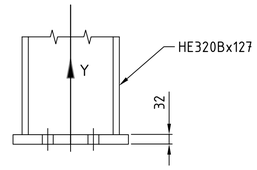
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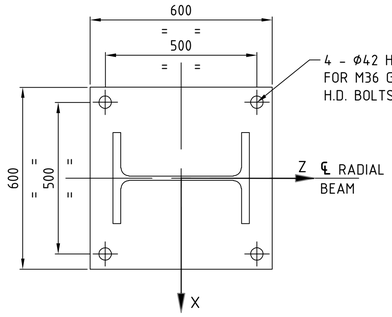
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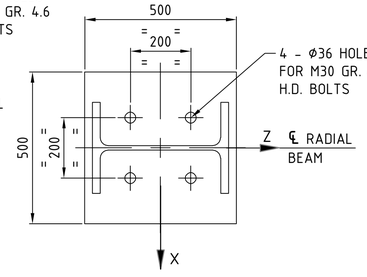
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CENTRE COLUMN
SCALE 1:50



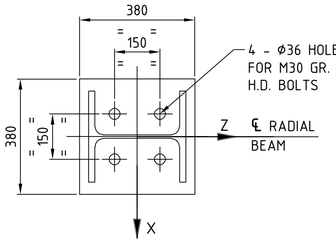
ELEVATION
INNER COLUMN



PLAN



PLAN



PLAN

		Tank Dead Load	Max Fluid Load SG 1.742		Normal Operating Fluid Load SG 1.493		Seismic Load (+/-)			Wind Load (+/-)			Bridge/Drive Dead Load (LP-A) *NOTE 9	Bridge/Drive Live Load (LP-A) *NOTE 9	Drive Torque (+/-)	
	Column PCD (mm)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Vertical (Y) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Torque (X,Z) (kNm)
Outer Columns	37 600	90	142	2 354	122	2 018	106	179	0	22	77	0	200	145	11	0
2nd Middle Columns	30 600	23	0	2 714	0	2 326	0	56	0	0	14	0	0	0	0	0
1st Middle Columns	23 300	24	0	2 440	0	2 091	0	6	0	0	3	0	0	0	0	0
Inner Columns	15 400	20	0	2 005	0	1 718	0	28	0	0	11	0	0	0	0	0
Central Column	0	962	0	23 413	0	20 066	1 547	0	6 463	388	0	1 331	1 936	1 753	0	2 875
Total Applied Load	-	4 730	0	251 725	0	215 744	2 759	0	36 313	554	0	8 295	2 336	2 044	0	7 666

NOTES:

1. ALL LOADS LISTED ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301
2. SEISMIC LOADS ARE CALCULATED CONSIDERING DYNAMIC SLOSHING EFFECTS OF NORMAL OPERATING SG LIQUIDS
3. COMBINATIONS INCLUDING SEISMIC LOAD TO USE NORMAL OPERATING FLUID LOAD
4. ALL HORIZONTAL REACTIONS ARE IN THE PLANE OF BRACING
5. TOTAL LOADS AND MOMENTS ARE CALCULATED AT THE UNDERSIDE OF BASEPLATE
6. COLUMN LOADS ARE GIVEN PER COLUMN
7. DIFFERENTIAL SETTLEMENT BETWEEN ANY ADJACENT COLUMN SHALL BE LIMITED TO L / 600
8. TILTING OF THE TANK DUE TO OUTER COLUMN DIFFERENTIAL SETTLEMENT SHALL BE LIMITED TO 15 mm
9. BRIDGE/DRIVE LOADS REACTED AT LOAD POINTS A (LP-A) ONLY. LP-A IS ADJACENT TO BRIDGE CENTER LINE AND AT CENTER COLUMN.
10. LENGTH OF ANCHOR BOLTS TO BE DEFINED AS A FUNCTION OF CIVIL ENGINEERING.

SEISMIC DATA TO SBC301

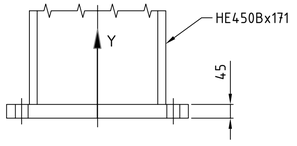
Ss, S1:	0.10, 0.02
Importance Factor:	1
Site Class:	B

WIND DATA TO SBC301

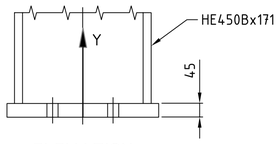
V:	43 m/s
Exposure Class:	C

FOUNDATION TOLERANCES

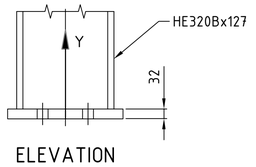
HEIGHT OF COLUMN FOUNDATIONS	± 10 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT IN A GROUP	± 3 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT GROUP	± 6 mm
ANCHOR BOLTS - MAX TOLERANCE ACCUMULATION BETWEEN BOLT GROUPS	± 6 mm



ELEVATION
OUTER COLUMN



ELEVATION
SECOND MIDDLE COLUMN



ELEVATION
FIRST MIDDLE COLUMN

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-ME-DED-1001	GENERAL NOTES								
MD-216-3000-EG-ME-SPC-3001	TECHNICAL SPECIFICATION								
MD-216-3710-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3710-EG-ME-GAD-1001	EQUIPMENT GENERAL ARRANGEMENT DRAWING	A01	14.11.2019	ISSUED FOR APPROVAL	MB	EJ	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-D-03171-001 | REVISION: 0

OUTOTEC DRAWING NUMBER: 900707__GAU01__ST03__MLD15__37101 | DOCUMENT ID: 0U601257563 | REVISION: 1

LARSEN & TOUBRO

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DRAWN	micbro	13.11.19	
CHECKED	edwjar	13.11.19	
ENGINEER	soucha	14.11.19	
APPROVED	ricleh	14.11.19	
EMPLOYER			
SCALE	1:100		A1

MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
TAILINGS PASTE THICKENER
3710-THK-001 T10-TAILS PASTE THK, 40m

DRAWING NUMBER: MD-216-3710-EG-ME-GAD-1002 | REV: A01

ATTACHMENT 2 STAAD OUTPUT

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*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR  6, 2020                *
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     2. START JOB INFORMATION
     3. ENGINEER DATE 21-JAN-20
     4. JOB NAME MANSOURAH MASSARAH GOLD PROJECT
     5. JOB CLIENT MA'ADEN
     6. JOB NO 3710
     7. ENGINEER NAME AVG
     8. CHECKER NAME SJM
     9. APPROVED NAME AG
    10. END JOB INFORMATION
    11. INPUT WIDTH 79
    12. UNIT METER KN
    13. JOINT COORDINATES
    14. 1 0 0 0; 2 1.65 0 0; 3 5.15 0 0; 4 8.8 0 0; 5 12.75 0 0; 6 13.5 0 0
    15. MEMBER INCIDENCES
    16. 1 1 2; 2 2 3; 3 3 4; 4 4 5; 5 5 6
    17. DEFINE MATERIAL START
    18. ISOTROPIC CONCRETE
    19. E 2.17185E+007
    20. POISSON 0.17
    21. DENSITY 23.5616
    22. ALPHA 1E-005
    23. DAMP 0.05
    24. TYPE CONCRETE
    25. STRENGTH FCU 27579
    26. END DEFINE MATERIAL
    27. MEMBER PROPERTY AMERICAN
    28. 1 TO 5 PRIS YD 0.8 ZD 1.8
    29. CONSTANTS
    30. MATERIAL CONCRETE ALL
    31. SUPPORTS
    32. 2 TO 5 FIXED BUT MY MZ KFY 0.2
    33. LOAD 301 LOADTYPE NONE  TITLE 1.4 ( D + F1)
    34. MEMBER LOAD
    35. 1 TRAP GY 1160.93 1136.33
    36. 2 TRAP GY 1136.33 1084.15
    37. 3 TRAP GY 1084.15 1029.74
    38. 4 TRAP GY 1029.74 970.851
```

```
39. 5 TRAP GY 970.851 959.67
40. JOINT LOAD
41. 2 FY -3421.6 MZ 407.54
42. 3 FY -3831.8
43. 4 FY -3449.6
44. 5 FY -2835
45. MEMBER LOAD
46. 1 TO 5 UNI GY -61.2
47. LOAD 302 LOADTYPE NONE  TITLE 1.2( D + F1 ) + 1.6W
48. JOINT LOAD
49. 2 FY -3009.8 MZ 421.48
50. 3 FY -3298.4
51. 4 FY -2959.8
52. 5 FY -2441
53. MEMBER LOAD
54. 1 TO 5 UNI GY -61.2
55. 1 TRAP GY 1032.84 1007.4
56. 2 TRAP GY 1007.4 953.438
57. 3 TRAP GY 953.438 897.164
58. 4 TRAP GY 897.164 836.263
59. 5 TRAP GY 836.263 824.7
60. LOAD 303 LOADTYPE NONE  TITLE 1.2( D + F2 ) + E
61. MEMBER LOAD
62. 1 TRAP GY 943.83 912.601
63. 2 TRAP GY 912.601 846.358
64. 3 TRAP GY 846.358 777.275
65. 4 TRAP GY 777.275 702.515
66. 5 TRAP GY 702.515 688.32
67. JOINT LOAD
68. 2 FY -2708.6 MZ 517.42
69. 3 FY -2874.8
70. 4 FY -2544
71. 5 FY -2113.6
72. MEMBER LOAD
73. 1 TO 5 UNI GY -61.2
74. LOAD 304 LOADTYPE NONE  TITLE 0.9 D + 1.6W
75. JOINT LOAD
76. 2 FY -158 MZ 72.16
77. 3 FY -34.7
78. 4 FY -24.6
79. 5 FY -29
80. MEMBER LOAD
81. 1 TO 5 UNI GY -61.2
82. 1 TRAP GY 97.5 93.1452
83. 2 TRAP GY 93.1452 83.9078
84. 3 TRAP GY 83.9078 74.2745
85. 4 TRAP GY 74.2745 63.8494
86. 5 TRAP GY 63.8494 61.87
87. LOAD 305 LOADTYPE NONE  TITLE 0.9 ( D+ F2 ) + E
88. MEMBER LOAD
89. 1 TRAP GY 740.65 713.948
90. 2 TRAP GY 713.948 657.308
91. 3 TRAP GY 657.308 598.24
92. 4 TRAP GY 598.24 534.317
93. 5 TRAP GY 534.317 522.18
94. JOINT LOAD
```

```
 95. 2 FY -2076.2 MZ 442.39
 96. 3 FY -2170.1
 97. 4 FY -1909.5
 98. 5 FY -1592.2
 99. MEMBER LOAD
100. 1 TO 5 UNI GY -61.2
101. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**6**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**4**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 9 DOF
TOTAL PRIMARY LOAD CASES = 5, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 50923.6 MB

```
**NOTE - NEAR INSTABILITY AT JOINT     6   DIRECTION = FY
 PROBABLE CAUSE  - NEAR SINGULAR K-MATRIX
 K-MATRIX DIAG=  6.5561693E+04 L-MATRIX DIAG=  4.5678681E-03 EQN NO      20

  **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
           DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
           BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
           STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
           IS   1.000E-09

           THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
           ELEMENTS AS WELL AS TRUE SINGULARITIES.


 102. FINISH
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********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 6,2020 TIME= 14:41:39 ****

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* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
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*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR  6, 2020                *
*           Time=    11:42:49                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
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     2. START JOB INFORMATION
     3. ENGINEER DATE 21-JAN-20
     4. JOB NAME MANSOURAH MASSARAH GOLD PROJECT
     5. JOB CLIENT MA'ADEN
     6. JOB NO 3710
     7. ENGINEER NAME AVG
     8. CHECKER NAME SJM
     9. APPROVED NAME AG
    10. END JOB INFORMATION
    11. INPUT WIDTH 79
    12. UNIT METER KN
    13. JOINT COORDINATES
    14. 1 0 0 0; 2 1.85 0 0; 3 5.35 0 0; 4 9 0 0; 5 12.95 0 0; 6 13.7 0 0
    15. MEMBER INCIDENCES
    16. 1 1 2; 2 2 3; 3 3 4; 4 4 5; 5 5 6
    17. DEFINE MATERIAL START
    18. ISOTROPIC CONCRETE
    19. E 2.17185E+007
    20. POISSON 0.17
    21. DENSITY 23.5616
    22. ALPHA 1E-005
    23. DAMP 0.05
    24. TYPE CONCRETE
    25. STRENGTH FCU 27579
    26. END DEFINE MATERIAL
    27. MEMBER PROPERTY AMERICAN
    28. 1 TO 5 PRIS YD 0.75 ZD 1.8
    29. CONSTANTS
    30. MATERIAL CONCRETE ALL
    31. SUPPORTS
    32. 2 TO 5 FIXED BUT MY MZ KFY 0.2
    33. LOAD 301 LOADTYPE NONE  TITLE 1.4 ( D + F1)
    34. JOINT LOAD
    35. 2 FY -3701.6 MZ 407.54
    36. 3 FY -3831.8
    37. 4 FY -3449.6
    38. 5 FY -2835
```

```
39. MEMBER LOAD
40. 1 TO 5 UNI GY -61.2
41. 1 TRAP GY 1164.56 1137.78
42. 2 TRAP GY 1137.78 1087.11
43. 3 TRAP GY 1087.11 1034.28
44. 4 TRAP GY 1034.28 977.097
45. 5 TRAP GY 977.097 966.24
46. LOAD 302 LOADTYPE NONE  TITLE 1.2( D + F1 ) + 1.6 W
47. JOINT LOAD
48. 2 FY -3249.8 MZ 421.48
49. 3 FY -3298.4
50. 4 FY -2959.8
51. 5 FY -2441
52. MEMBER LOAD
53. 1 TO 5 UNI GY -61.2
54. 1 TRAP GY 1037.56 1009.86
55. 2 TRAP GY 1009.86 957.466
56. 3 TRAP GY 957.466 902.823
57. 4 TRAP GY 902.823 843.688
58. 5 TRAP GY 843.688 832.46
59. LOAD 303 LOADTYPE NONE  TITLE 1.2( D + F2 ) + E
60. JOINT LOAD
61. 2 FY -2949.6 MZ 517.42
62. 3 FY -2874.8
63. 4 FY -2544
64. 5 FY -2113.6
65. MEMBER LOAD
66. 1 TO 5 UNI GY -61.2
67. 1 TRAP GY 947.72 913.719
68. 2 TRAP GY 913.719 849.393
69. 3 TRAP GY 849.393 782.31
70. 4 TRAP GY 782.31 709.714
71. 5 TRAP GY 709.714 695.93
72. LOAD 304 LOADTYPE NONE  TITLE 0.9 D + 1.6 W
73. JOINT LOAD
74. 2 FY -338 MZ 72.16
75. 3 FY -34.7
76. 4 FY -24.6
77. 5 FY -29
78. MEMBER LOAD
79. 1 TO 5 UNI GY -61.2
80. 1 TRAP GY 111.49 106.748
81. 2 TRAP GY 106.748 97.7753
82. 3 TRAP GY 97.7753 88.4185
83. 4 TRAP GY 88.4185 78.2926
84. 5 TRAP GY 78.2926 76.37
85. LOAD 305 LOADTYPE NONE  TITLE 0.9 ( D+ F2 ) + E
86. JOINT LOAD
87. 2 FY -2256.2 MZ 442.39
88. 3 FY -2170.1
89. 4 FY -1909.5
90. 5 FY -1592.2
91. MEMBER LOAD
92. 1 TO 5 UNI GY -61.2
93. 1 TRAP GY 743.11 714.041
94. 2 TRAP GY 714.041 659.045
```

STAAD SPACE -- PAGE NO. 3

95. 3 TRAP GY 659.045 601.692
96. 4 TRAP GY 601.692 539.625
97. 5 TRAP GY 539.625 527.84
98. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 6 NUMBER OF MEMBERS 5
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 4

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 9 DOF
TOTAL PRIMARY LOAD CASES = 5, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 50923.7 MB

**NOTE - NEAR INSTABILITY AT JOINT 6 DIRECTION = FY
PROBABLE CAUSE - NEAR SINGULAR K-MATRIX
K-MATRIX DIAG= 5.9481046E+04 L-MATRIX DIAG= 4.5678152E-03 EQN NO 20

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

99. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 6,2020 TIME= 11:42:51 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
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****

ATTACHMENT 3 FOUNDATION SETTLEMENT ANALYSIS

Foundation settlement Analysis for Tailings Paste Thickener 3710/3710A

for MANSOURAH - MASSARAH GOLD PROJECT

February 2020

Debi Prasad Ghosh, PhD
Deputy General Manager (Civil)

Larsen & Toubro Ltd
11th Floor, Tower - 2, DP-5,
Sector - V, Salt Lake City,
West Bengal
Kolkata- 700091

Summary

The main purpose of these analyses is to investigate if the soil pressures under the foundations exceed the design limit due to the consideration of flexible foundation. Two types of foundations are analyzed considering without bridge load with smaller foundation type and with bridge Load for larger foundation type. Finite element analyses are done for both types of foundations. Maximum pressure under smaller foundation type is 0.537 MPa and under larger foundation type is 0.524 MPa, which are less than the net bearing capacity of the soil, 0.950 MPa. So, Foundations are in safe zone.

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   - 2.2. Material Properties 4
   - 2.3. Soil Properties 4
   - 2.4. Solid Modelling of Foundations 5
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   - 3.2. Boundary Conditions 7
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   - 3.4. Maximum Fluid Loads (F1) 10
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   - 5.2. Maximum Fluid Loads (F1) 21
   - 5.3. Normal Operating Fluid Loads (F2) 26
   - 5.4. Seismic Loads (E) 31
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1. Introduction

This report is a verification of the pressure under the foundation of the Tailings Paste Thickening Foundation system, using finite element analysis [1]. Different primary load cases and their combinations are considered for these analyses.

The main steps of these analyses are to calculate:

- Vertical deflections and pressures at the four bottom corners of the foundation for each primary load cases.
- Combination of primary load cases to get combined analyses results assuming principle of superposition.

Two sets of analyses are performed for two types of foundations: smaller (3710) and larger (3710A) Foundations.

2. Design Basis

2.1. Geometry

Tailings Paste Thickening Foundations consists of the foundation raft and four different column pedestals shown with plan & elevation in Figure 1.

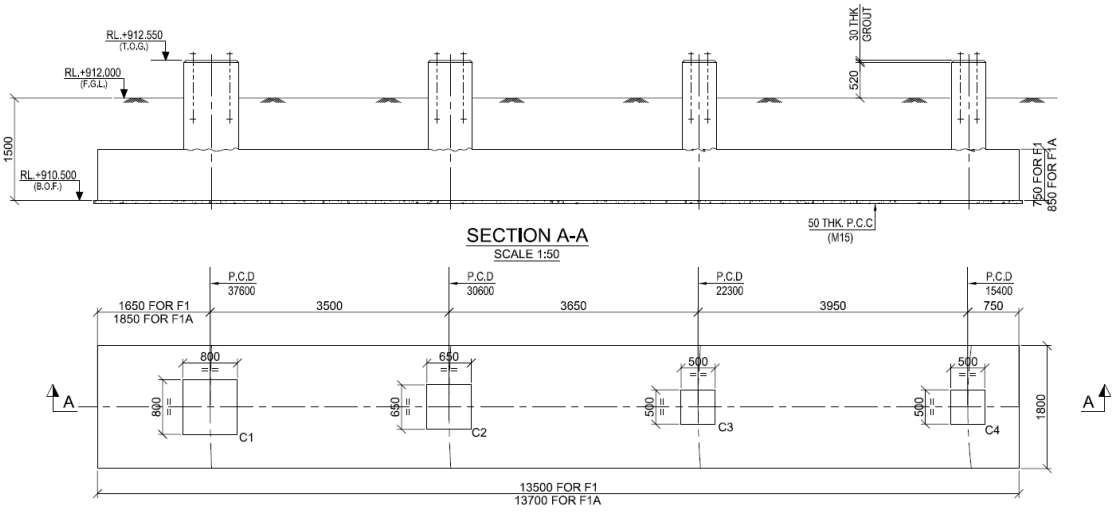


Figure 1. Dimensions of Tailings Paste Thickening Foundation: F1 for 3710 & F1A for 3710A

2.2. Material Properties

The material properties of concrete used in the model are taken from ANSYS material data base (Young's Modulus=3E+10 Pa, Poisson's Ratio=0.18, Shear Modulus=1.2712E+10 Pa).

2.3. Soil Properties

Soil properties are taken from Geotechnical Report (Table 6 - BH-14 of Geotechnical Report (MD- 216-3000-EG-CV-RPT-1008) [2]), shown in Table-1.

Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
					Settlement	Shear Failure				
			(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
BH-02, 12 to 15, & 20	-	Isolated Square	1.20	1.0 x 1.0	1600	570	570	25	68,400	32,200
	-		1.50		2700	850	850	25	102,000	38,900
	-		1.20	2.0 x 2.0	1250	750	750	25	90,000	37,500
	-		1.50		1800	980	980	25	117,600	44,200
	-		1.20	3.0 x 3.0	1100	900	900	25	108,000	42,800
	-		1.50		1500	1120	1120	25	134,400	49,500
	-		1.20	4.0 x 4.0	1000	1040	1000	25	120,000	48,100
	-		1.50		1300	1250	1250	25	150,000	54,800
	-		1.20	5.0 x 5.0	900	1172	900	25	108,000	53,400
	-		1.50		1200	1342	1200	25	144,000	60,200
BH-02	Ball Mill	Raft	2.50	18.0 x 25.0	1300	*	1300	50	117,000	137,000
			3.00		1350	*	1350	50	121,500	143,000
BH-12	CCD	Rectangular	1.20	3.0 x 4.0	1000	900	900	25	81,000	42,000
			1.50		1400	1120	1120	25	100,800	49,000
BH-13	Neutralization and thickening	combined raft	1.20	17.0 x 23.0	1000	*	1000	50	90,000	130,000
			1.50		1200	*	1200	50	108,000	136,000
BH-14	Tailing Paste Thickening	Ring wall	1.20	Dia. = 40m	800	*	800	50	72,000	235,000
			1.50		950	*	950	50	85,500	240,000

Table 1. Subgrade Reaction and bearing capacity of Soil (taken from [2])

Soil properties are taken as:

- The net safe bearing capacity of soil as 950 k-N/m^2
- Vertical Subgrade modulus 85500 k-N/m^3

2.4. Solid Modelling of Foundations

Complete models of the Foundations are shown in Figure 2.

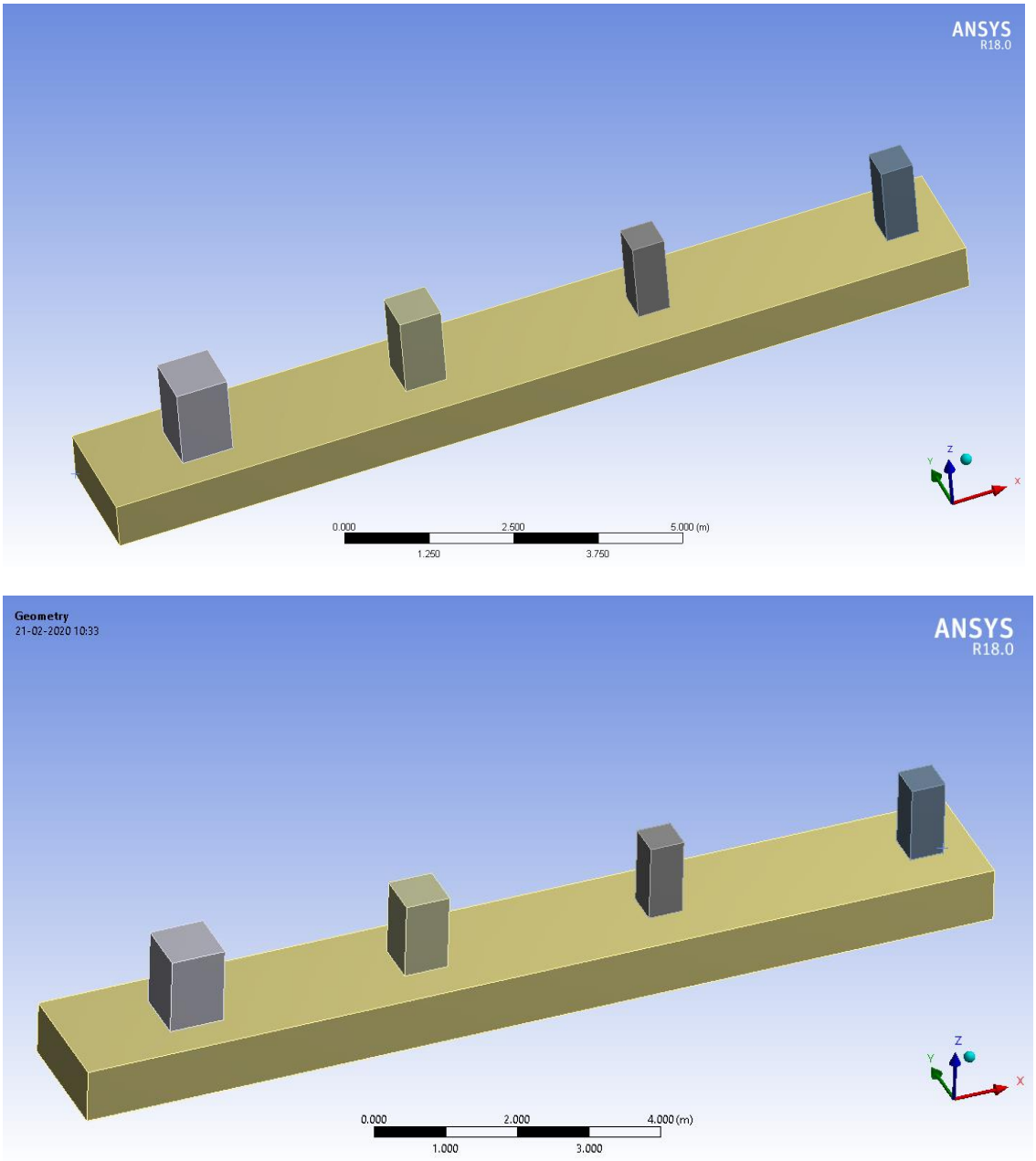


Figure 2. Solid Model of the Tailings Paste Thickening Foundation, above: smaller foundation (3710) and below: larger foundation (3710A)

3. Finite Element Modelling

3.1. Mesh

Figure 3 shows the finite element meshes. The number of elements and nodes are listed in Table 2.

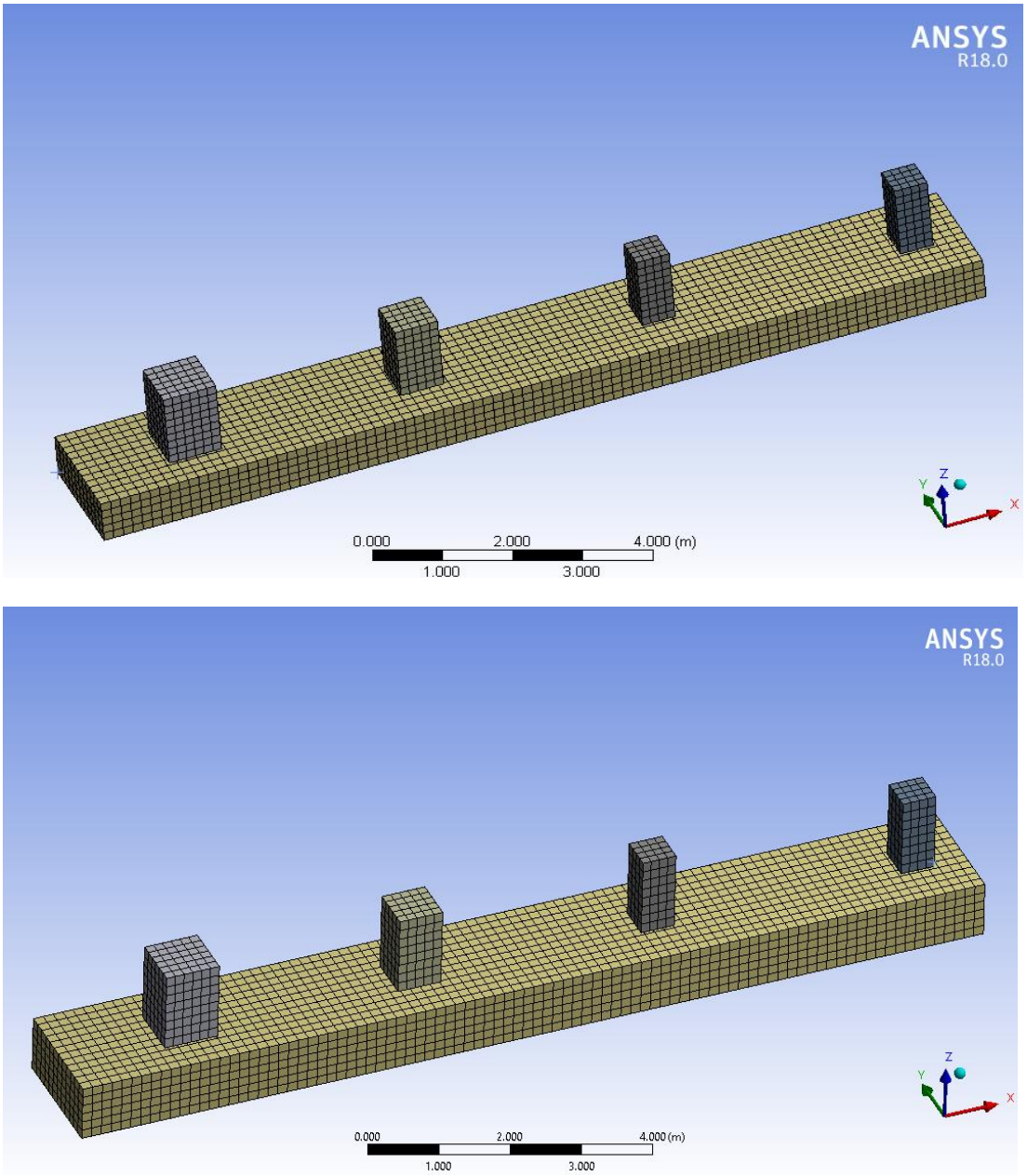


Figure 3. Finite Element Mesh, above: smaller foundation (3710) and below: larger foundation (3710A)

Statistics	
Nodes	31390
Elements	6237

Statistics	
Nodes	36212
Elements	7368

Table 2. Finite Element Mesh Statistics, left: smaller foundation (3710) and right: larger foundation (3710A)

3.2. Boundary Conditions

The models are attached with elastic support (springs) at the bottom of the foundations, shown in the Figure-4. Displacements at the bottom of the foundation for other than vertical directions are restrained.

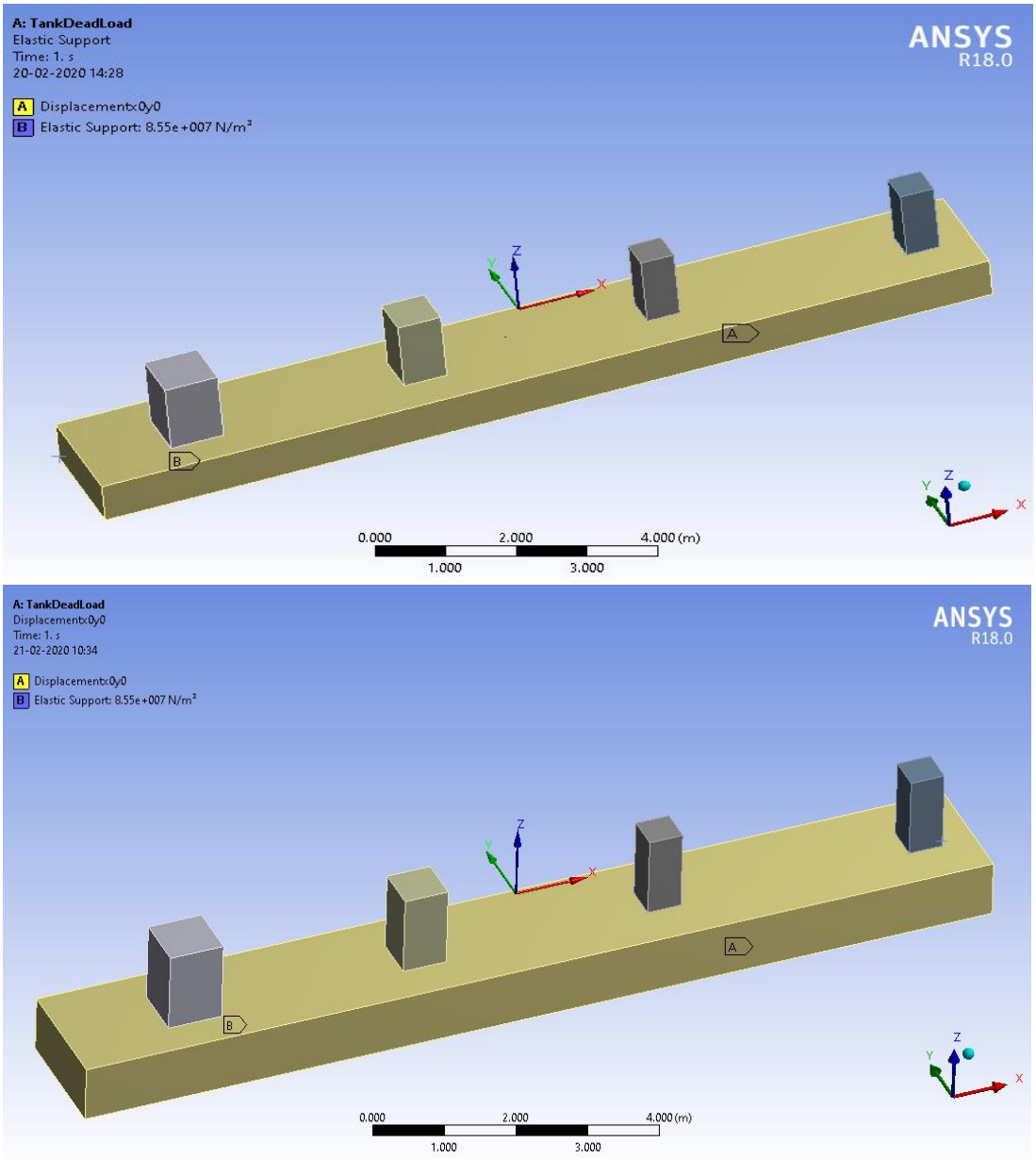


Figure 4. Support at bottom of foundation, above: smaller foundation (3710) and below: larger foundation (3710A)

3.3. Dead Loads (D)

3.3.1. Self-weight of Concrete Foundation

Self-weight of foundation is incorporated in the model by inserting standard gravity of the acceleration.

3.3.2. Soil Weight Above Foundation

Soil weights are applied at the top face of the foundation and shown in Table-3.

Soil Wt:		
Soil Density	18	kN/m3
soil Height	0.75	m
Pressure	13.5	kN/m2
Area of Foundation	24.3	m2
Load of Soil	328.05	kN

Soil Wt:		
Soil Density	18	kN/m3
soil Height	0.65	m
Pressure	11.7	kN/m2
Area of Foundation	24.3	m2
Load of Soil	284.31	kN

Table 3. Soil weight above foundation, left: smaller foundation (3710) and right: larger foundation (3710A)

3.3.3. Tank Dead Loads

Tank dead loads are applied at the top the column pedestals (taken from [3]).

3.3.4. Total Dead Loads

Total dead loads are coming from (1) Self-weight of the Concrete foundation, (2) Soil weights above foundations, and (3) Tank Dead loads. Total Dead loads as primary load cases are shown in Figure-5.

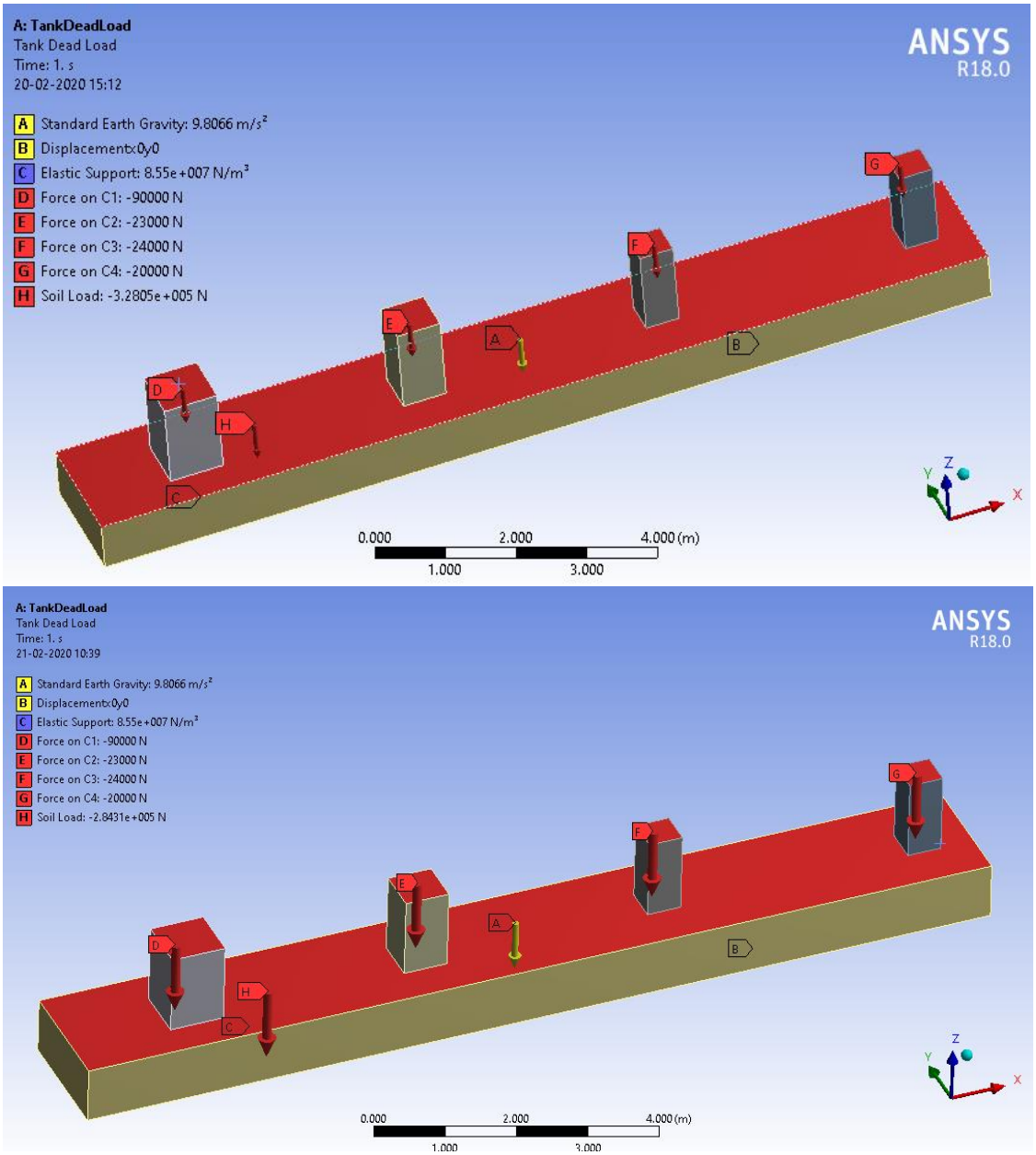


Figure 5. Dead load case, above: smaller foundation (3710) and below: larger foundation (3710A)

3.4. Maximum Fluid Loads (F1)

Tank maximum fluid loads are taken from [3], shown in Figure-6.

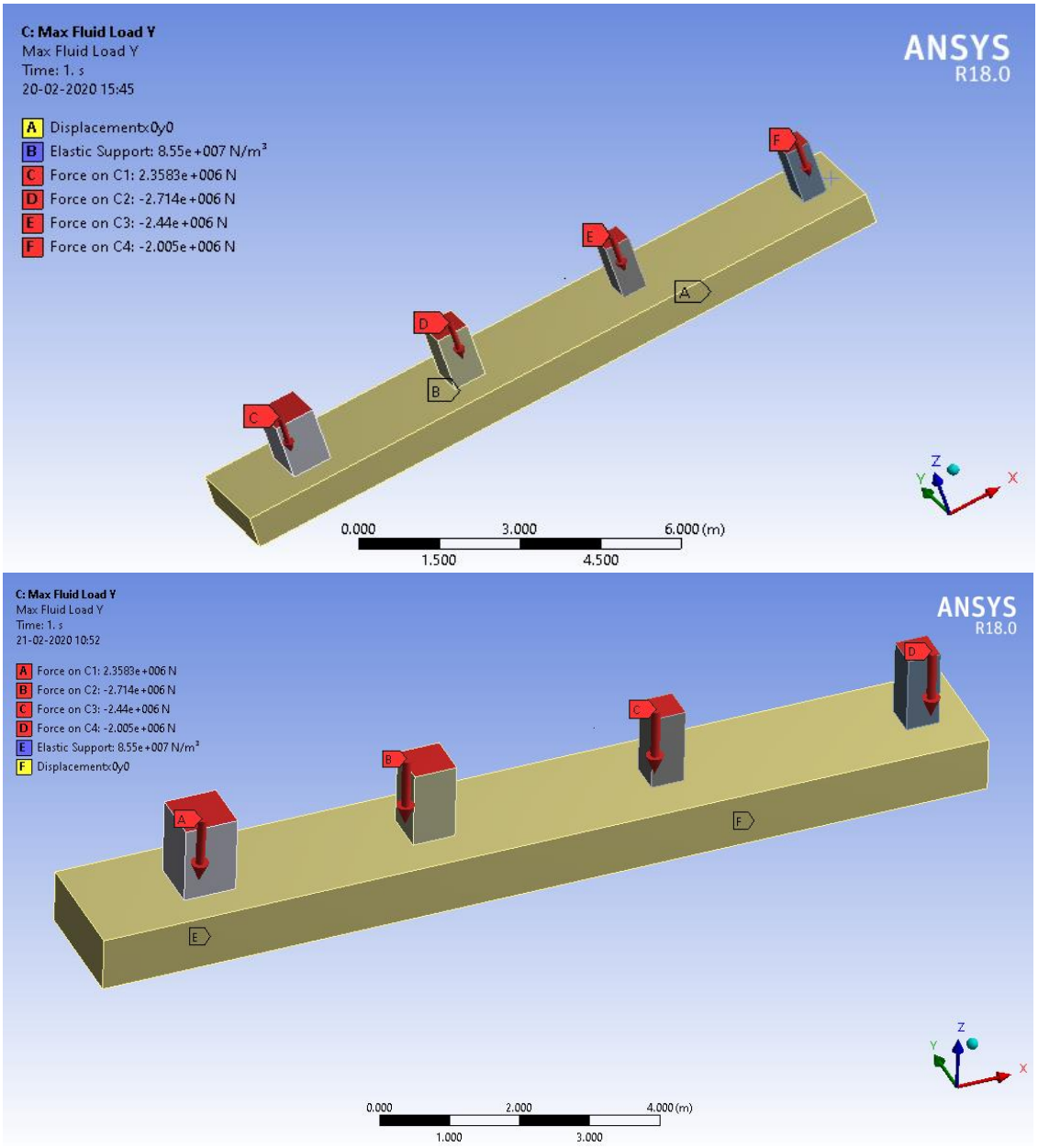


Figure 6. Maximum Fluid load cases, above: smaller foundation (3710) and below: larger foundation (3710A)

3.5. Normal Operating Fluid Loads (F2)

Tank normal operating fluid loads are taken from [3], shown in Figure-7.

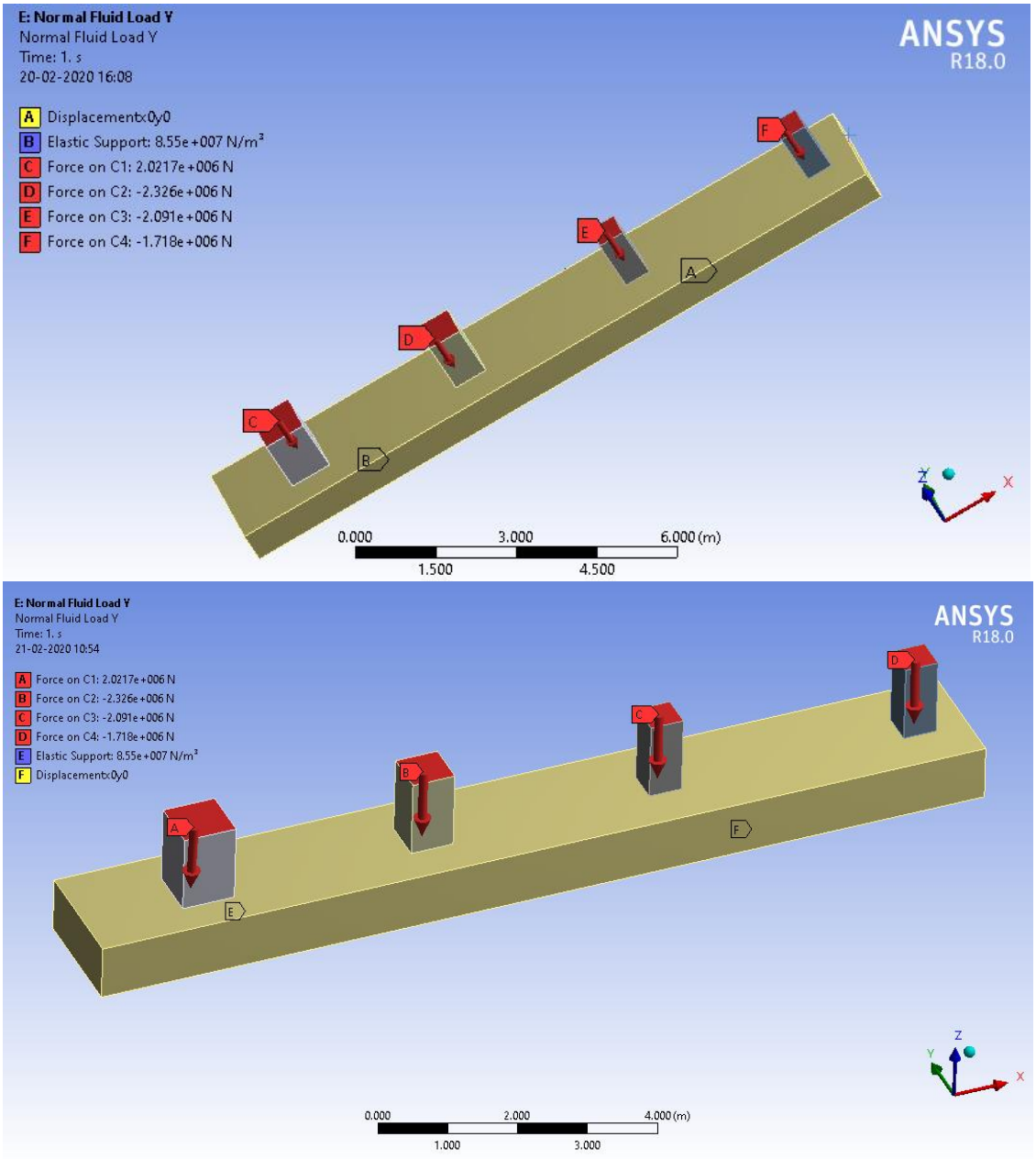


Figure 7. Normal Operation Fluid load cases, above: smaller foundation (3710) and below: larger foundation (3710A)

3.6. Seismic Loads (E)

Seismic loads are taken from [3], shown in Figure-8.

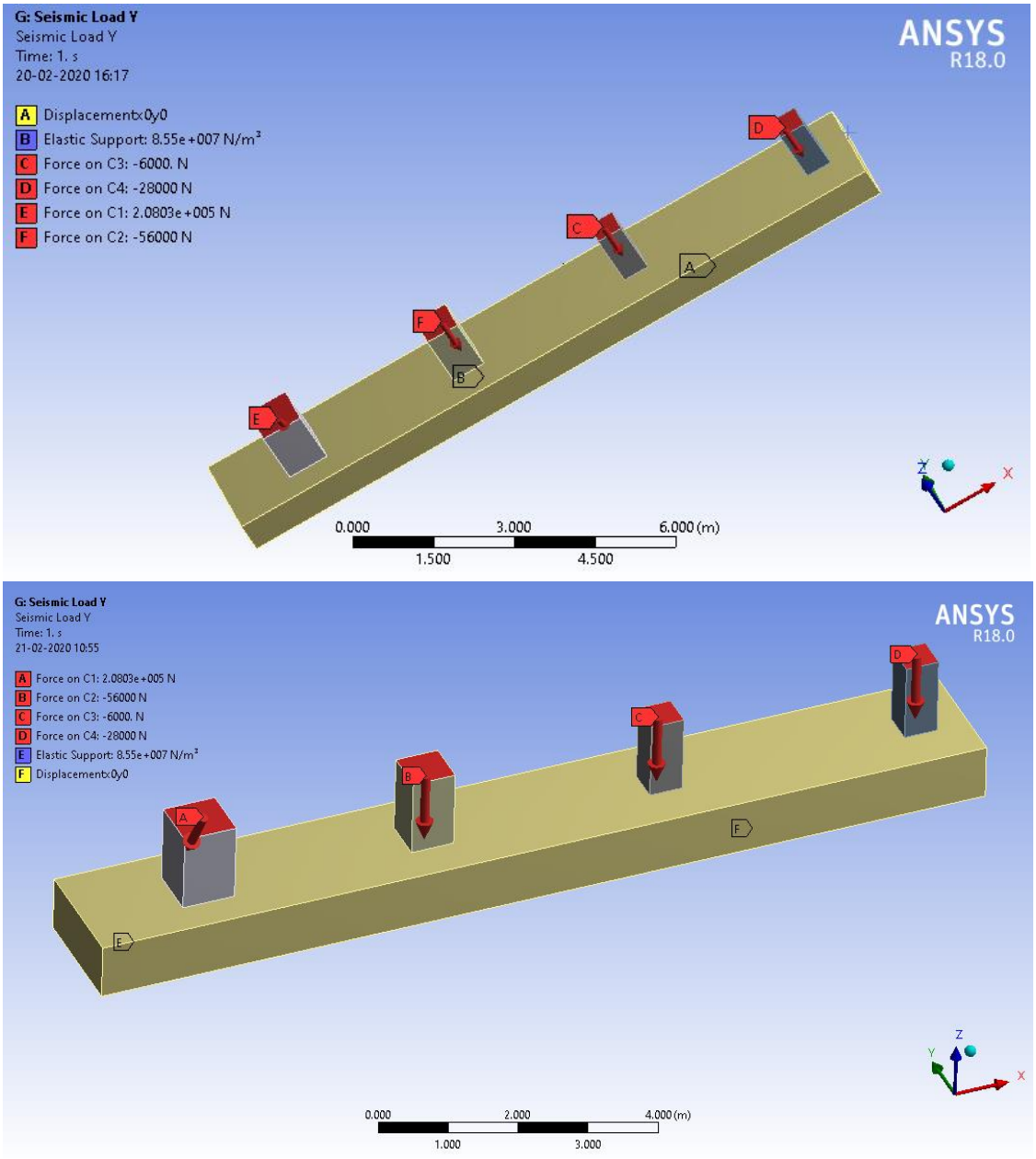


Figure 8. Seismic load cases, above: smaller foundation (3710) and below: larger foundation (3710A)

3.7. Wind Loads (W)

Wind loads are taken from [3], shown in Figure-9.

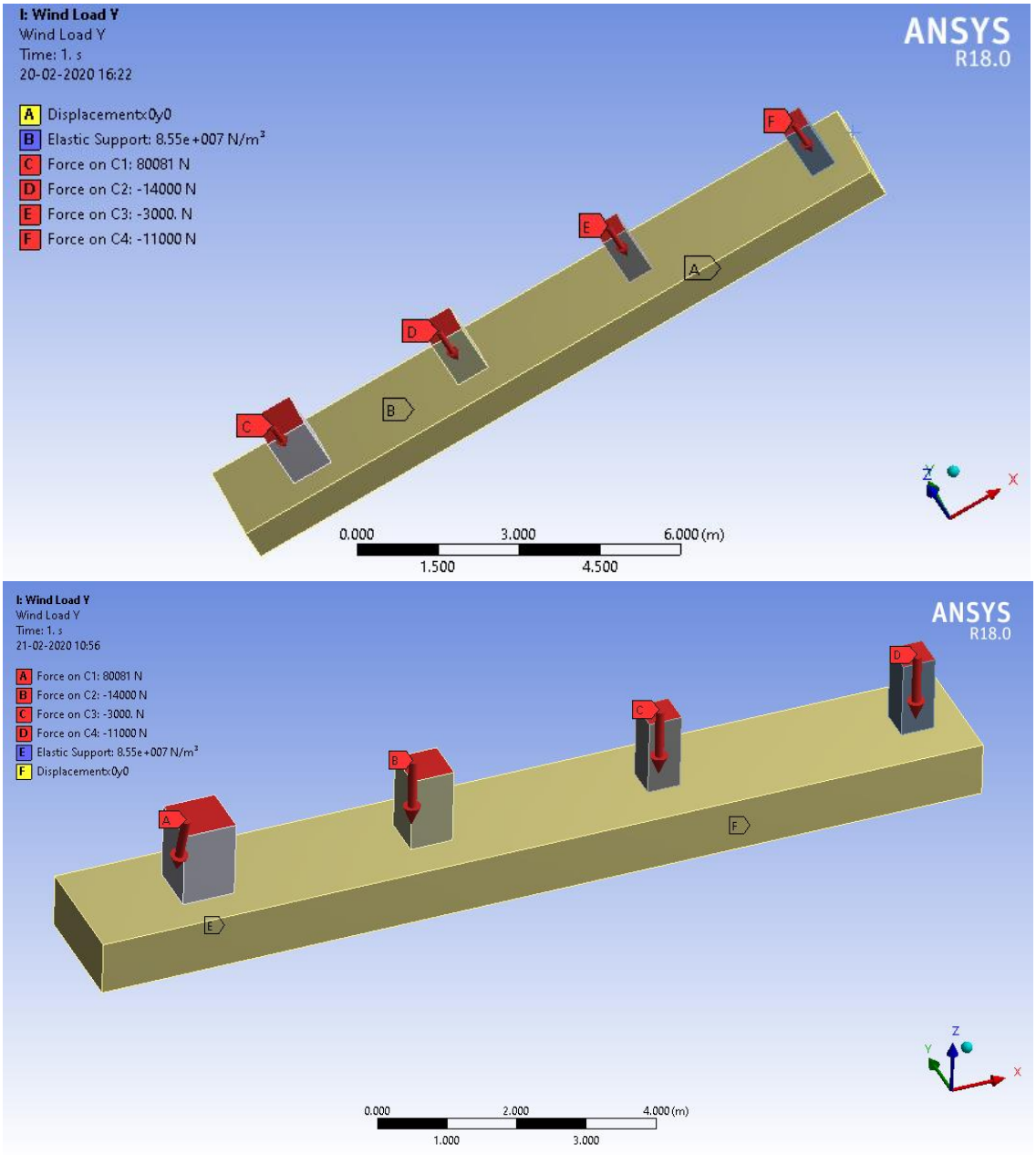


Figure 9. Wind load cases, above: smaller foundation (3710) and below: larger foundation (3710A)

3.8. Bridge Loads (B)

Bridge loads are only applied in the larger foundation type (3710A), shown in Figure-10.

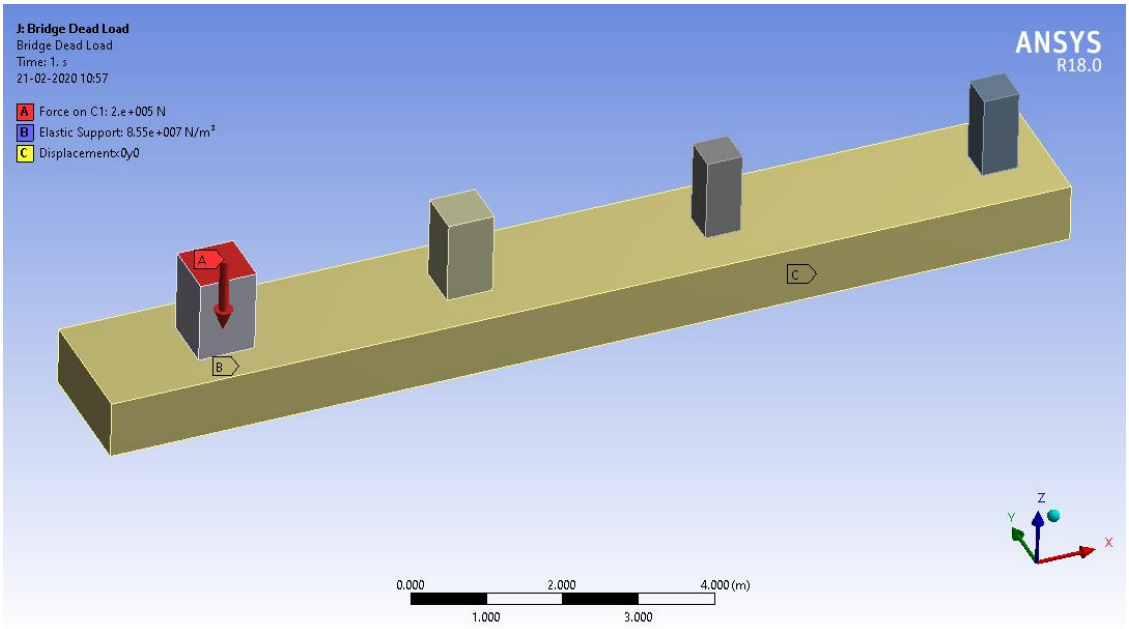


Figure 10. Bridge load cases, above: smaller foundation (3710) and below: larger foundation (3710A)

4. Analysis

Main purpose of the analysis is to ensure that the pressures under the foundations are within the design limit.

4.1. Static Analysis: Primary Load Cases

Following five different primary load cases [3] are analyzed in ANSYS [1]:

1. Tank Dead Load (D)
2. Max. Fluid Load (F1)
3. Normal Operational Fluid Load (F2)
4. Seismic Load (E)
5. Wind Load (W)

Extra primary load case, Bridge Load (B) is considered only for larger foundation type (3710A).

4.2. Static Analysis: Load Combinations

Load combination cases are factored superpositions of primary load cases. Three separate load combinations are considered with different multiplication coefficients, shown in Table 4.

Load Case	Load Combination
201	D + F1
202	D + F1 + W
203	D + F2+ 0.7 E

Table 4. Load Combination Table, taken from [3]

4.3. Location of Study

Four outer corners of the bottom of the foundations (P1, P2, P3, P4) are considered for study (shown in Figure-11).

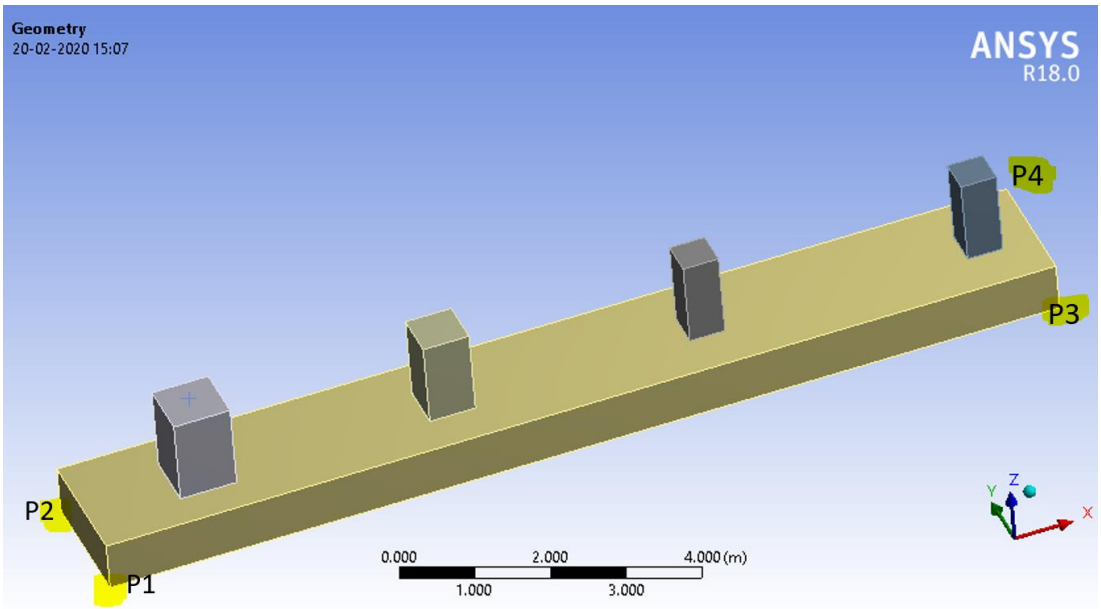


Figure 11. Location of Study

5. Results

5.1. Dead Loads (D)

5.1.1. Vertical Displacement for Dead Load Cases

Vertical displacements of the foundations due to dead loads are shown in Figure-12.

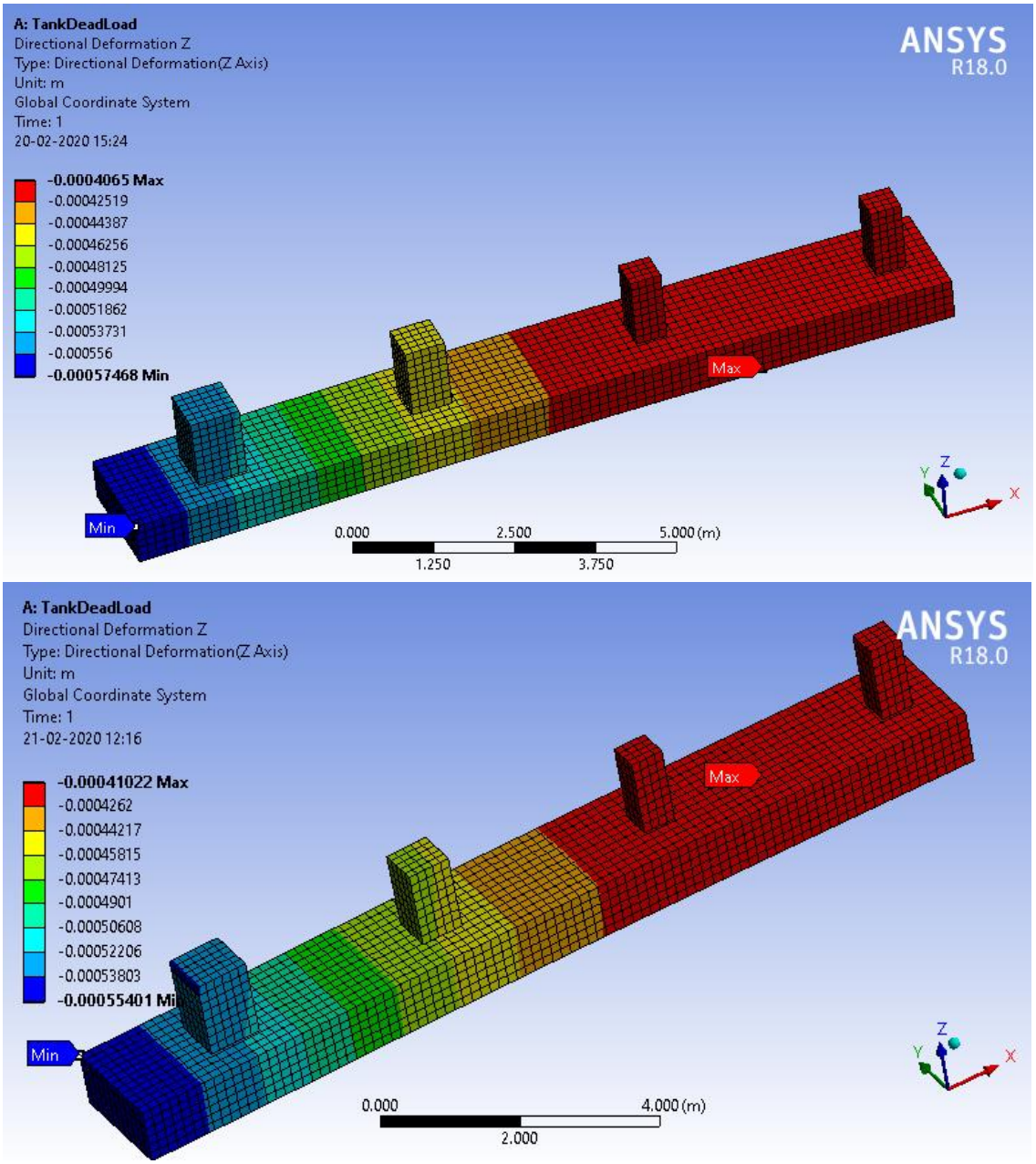


Figure 12. Vertical Deformation for dead load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.1.2. Vertical Settlement at P1 for Dead Load Cases

Vertical displacements at P1 corner of the foundations due to dead loads are shown in Figure-13.

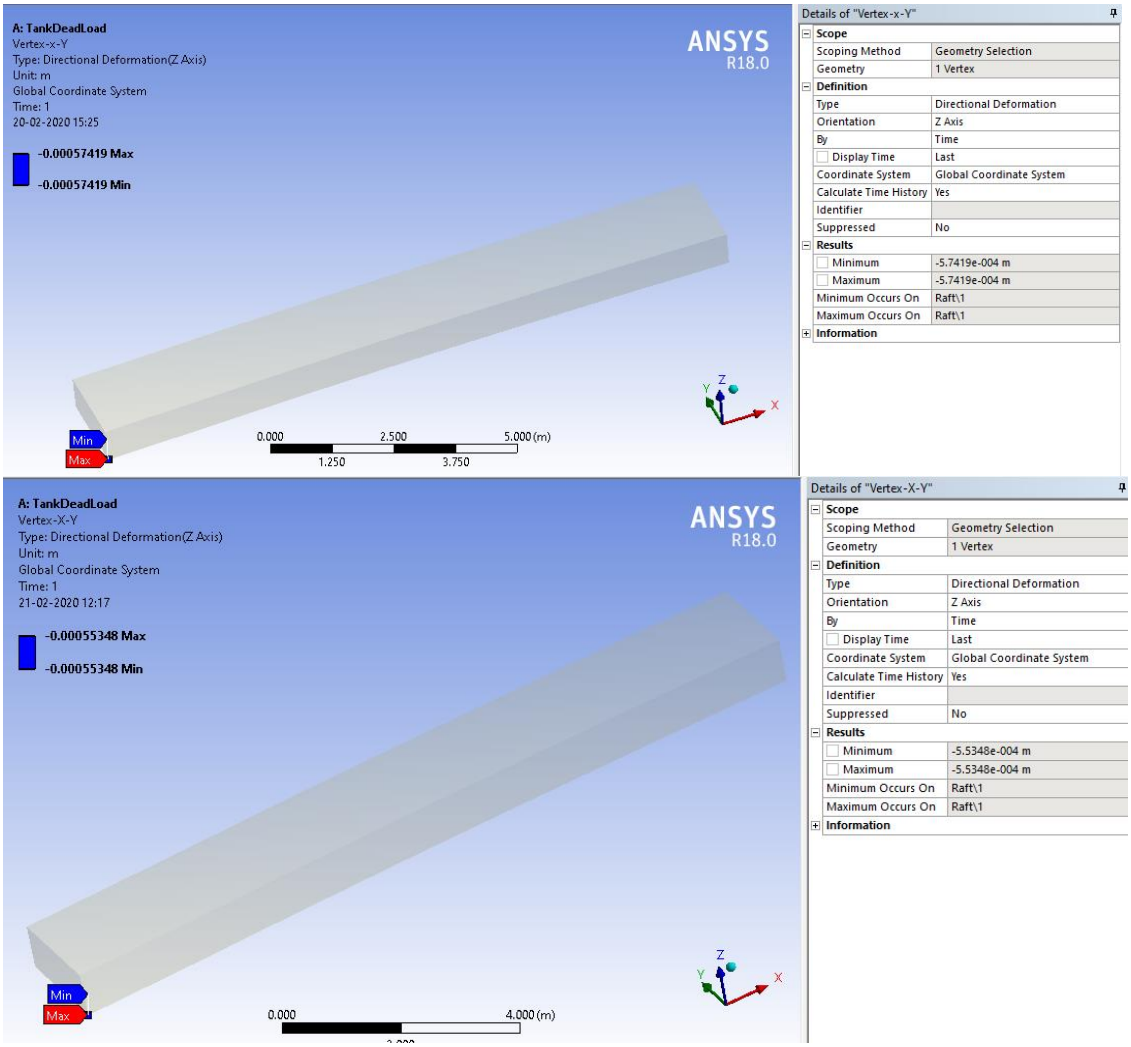


Figure 13. Vertical Deflection at P1 for Dead load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.1.3. Vertical Settlement at P2 for Dead Load Cases

Vertical displacements at P2 corner of the foundations due to dead loads are shown in Figure-14.

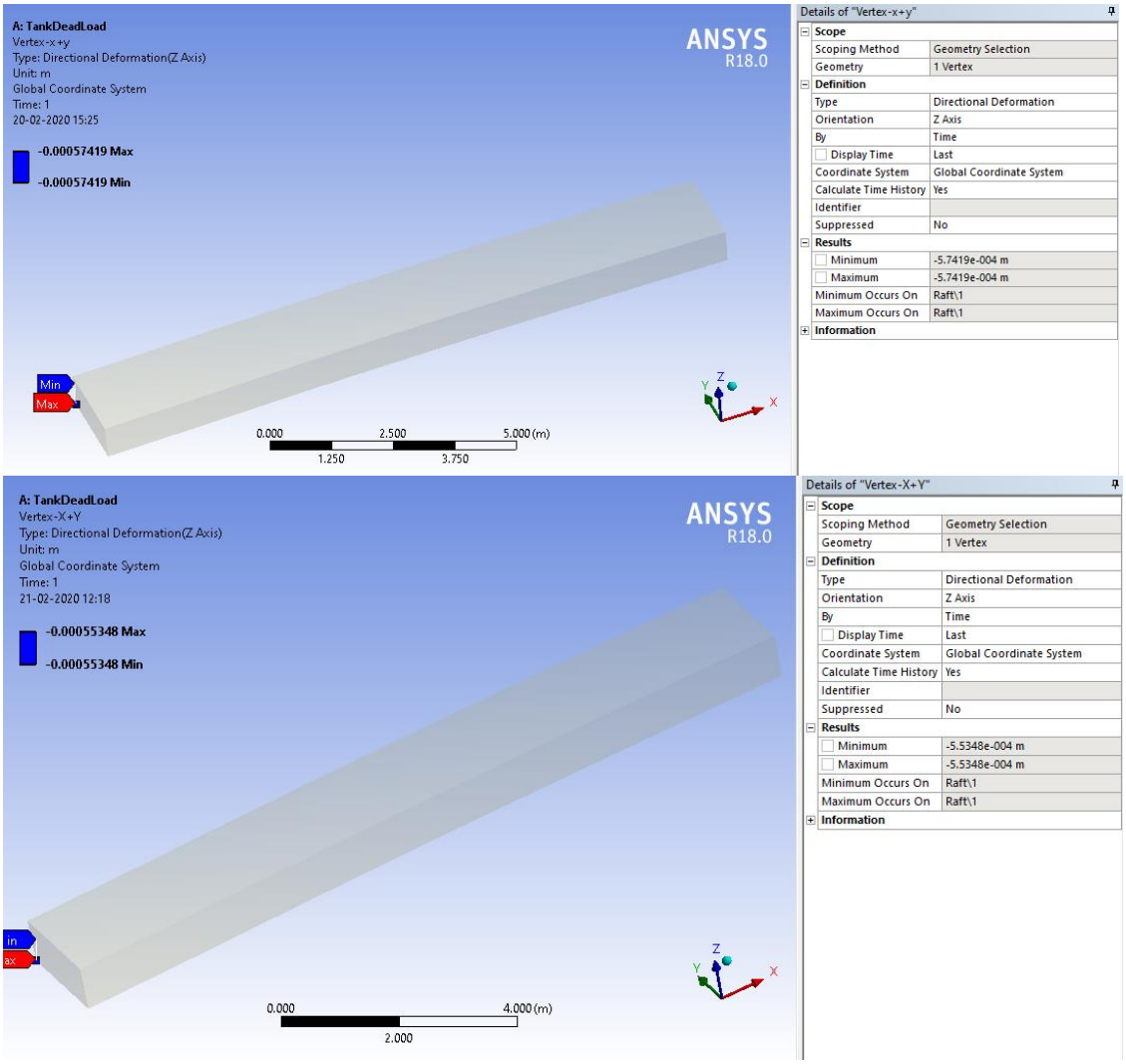


Figure 14. Vertical Deflection at P1 for Dead load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.1.4. Vertical Settlement at P3 for Dead Load Cases

Vertical displacements at P3 corner of the foundations due to dead loads are shown in Figure-15.

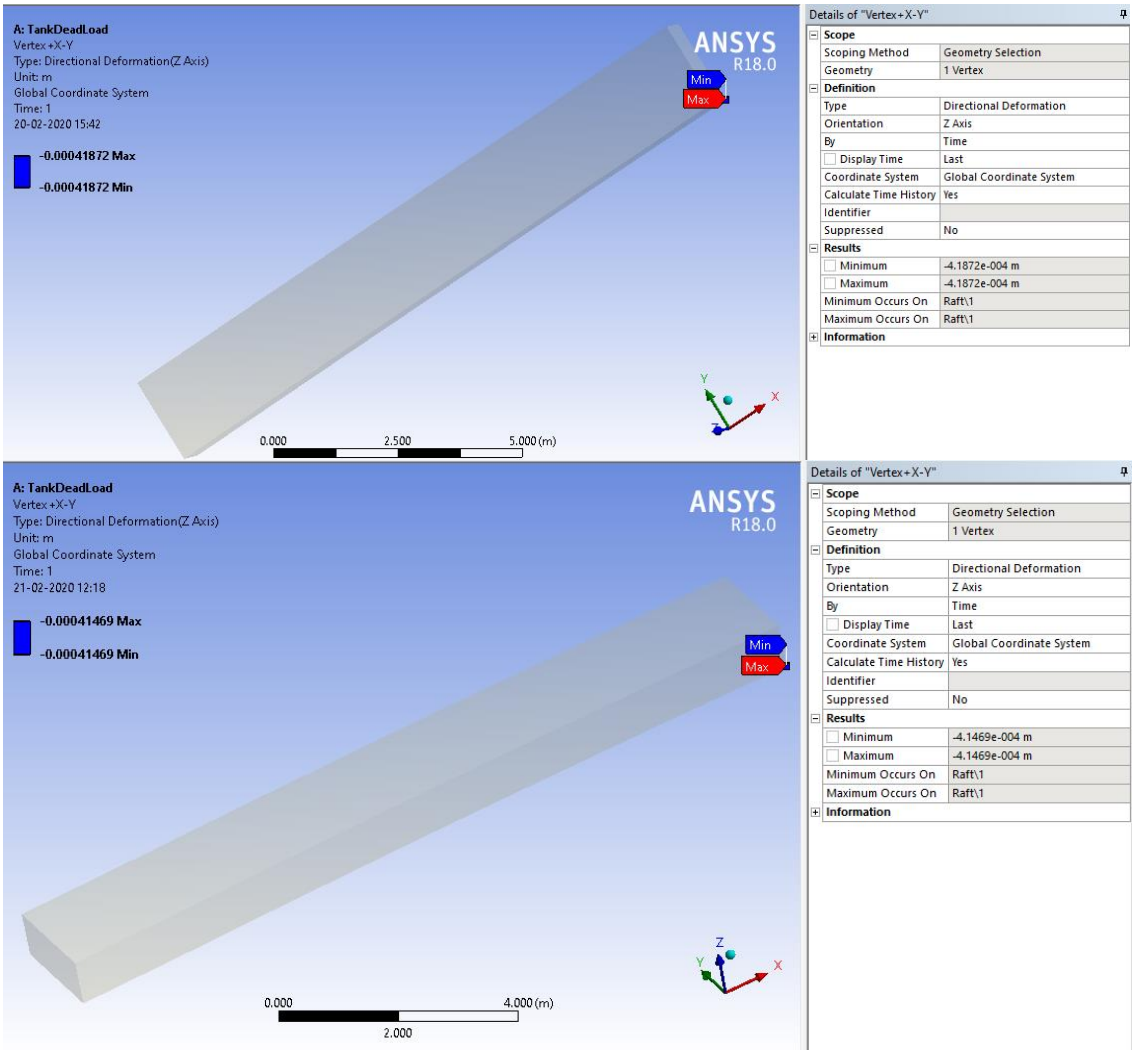


Figure 15. Vertical Deflection at P1 for Dead load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.1.5. Vertical Settlement at P4 for Dead Load Cases

Vertical displacements at P4 corner of the foundations due to dead loads are shown in Figure-16.

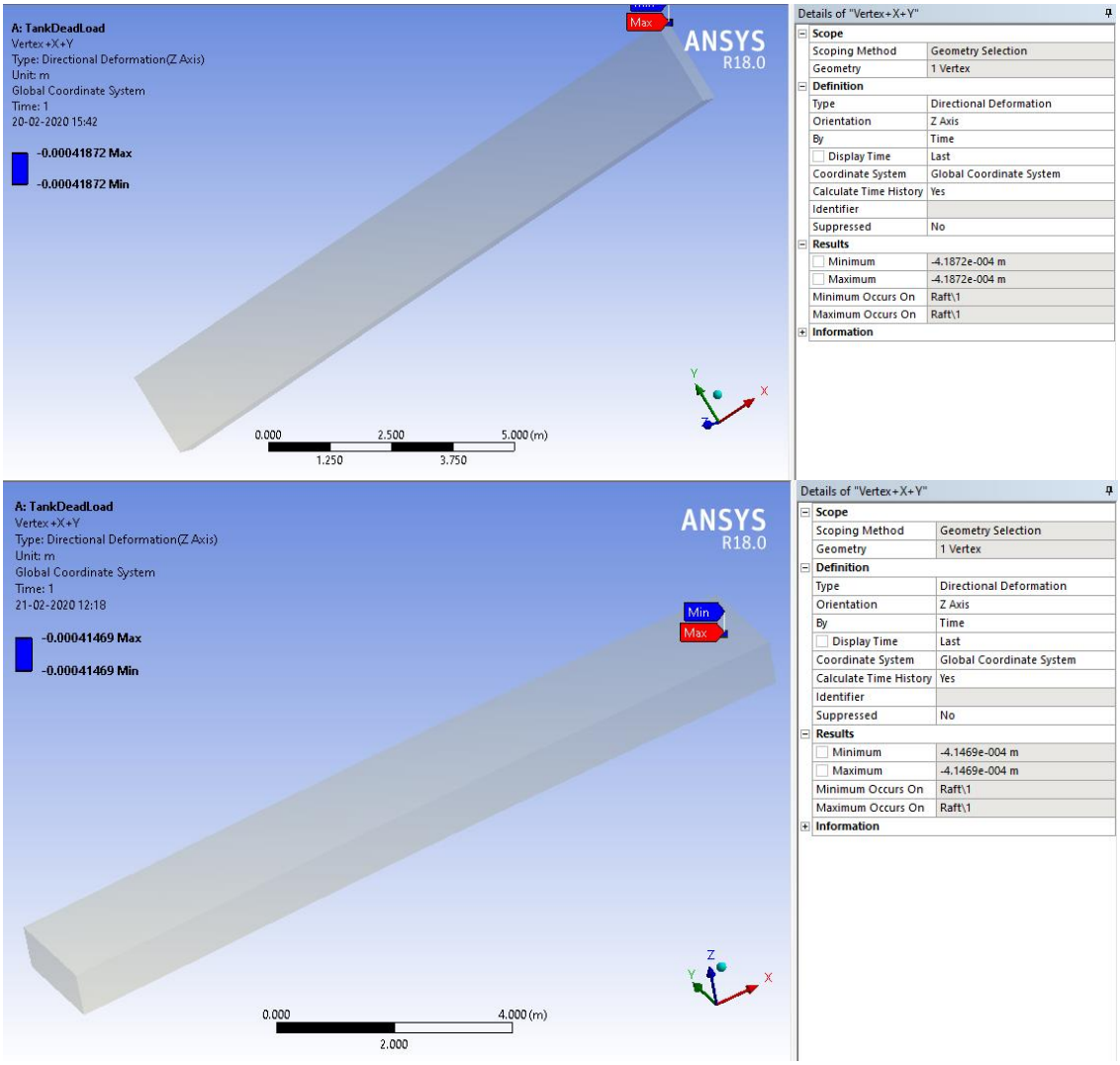


Figure 16. Vertical Deflection at P1 for Dead load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.2. Maximum Fluid Loads (F1)

5.2.1. Vertical Displacement for Maximum Fluid Load Cases

Vertical displacements of the foundations due to maximum fluid loads are shown in Figure-17.

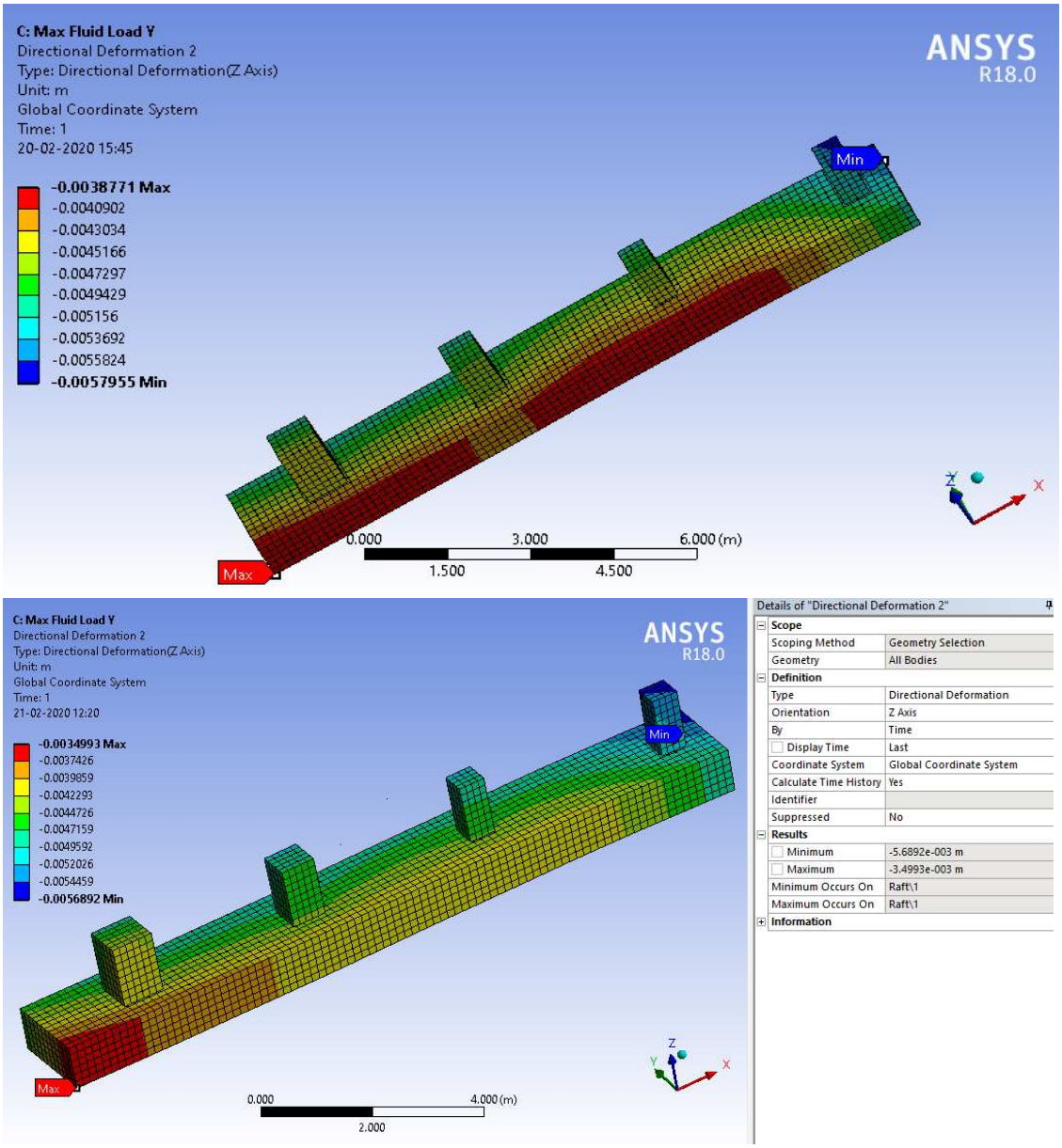


Figure 17. Vertical Deformation due to maximum fluid load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.2.2. Vertical Settlement at P1 for Maximum Fluid Load Cases

Vertical displacements at P1 corner of the foundations due to maximum fluid loads are shown in Figure-18.

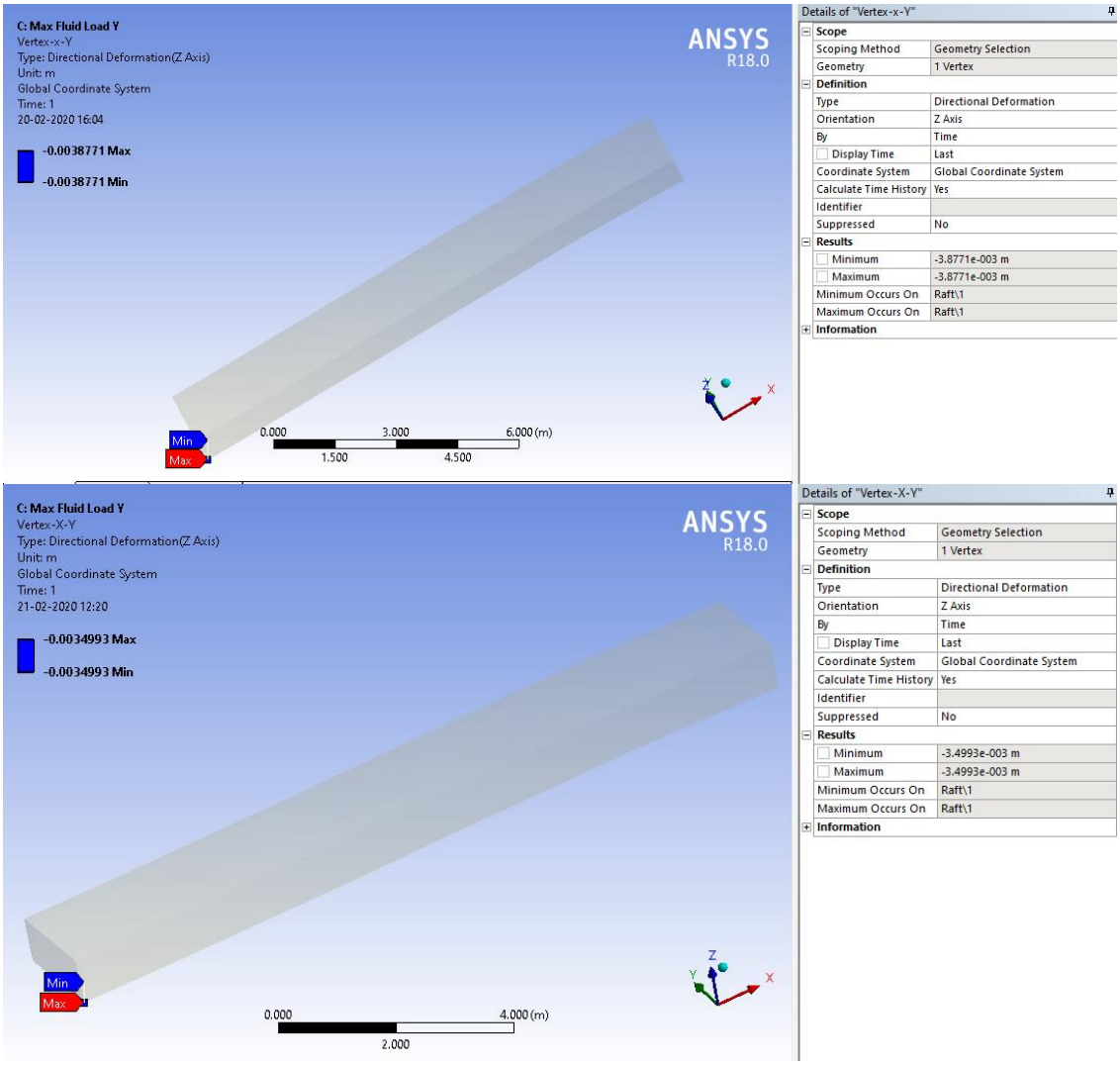


Figure 18. Vertical Deflection at P1 due to maximum fluid load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.2.3. Vertical Settlement at P2 for Maximum Fluid Load Cases

Vertical displacements at P2 corner of the foundations due to maximum fluid loads are shown in Figure-19.

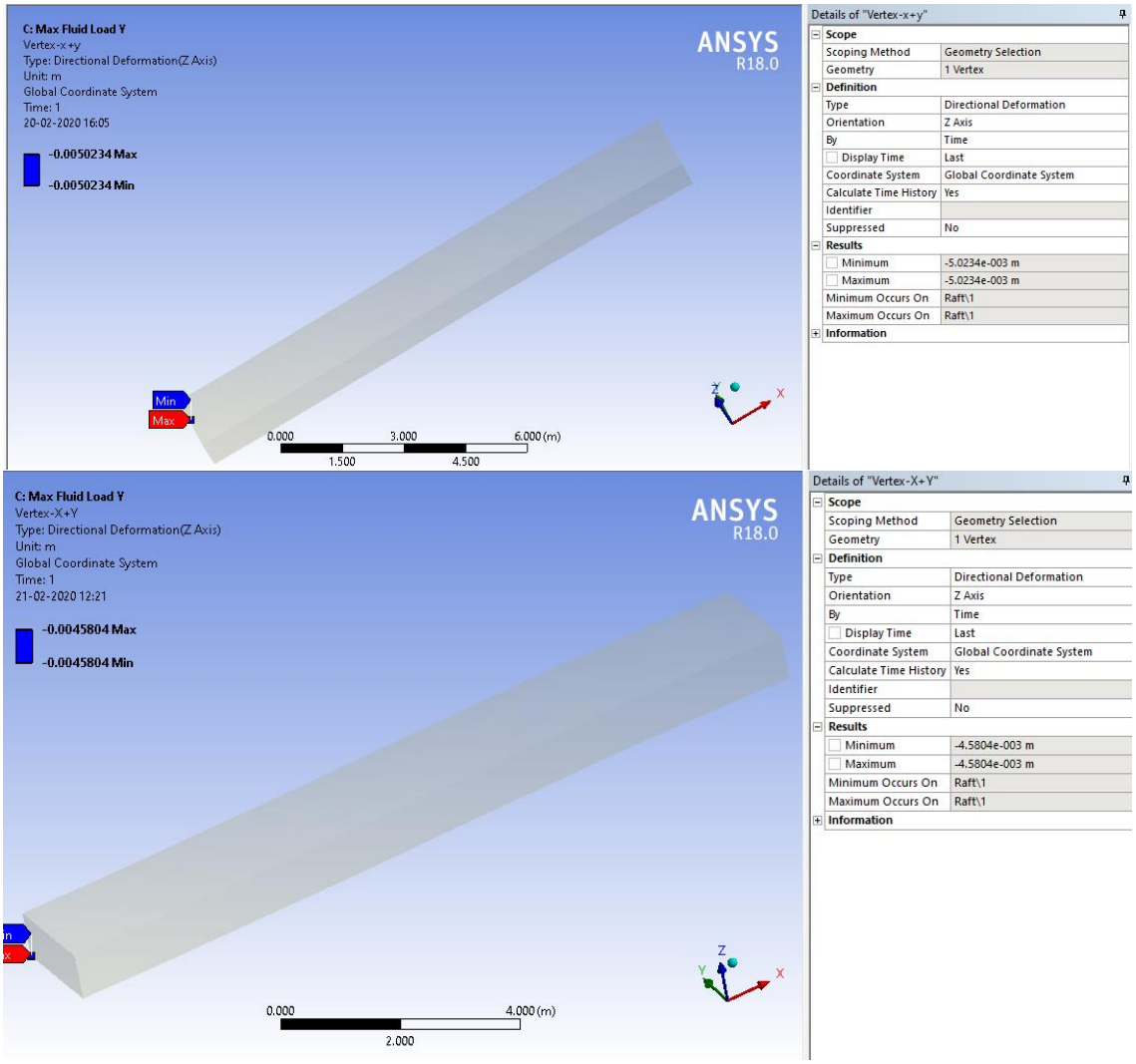


Figure 19. Vertical Deflection at P1 due to maximum fluid load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.2.4. Vertical Settlement at P3 for Maximum Fluid Load Cases

Vertical displacements at P3 corner of the foundations due to maximum fluid loads are shown in Figure-20.

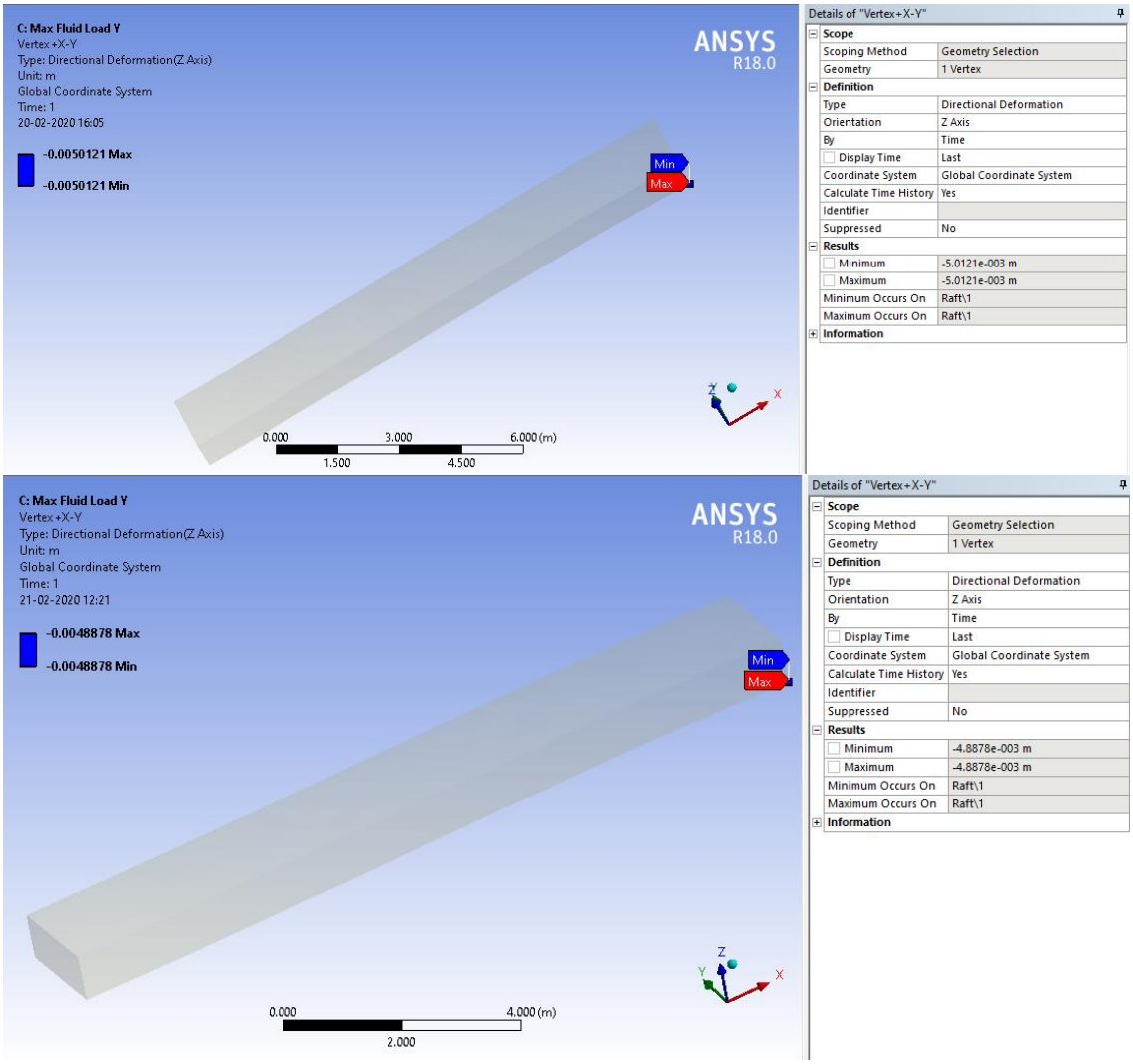


Figure 20. Vertical Deflection at P1 due to maximum fluid load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.2.5. Vertical Settlement at P4 for Maximum Fluid Load Cases

Vertical displacements at P4 corner of the foundations due to maximum fluid loads are shown in Figure-21.

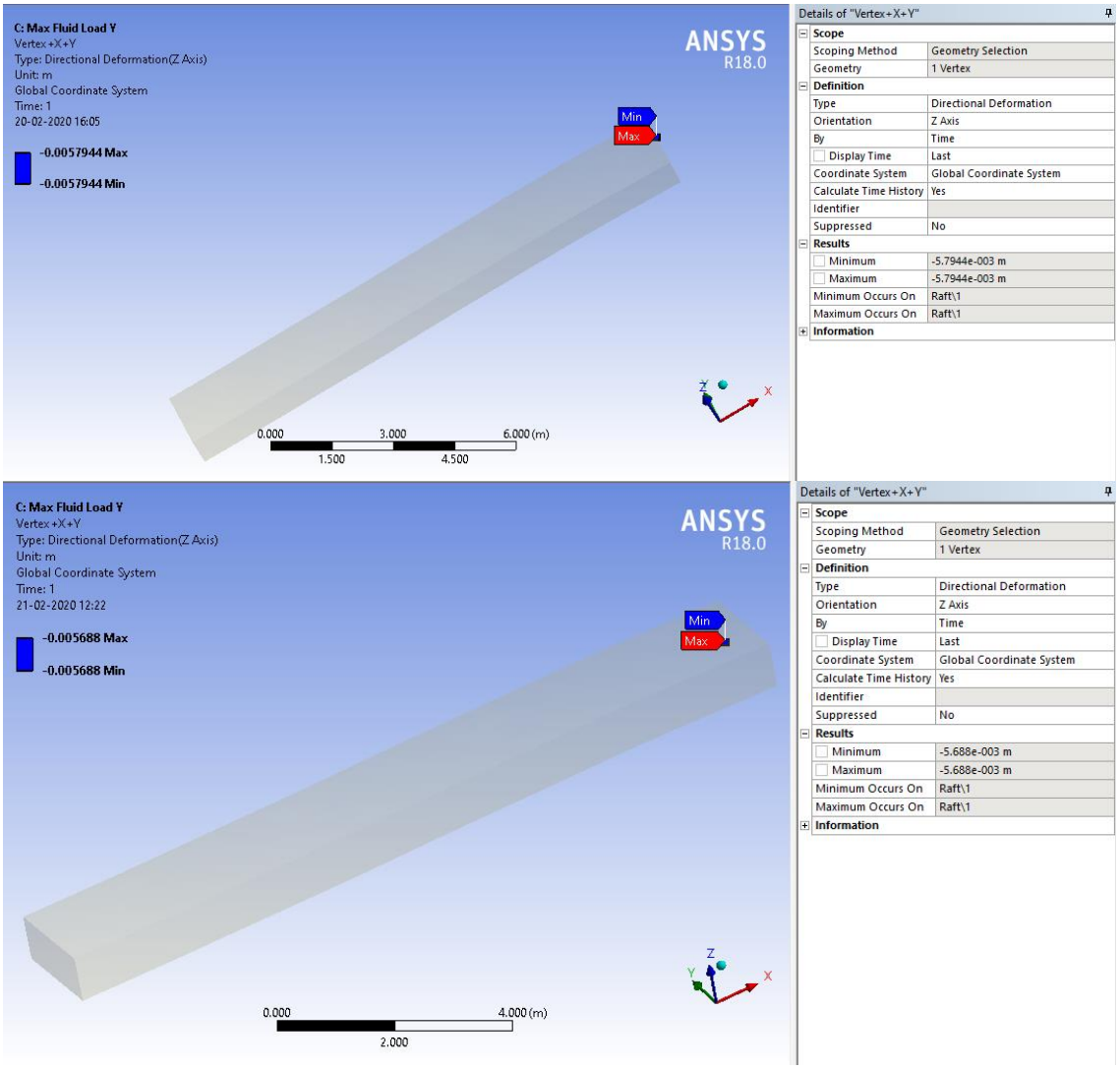


Figure 21. Vertical Deflection at P1 due to maximum fluid loads case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.3. Normal Operating Fluid Loads (F2)

5.3.1. Vertical Displacement for Normal Operating Fluid Load Cases

Vertical displacements of the foundations due to normal operating fluid loads are shown in Figure-22.

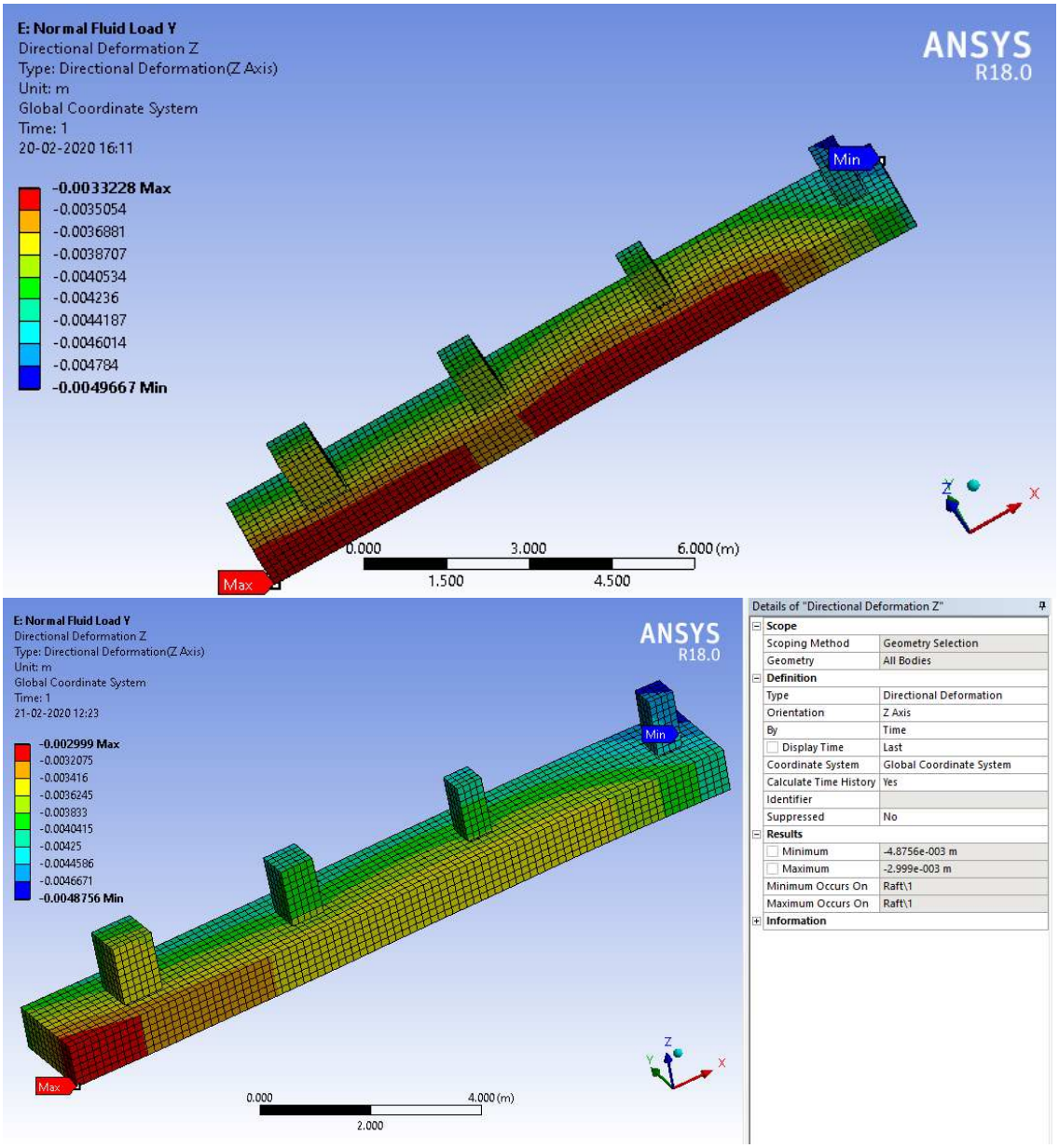


Figure 22. Vertical Deformation due to normal operating fluid load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.3.2. Vertical Settlement at P1 for Normal Operational Fluid Load Cases

Vertical displacements at P1 corner of the foundations due to normal operating fluid loads are shown in Figure-23.

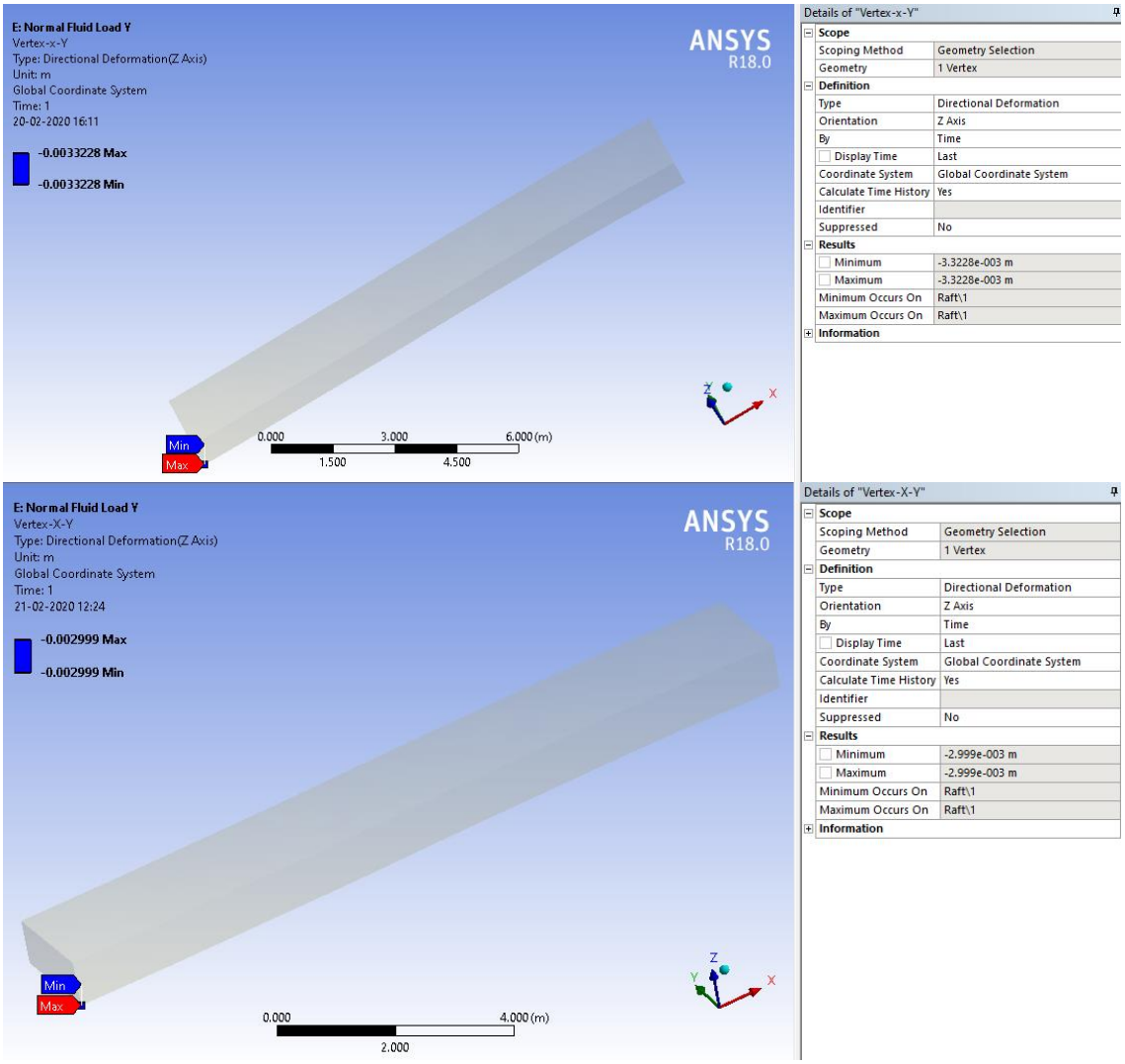


Figure 23. Vertical Deflection at P1 due to normal operating fluid load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.3.3. Vertical Settlement at P2 for Normal Operational Fluid Load Cases

Vertical displacements at P2 corner of the foundations due to normal operating fluid loads are shown in Figure-24.

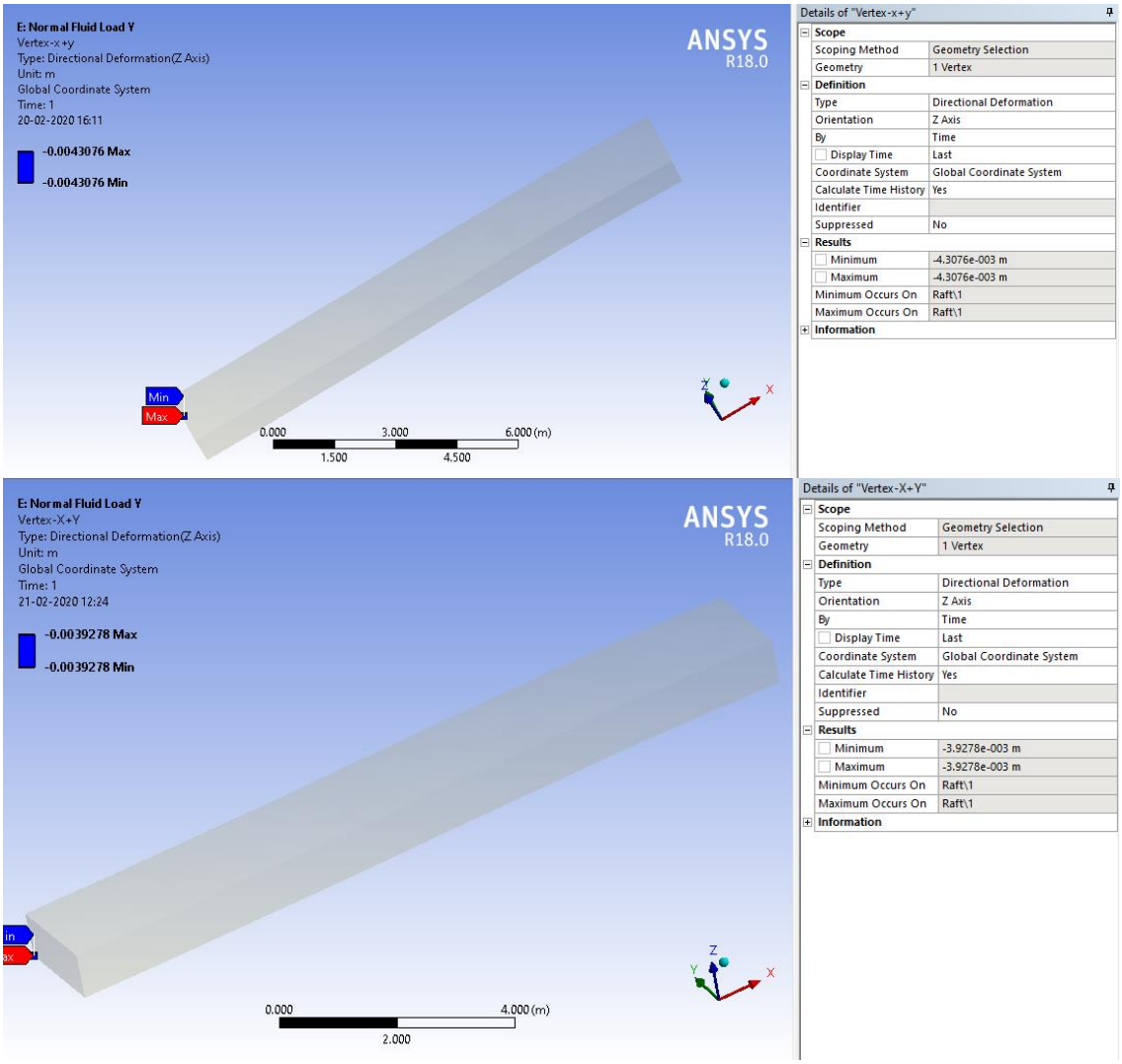


Figure 24. Vertical Deflection at P1 for due to normal operating fluid load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.3.4. Vertical Settlement at P3 for Normal Operational Fluid Load Cases

Vertical displacements at P3 corner of the foundations due to normal operating fluid loads are shown in Figure-25.

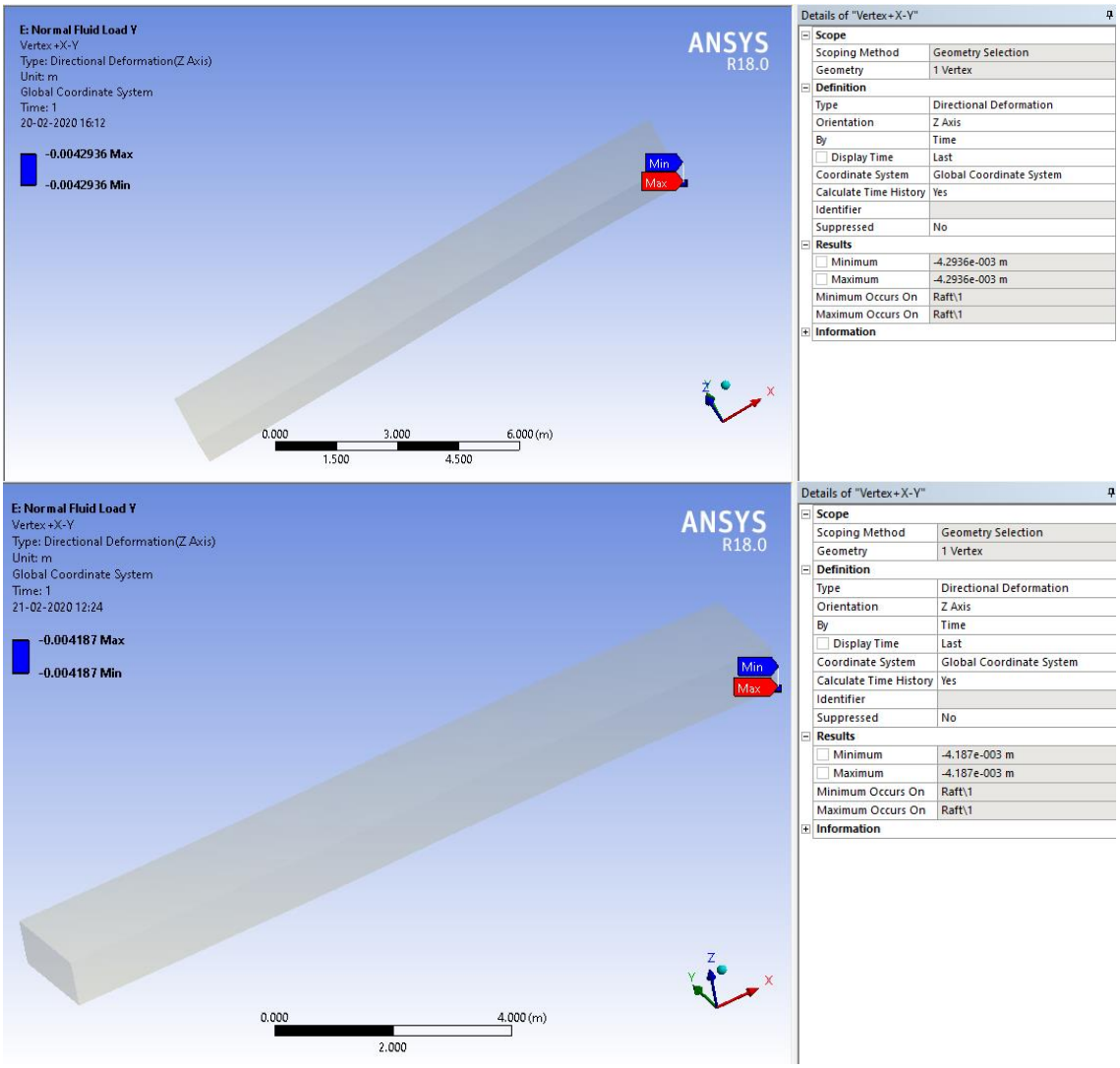


Figure 25. Vertical Deflection at P1 due to normal operating fluid loads case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.3.5. Vertical Settlement at P4 for Normal Operational Fluid Load Cases

Vertical displacements at P4 corner of the foundations due to normal operating fluid loads are shown in Figure-26.

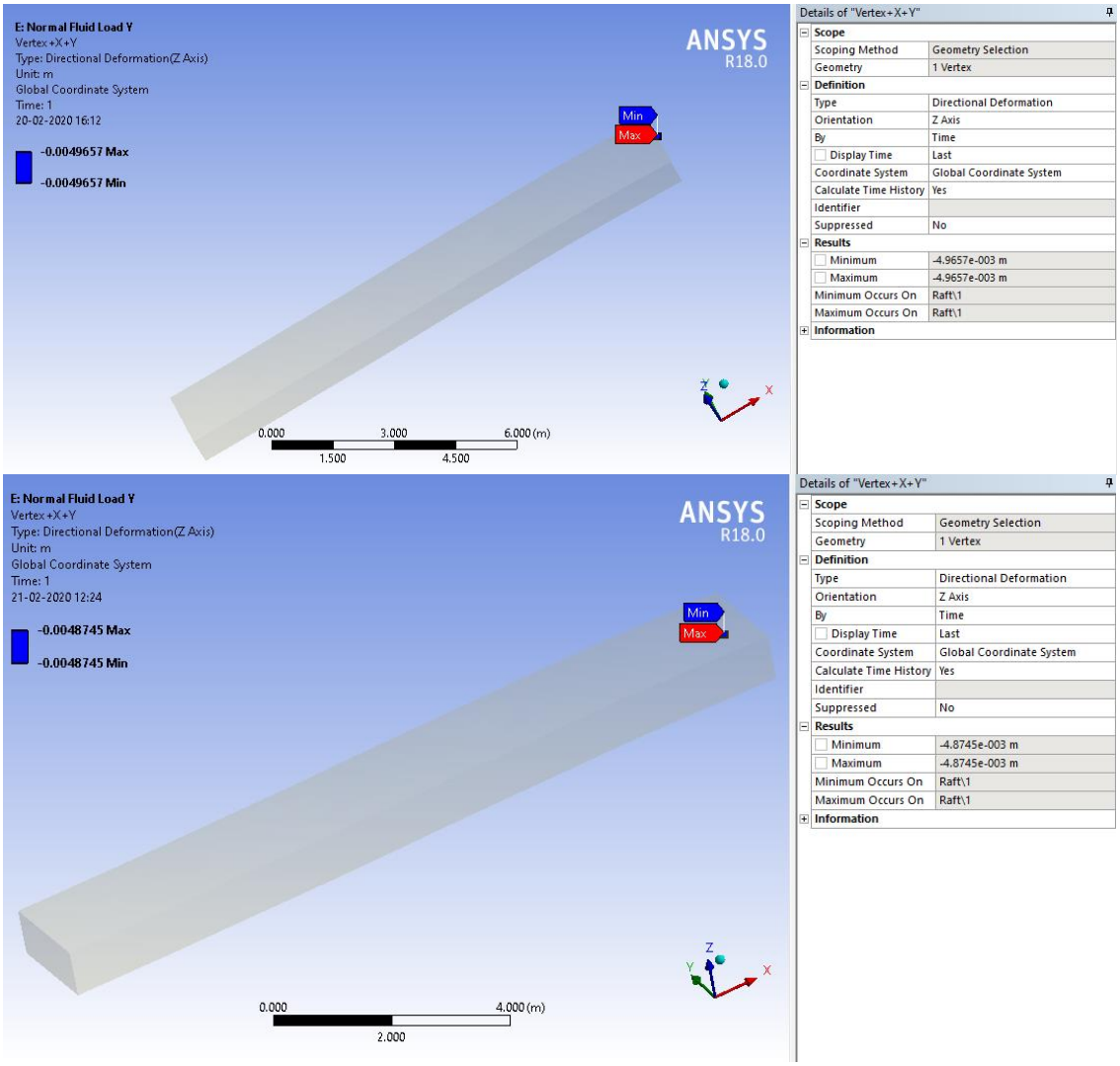


Figure 26. Vertical Deflection at P1 due to normal operating fluid loads case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.4. Seismic Loads (E)

5.4.1. Vertical Displacement for Seismic Load Cases

Vertical displacements of the foundations due to seismic loads are shown in Figure-27.

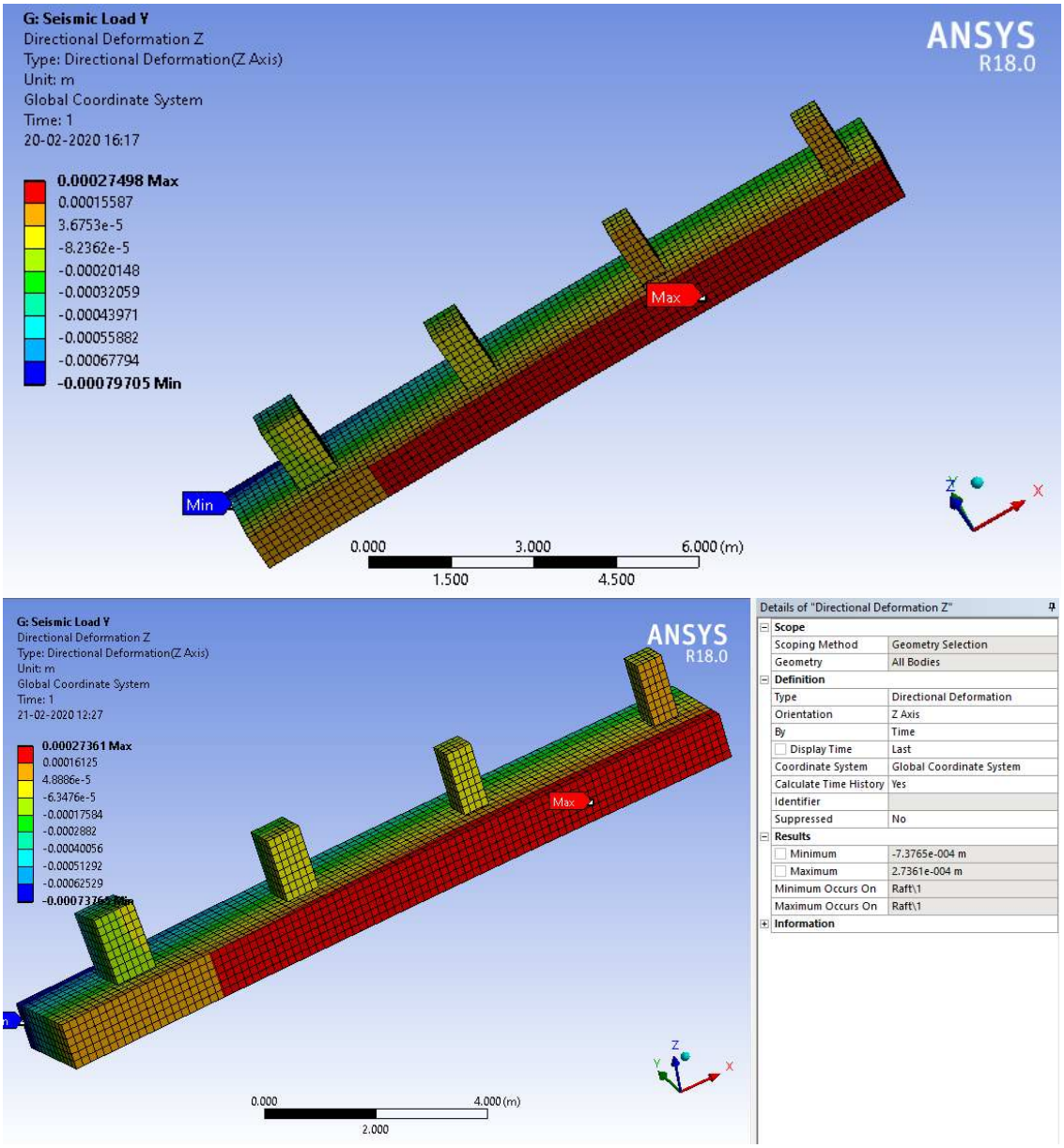


Figure 27. Vertical Deformation due to seismic loads case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.4.2. Vertical Settlement at P1 for Seismic Load Cases

Vertical displacements at P1 corner of the foundations due to seismic loads are shown in Figure-28.

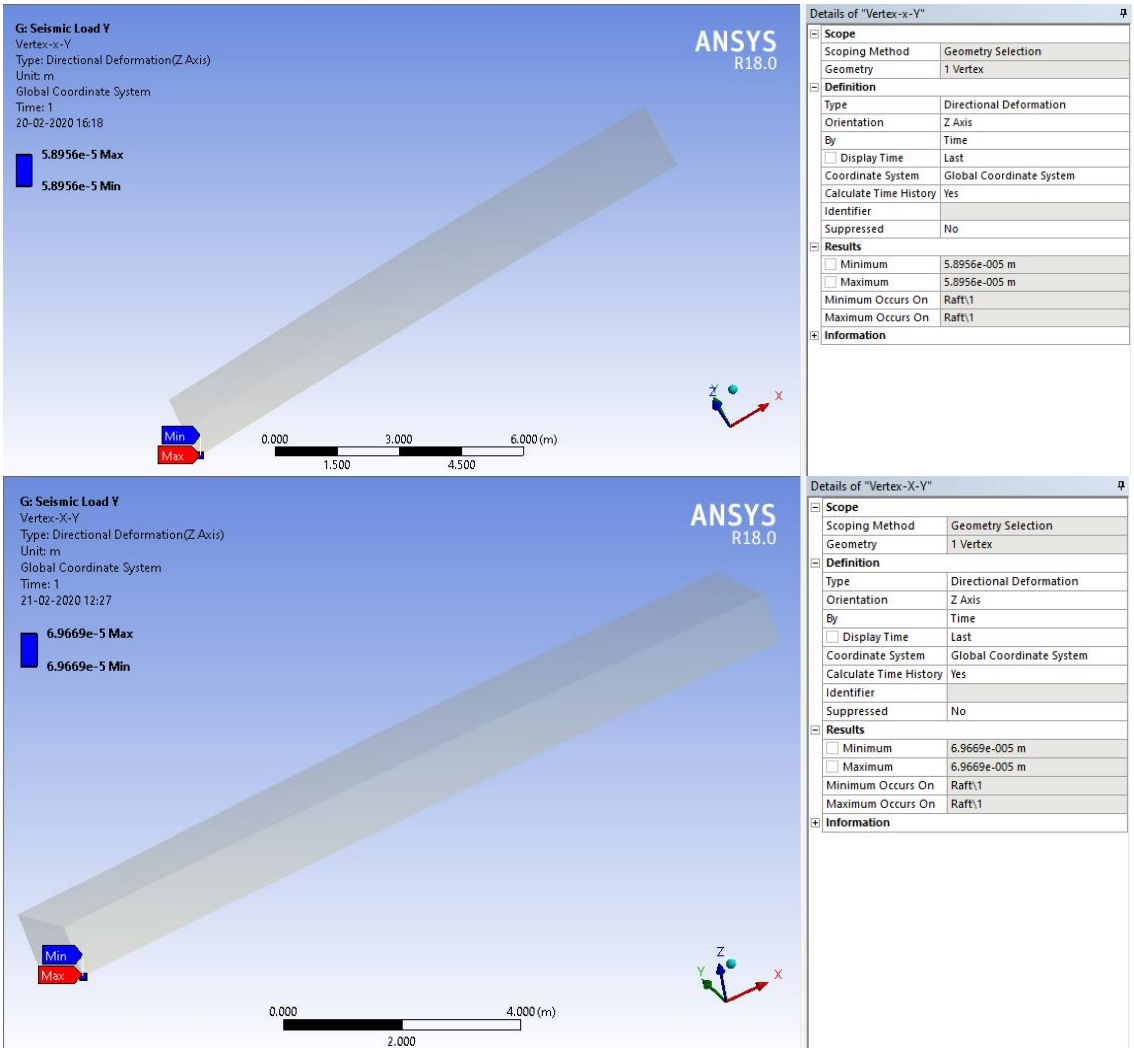


Figure 28. Vertical Deflection at P1 due to seismic loads case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.4.3. Vertical Settlement at P2 for Seismic Load Cases

Vertical displacements at P2 corner of the foundations due to seismic loads are shown in Figure-29.

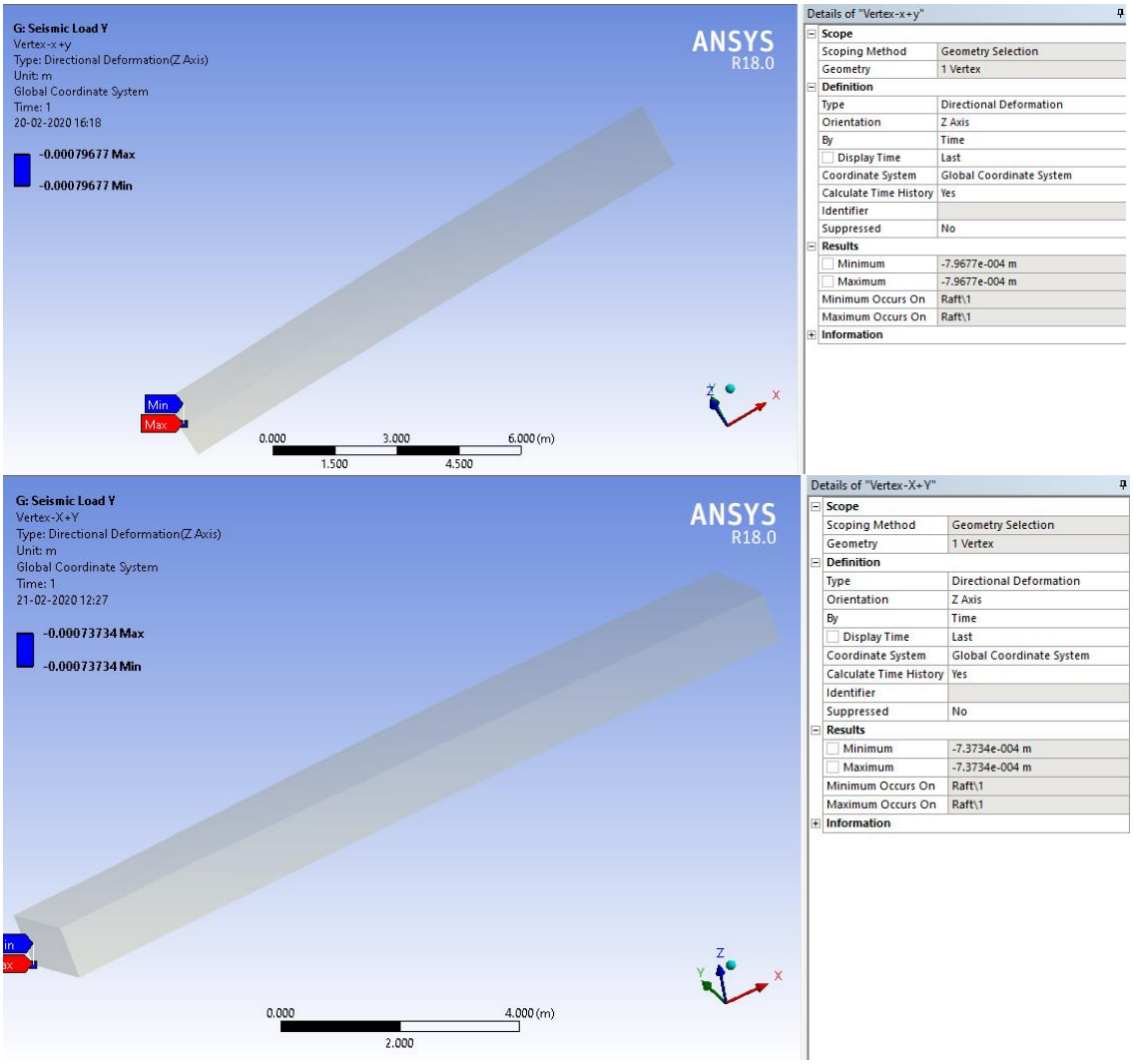


Figure 29. Vertical Deflection at P1 due to seismic load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.4.4. Vertical Settlement at P3 for Seismic Load Cases

Vertical displacements at P3 corner of the foundations due to seismic loads are shown in Figure-30.

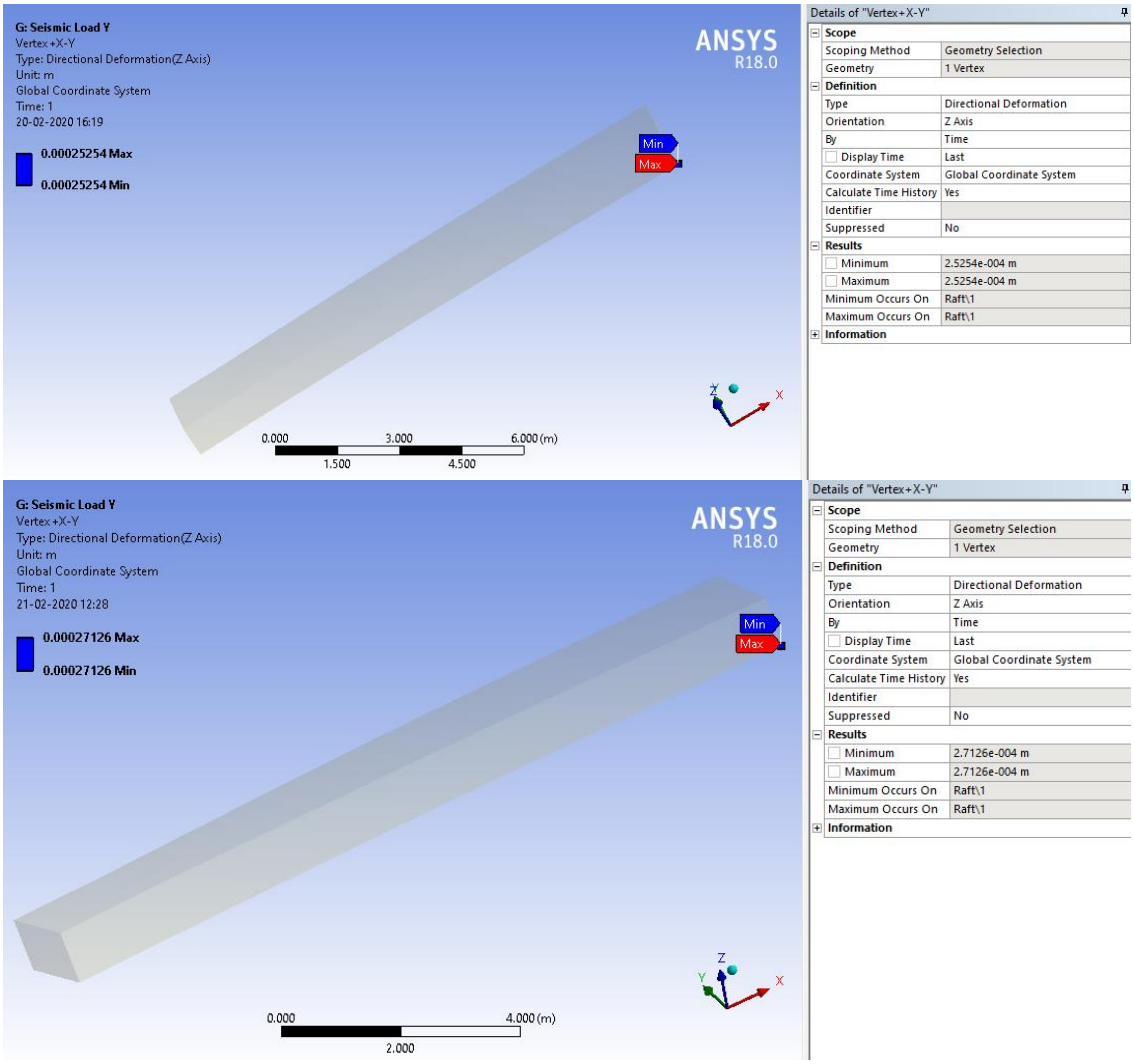


Figure 30. Vertical Deflection at P1 due to seismic loads case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.4.5. Vertical Settlement at P4 for Seismic Load Cases

Vertical displacements at P4 corner of the foundations due to seismic loads are shown in Figure-31.

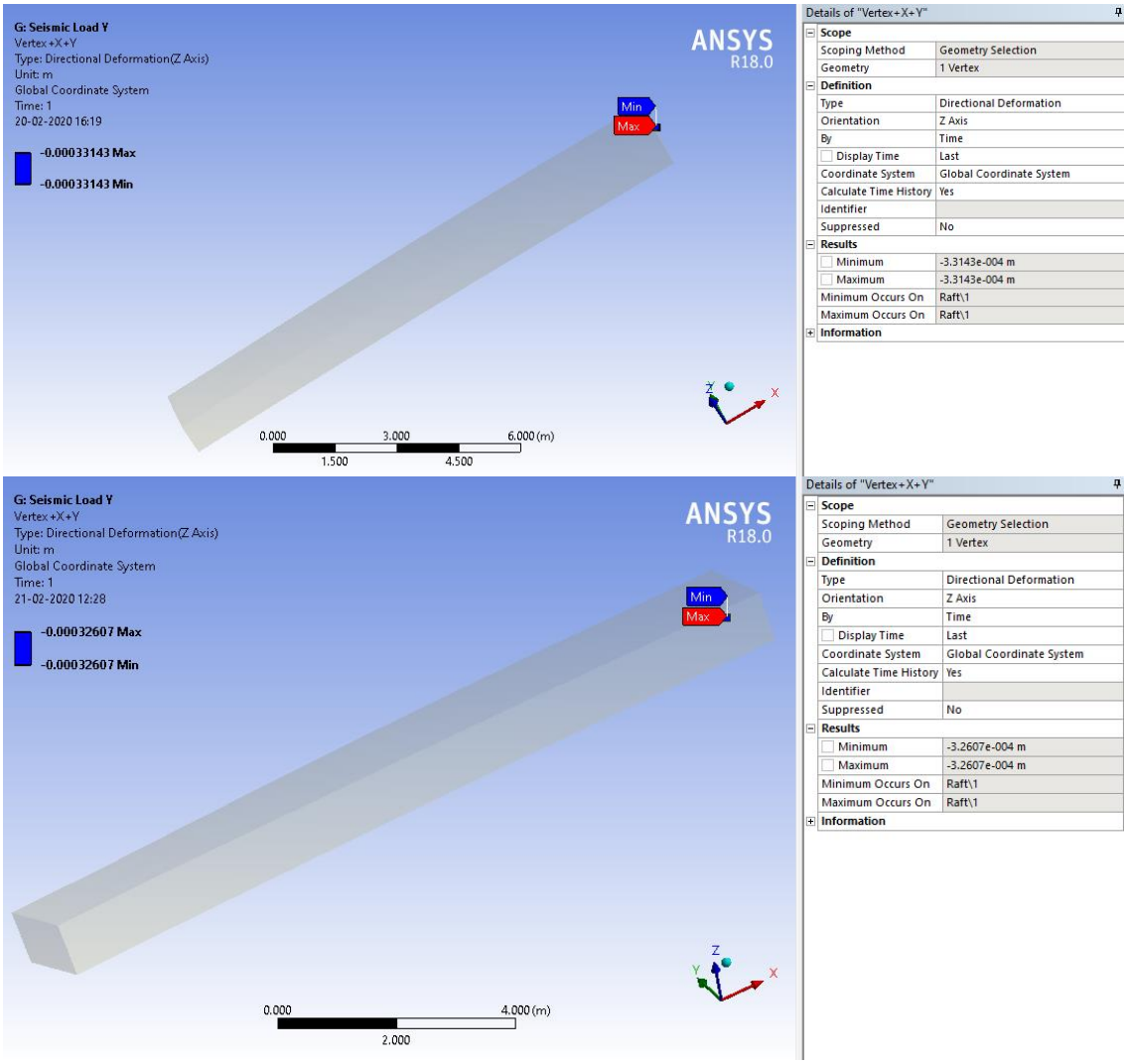


Figure 31. Vertical Deflection at P1 due to seismic loads case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.5. Wind Loads (W)

5.5.1. Vertical Displacement for Wind Load Cases

Vertical displacements of the foundations due to wind loads are shown in Figure-32.

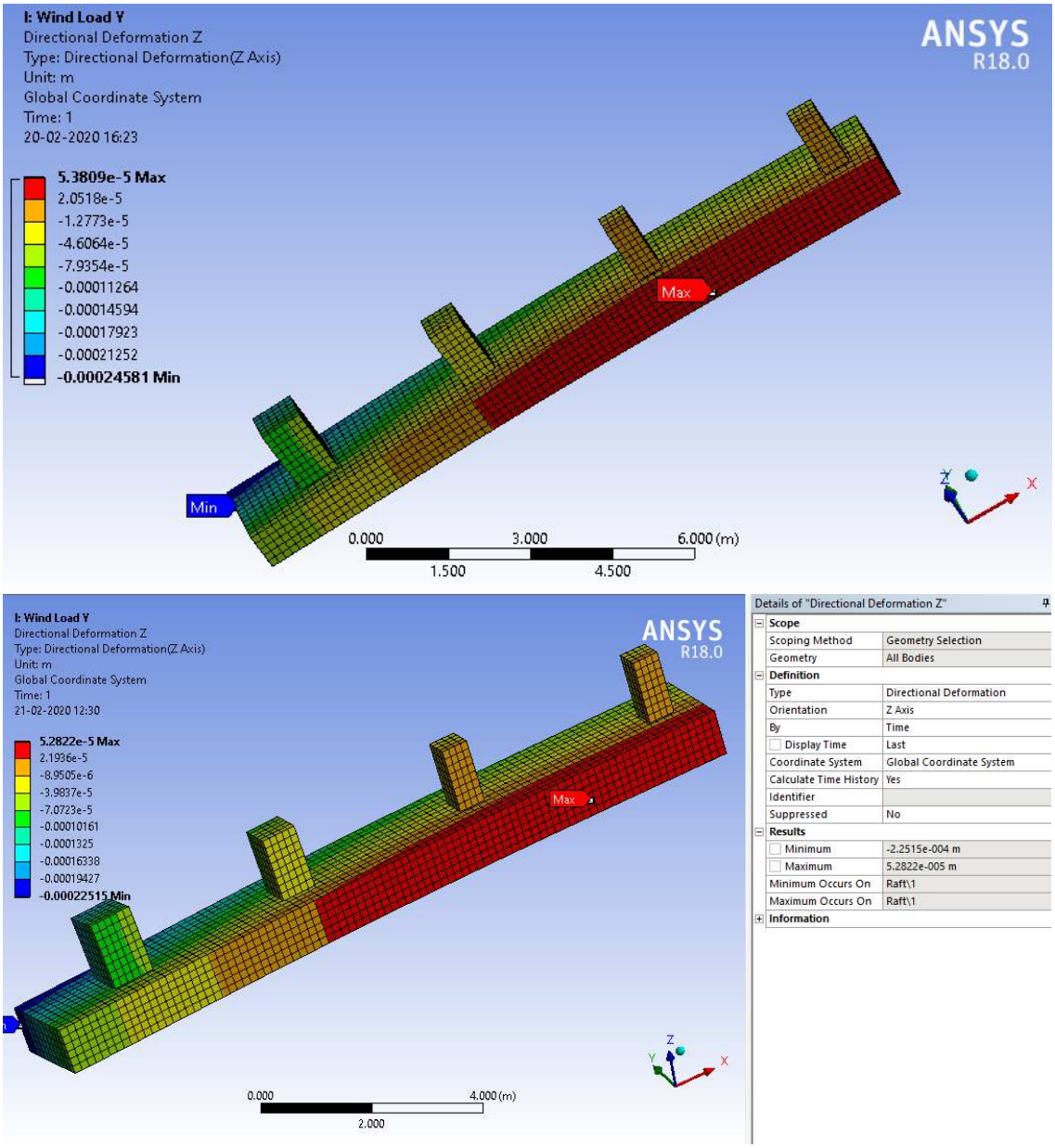


Figure 32. Vertical Deformation due to wind load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.5.2. Vertical Settlement at P1 for Wind Load Cases

Vertical displacements at P1 corner of the foundations due to wind loads are shown in Figure-33.



Figure 33. Vertical Deflection at P1 due to wind load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.5.3. Vertical Settlement at P2 for Wind Load Cases

Vertical displacements at P2 corner of the foundations due to wind load are shown in Figure-34.

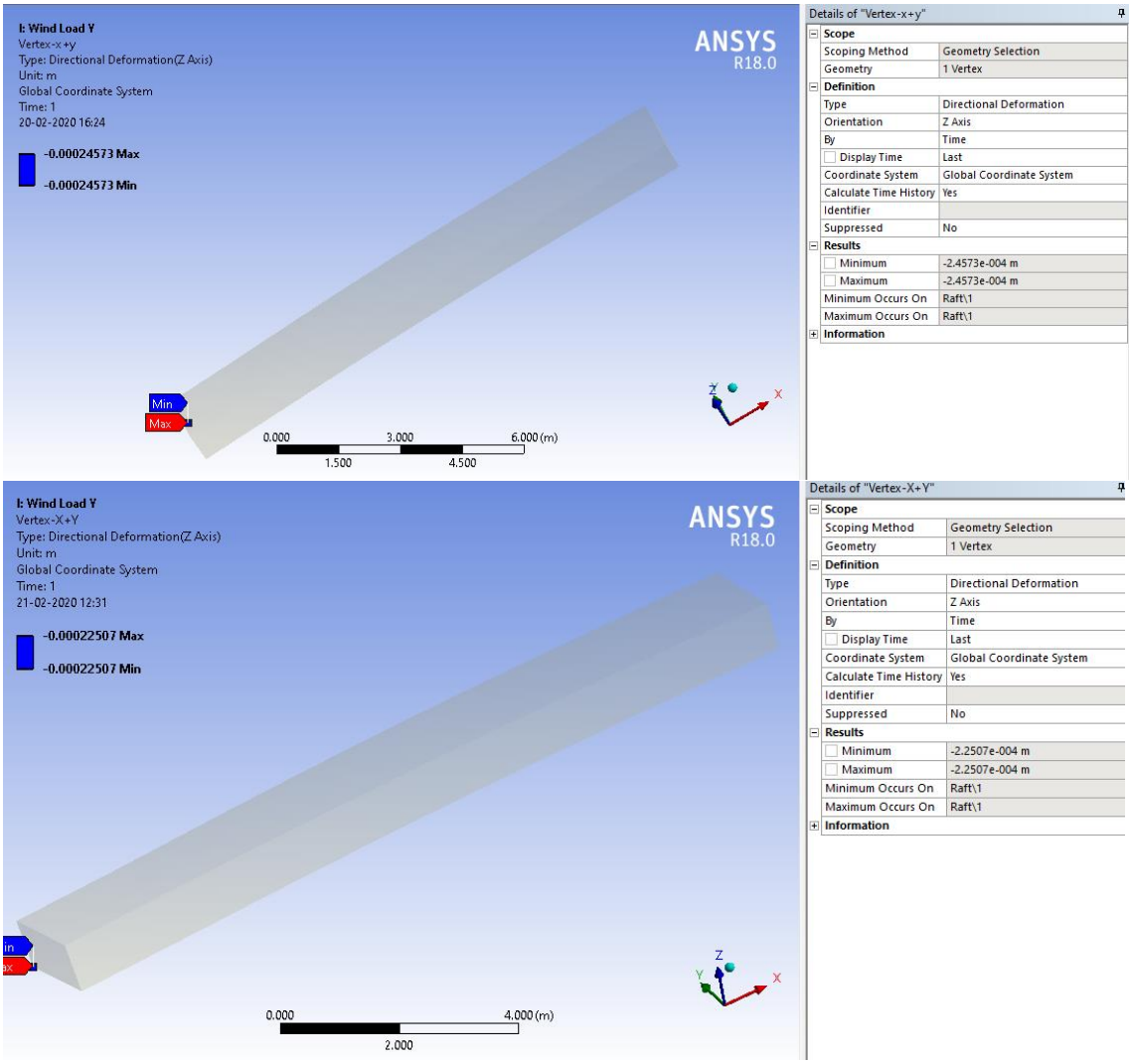


Figure 34. Vertical Deflection at P1 due to wind load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.5.4. Vertical Settlement at P3 for Wind Load Cases

Vertical displacements at P3 corner of the foundations due to wind loads are shown in Figure-35.

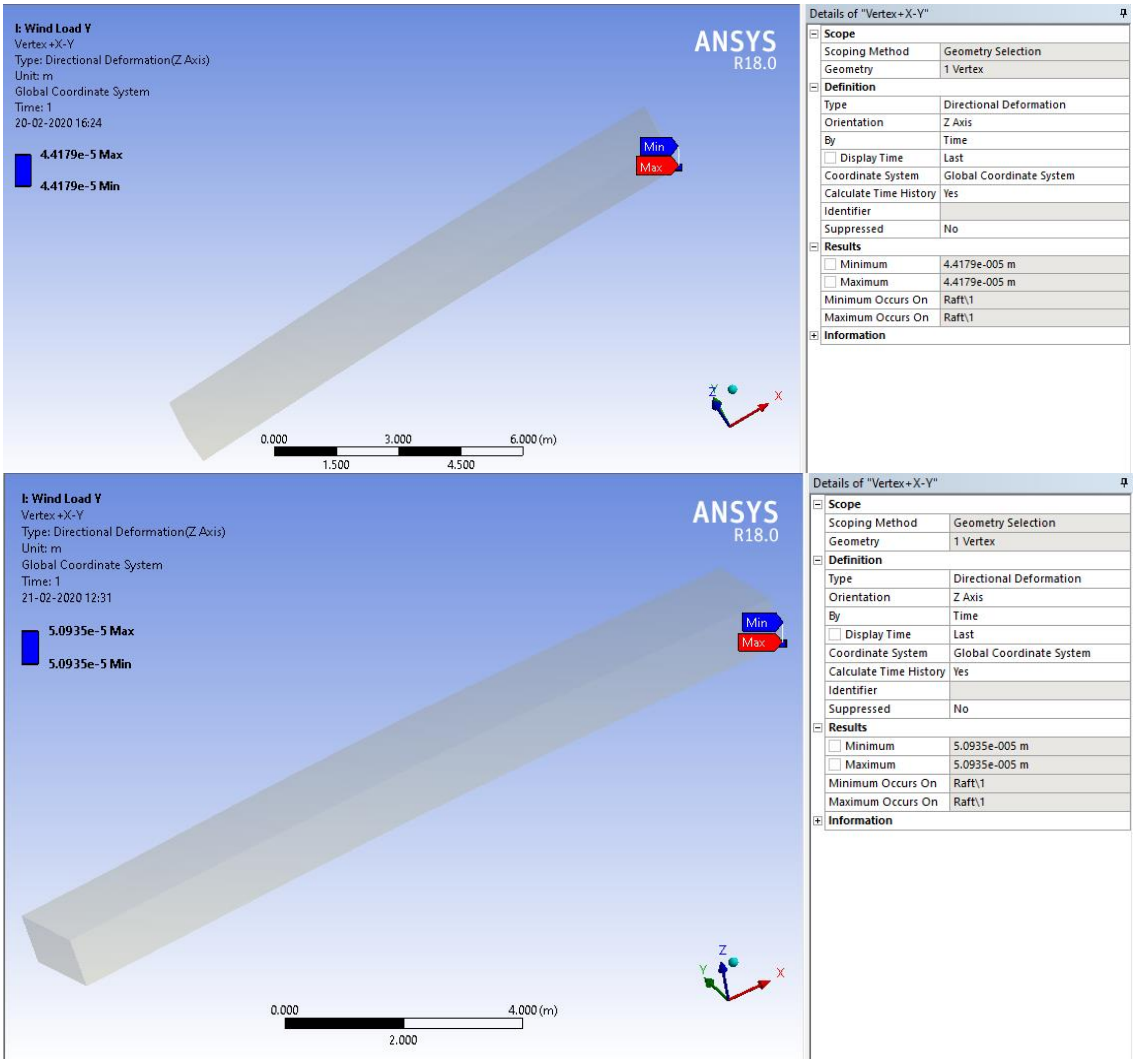


Figure 35. Vertical Deflection at P1 due to wind load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.5.5. Vertical Settlement at P4 for Wind Load Cases

Vertical displacements at P4 corner of the foundations due to wind loads are shown in Figure-36.

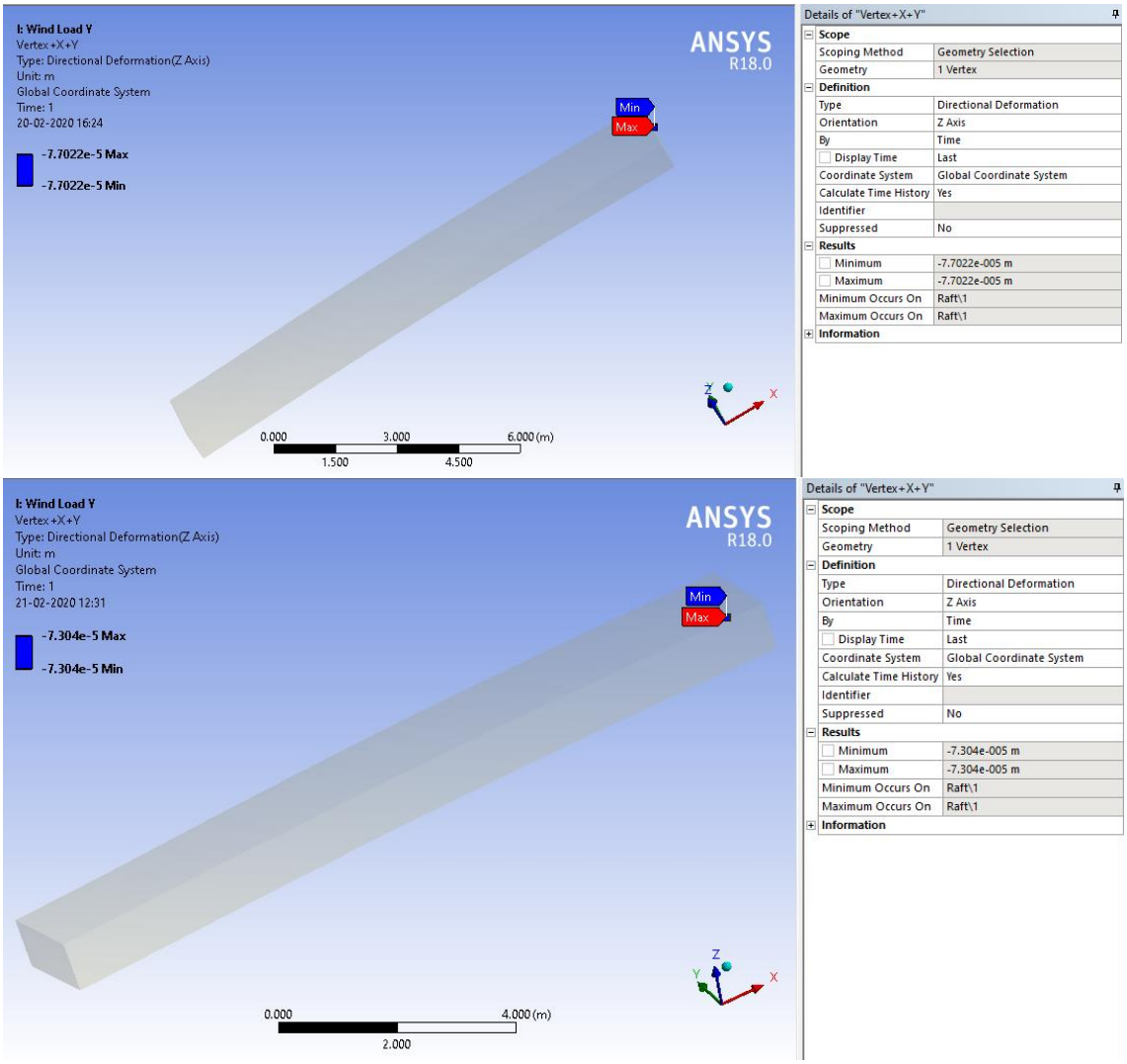


Figure 36. Vertical Deflection at P1 due to wind load case, above: smaller foundation (3710) and below: larger foundation (3710A)

5.6. Bridge Loads (B)

5.6.1. Vertical Displacement for Bridge Load Cases

Vertical displacements of the larger foundation due to bridge loads are shown in Figure-37.

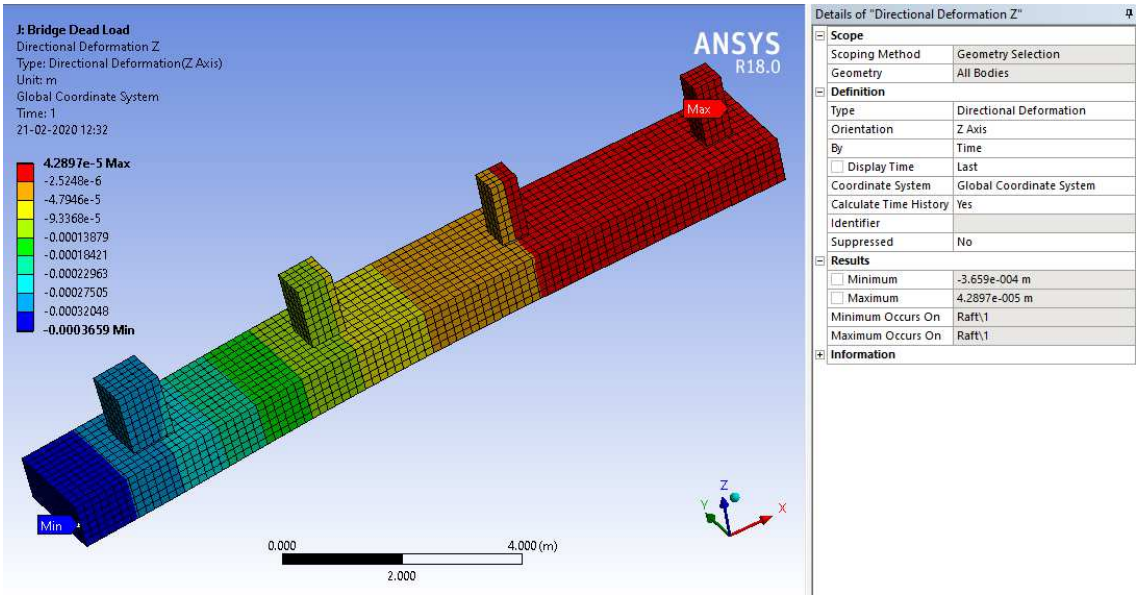


Figure 37. Vertical Deformation due to bridge load case, larger foundation (3710A)

5.6.2. Vertical Settlement at P1 for Bridge Load Case

Vertical displacement at P1 corner of the foundation for bridge load is shown in Figure-38.

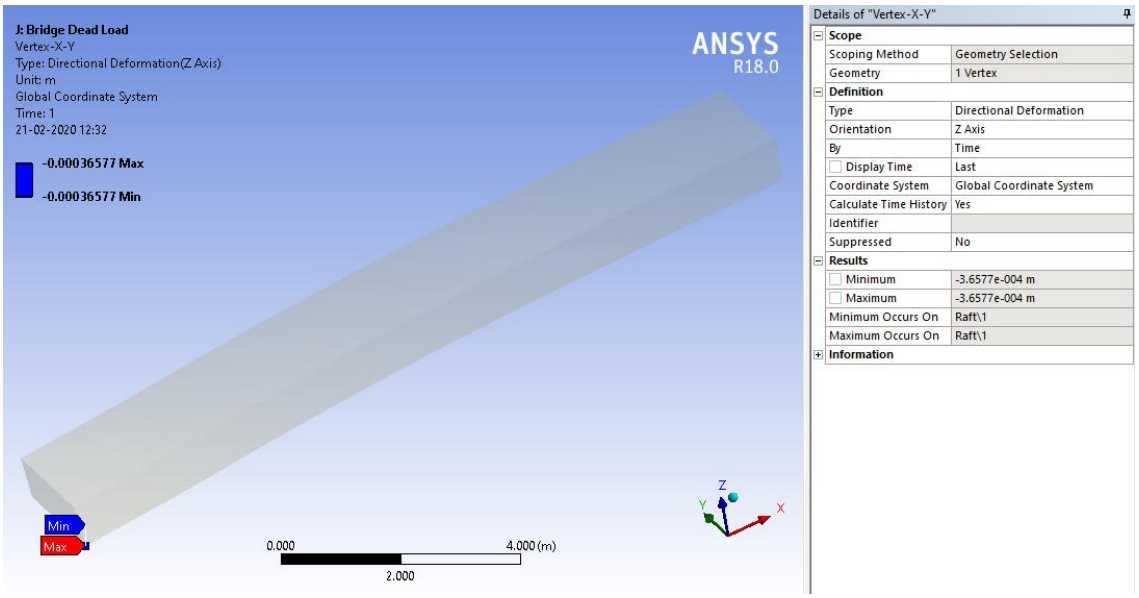


Figure 38. Vertical Deflection at P1 due to bridge load case, larger foundation (3710A)

5.6.3. Vertical Settlement at P2 for Bridge Load Cases

Vertical displacement at P2 corner of the foundation for bridge load is shown in Figure-39.

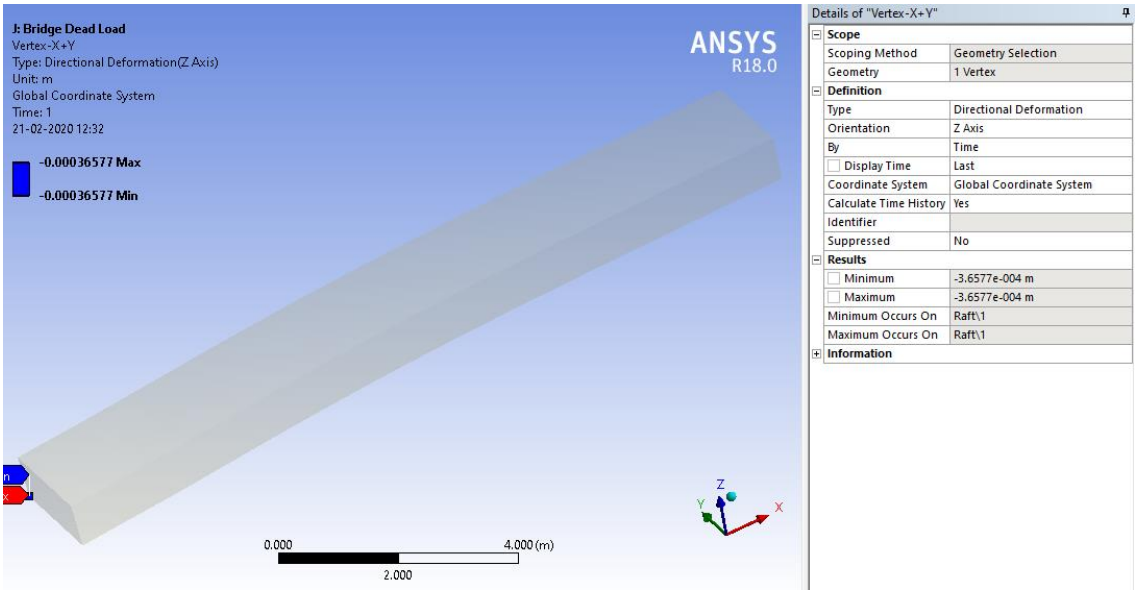


Figure 39. Vertical Deflection at P1 due to bridge load case, larger foundation (3710A)

5.6.4. Vertical Settlement at P3 for Bridge Load Cases

Vertical displacement at P3 corner of the foundation for bridge load is shown in Figure-40.

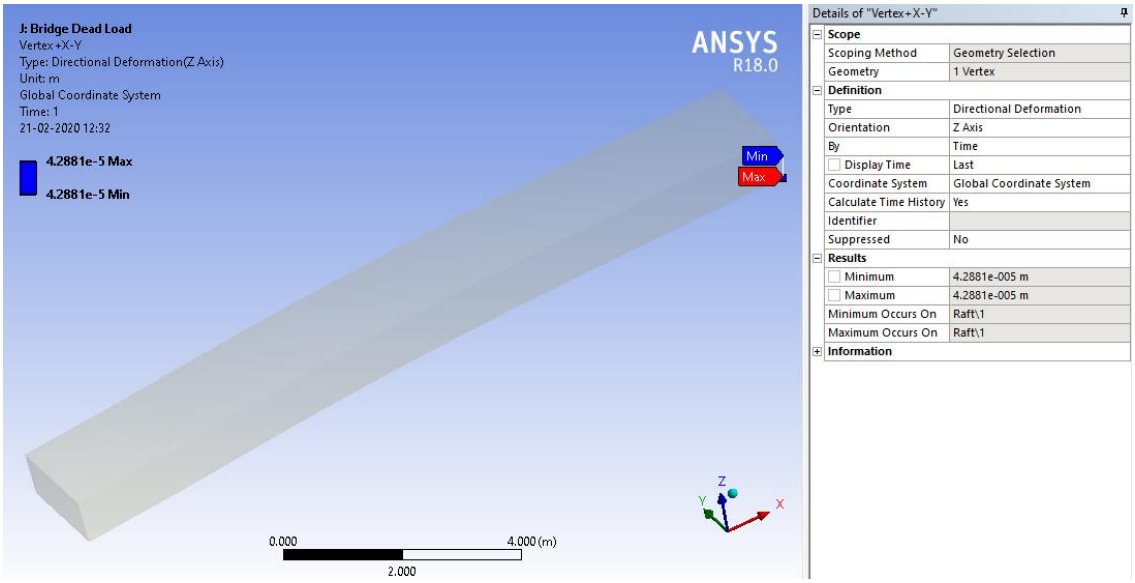


Figure 40. Vertical Deflection at P1 due to bridge load case, larger foundation (3710A)

5.6.5. Vertical Settlement at P4 for Bridge Load Cases

Vertical displacement at P4 corner of the foundation for bridge load is shown in Figure-41.

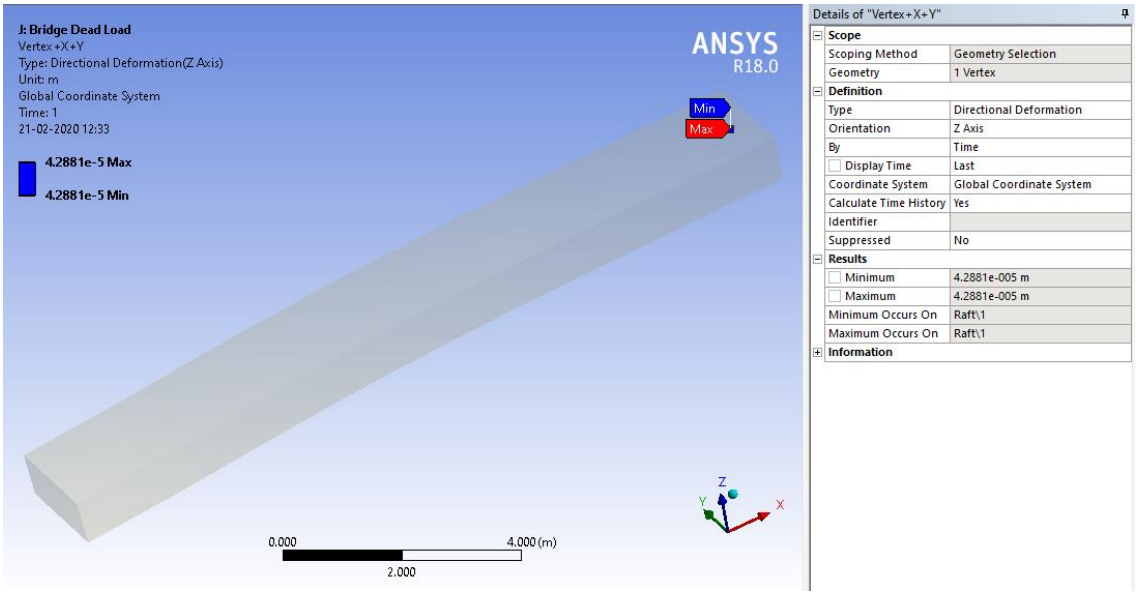


Figure 41. Vertical Deflection at P1 due to bridge load case, larger foundation (3710A)

5.7. Pressure for Primary Load Cases

	Deflection (micron)				Pressure (Pa)			
Load Cases	**P1**	**P2**	**P3**	**P4**	**P1**	**P2**	**P3**	**P4**
Self wt + Soil load + Tank Dead Load (D)	574	574	419	419	49077	49077	35825	35825
Max Fluid Load (F1)	3877	5023	5012	5794	331484	429467	428526	495387
Normal Fluid Load (F2)	3323	4308	4294	4966	284117	368334	367137	424593
Seismic Load (E)	-58	796	-252	331	-4959	68058	-21546	28301
Wind Load (W)	68	245	-44	77	5814	20948	-3762	6584

	Deflection (micron)				Pressure (Pa)			
Load Cases	**P1**	**P2**	**P3**	**P4**	**P1**	**P2**	**P3**	**P4**
Self wt + Soil load + Tank Dead Load (D)	554	554	414	414	47367	47367	35397	35397
Max Fluid Load (F1)	3499	4580	4888	5688	299165	391590	417924	486324
Normal Fluid Load (F2)	2999	3927	4187	4875	256415	335759	357989	416813
Seismic Load (E)	-69	737	-271	326	-5900	63014	-23171	27873
Wind Load (W)	57	225	-51	73	4874	19238	-4361	6242
Bridge Dead (B)	366	366	-43	-43	31293	31293	-3677	-3677

Table 5. Deflection & Pressure due to Primary Load cases, above: smaller foundation (3710) and below: larger foundation (3710A)

5.8. Pressure for Load Combinations

Load Case	Deflection (micron)				Pressure (Pa)			
	P1	P2	P3	P4	P1	P2	P3	P4
D + F1	4451	5597	5431	6213	380561	478544	464351	531212
D + F2 + W	4519	5842	5387	**6290**	386375	499491	460589	**537795**
D + F2 + 0.7 E	3856	5439	4537	5617	329722	465052	387879	480228

Load Case	Deflection (micron)				Pressure (Pa)			
	P1	P2	P3	P4	P1	P2	P3	P4
D + F1 + B	4419	5500	5259	6059	377825	470250	449645	518045
D + F2 + W + B	4476	5725	5208	**6132**	382698	489488	445284	**524286**
D + F2 + 0.7 E + B	3871	5363	4368	5474	330945	458528	373490	468044

Table 6. Deflections and Pressures for Load Combinations, above: smaller foundation (3710) and below: larger foundation (3710A)

6. Conclusions

The maximum vertical deflections for primary load cases are computed in the finite element analyses. Afterwards, principle of superposition is used for vertical deflections of load combination cases. The maximum deflection computed for smaller & larger foundations are 6290 & 6132 microns, respectively. The maximum Stresses under the foundations are 0.537 & 0.524 MPa.

Stress levels in the system are within 0.950 MPa, hence foundation is in safe zone.

7. References

[1] ANSYS Mechanical Software

[2] Table 6 - BH-14 of Geotechnical Report (MD- 216-3000-EG-CV-RPT-1008)

[3] Design calculation 3710 tailings paste thickening design of foundation (MD-216-3710-EG-CV-CAL-1391-A01)